Final project form

- Final prototype: demonstrable, runnable build implementing core MVP features.
- Backlog implementation traceability: explicit mapping of implemented backlog items to team members (who implemented what) and commits/pull requests or timestamps.
- Definition of done (DoD) applied: documented DoD used for each completed backlog item (e.g., code compiled, unit tests passing, peer review, updated docs); evidence that DoD was met.

Final reporting form			
This version must meet the definition of done; i.e.: code compiles and runs without errors, unit tests with decent coverage pass. Security issues are addressed.			
Team details			
Project name:	MatchPoint		
Team nickname	TLDR		
Team members (first/last name)	Thomas Kabalin	Luke Bryant	Danny Gebhard
Student numbers	KBLTHO001	BRYLUK001	GBHDAN001
Project description 1 or 2 lines	MatchPoint is a tennis statistics tracking that lets coaches capture live match data and turn it into player performance analytics.		
Git repo url	https://gitlab.cs.uct.ac.za/tldr/matchpoint		
Demo session			
Fill in the blue fields prior to the demo, fill in the green fields during or shortly after the demo, make sure to assign a member to take written notes during the demo			
Link to executable	https://matchpoint.thomaskabalin.com/		
Link to last commit	*chore: release version 1.0.0 (a266dbf5) · Commits · TLDR / matchpoint · GitLab*		
List of backlog items implemented	1. Data model 2. Player, match and point APIs 3. Accounts And Access 4. Scoring Engine 5. Match Data Capture 6. Statistics And Analytics 7. Offline Capture 8. Home Page & App Navigation Flow 9. Test data generation		
Demos will be in the week of 13 October: Monday to Friday.			

Backlog items implemented	
For each implemented backlog item, state what you actually implemented. Ergo, ament the planning document. Add implementation details in the blue fields.	
Scope adjustments?	
Description	Scope expanded in four main areas since the skateboard prototype: • Player accounts (Item 3): players can now sign in with a generated username and one-time password, and see their own scoped view of matches, stats and settings. This was not in the original plan. • Capture UI (Item 5): major UX redesign. The final ball position now infers the point winner (fewer taps), the court zooms into the region after placement, rally length uses a visual basket display, and match setup gained event naming and opponent suggestions. • Analytics (Item 6): added surface filtering, mean rally length by outcome, and last-five-match roster figures. • Navigation (Item 8): now role-aware (coach versus player views). The home page gained a start-match quick action and squad attention alerts; settings gained an FAQ, privacy policy, terms of service, and app version display. • Darkmode (item 3, 6 and 8) was removed from the scope due to the unforeseen effect on every page and design decision – an entire second visual redesign had to be redone for every graph and colour choice and this expanded the scope too large as seen fit by the team – this was not a requirement by the client in the first place so was deemed acceptable to remove. • Bar chart break downs retired as much as possible throughout the app in favour of cobweb diagrams, tennis court representation of data and scatter plots with profile pictures representing the players (item 6) as per request from the client after the demo • Rally intensity chart on rally tab (representing proportion of rallies longer than 5 shots) and match momentum graphs on matches cards (representing how dominant a win was during a game) added as per client request after demo (item 6) to bring more visualisation to the user. • Player sign in functionality (item 8) added to allow a player themselves to sign in and to see their own stats and only viewable by them – as per client request since the last demo. This includes a new separate sign in and new Bottom tab navigation with 3 tabs (Me, Matches, Settings) for logged in player. • Start match action added to every available screen besides squad stats (item 8) as per client request • Major redesigns to the Matches and player statistics cards. These mainly included removing drill down functionality (which was an early design decision) in favour for scrolling – as per the client request (item 8). • Fake data generation – more data than before was generated, with the addition of play pictures Scope reduced: MFA was dropped in Assignment 2 (OTP, rate limiting and the NIST password policy were deemed sufficient) and remains dropped.

Backlog item 1						
Name	Data model					
Description	Shippable increment: migrated PostgreSQL schema plus SQLAlchemy models and test factories					
Priority	High					
Main developer	Thomas Kabalin					
Pair programming?	0%					
Time spend	KBLTHO001	100%	BRYLUK001	0%	GBHDAN001	0%
Estimated time	*0.5 weeks*					
Acceptance criteria (DoD):						
Functionality	Schema and models must accurately represent coach, player, match, and point data, structured for use by the API layer. Must support input validation and access control at the database level to guard against SQL injection attacks and unauthorized data access					
Security issues	• Direct external access to the database bypassing the application's own access controls and business logic • Exposure of the database port to the internet or wider network, allowing unauthorised connection attempts • Credential reuse across environments, where a leak in one environment compromises others • Hardcoded credentials being exposed via source control history • SQL injection					
Test cases	• Given a migration is run, expect it to complete cleanly with no errors or data loss • Given the test factories are run, expect them to generate valid records for all entities (coach, player, match, point) • Given invalid or malformed input is submitted, expect it to be rejected at the model/DB layer with no injection-style input reaching raw queries • Given a request attempts direct/unauthenticated access to the database, expect it to be denied, with access restricted to the application layer only					
Client remarks						
For **implemented** backlog items fill in the below fields prior to the demo						
actual time	0.5 weeks					
Link to last git commit	*418385c · docs(data-model): write down the two capture fields the API drops*					
Commit history	*See Appendix C at the end of this document*					
Acceptance criteria	Working PostgreSQL schema created through versioned Alembic migrations, covering all entities needed by the API layer (coaches, players, matches, match sets, points, coach sessions, email					

	verifications, password resets, player transfer codes, player sessions, deleted-match tombstones). SQLAlchemy models mapping every table, with input validation and access constraints enforced at the database level. Test factories that generate valid records for all entities, alongside a schema-head test confirming migrations and models stay in sync and a round-trip test verifying every migration applies and rolls back cleanly. Every Alembic migration has an explicit downgrade test asserting that the schema reverts cleanly. Point model refined so that serve outcome and ending may be null, covering serves and endings that were not watched. Scope adjustment (minor): point model refined to accept null for unwatched serves and endings; migration downgrade testing added.
DoD checklist Do not fill in, the tutor will edit these during the final demo	
Compile compiles	*Y/N if no: list any errors here (reviewee has 1 hour to fix non-compiling code, if so attach commit report), compiles after last minute fixes Y/N*
Unit test coverage	*None/Some/All*
Unit tests succeed	*None/Some/All*
Unit tests report	*Add unit test report to appendix T*
Security done?	*y/ n/ N/A (database, web, input, ... cannot be N/A)*
Security actions?	*Describe security fixes implemented or planned*

Backlog item 2						
Name	Player, match and point APIs					
Description	As a coach, I want to manage players, create matches with their metadata, and have captured points accepted reliably.					
Priority	High					
Main developer	Thomas Kabalin					
Pair programming?	0%					
Time spend	KBLTHO001	100%	BRYLUK001	0%	GBHDAN001	0%
Estimated time	*0.7 weeks*					
Acceptance criteria (DoD):						
Functionality	Must reliably create/read/update players and matches and accept point data with server-side validation. Every endpoint must require authentication and only expose a coach's own data.					
Security issues	• Cross-site request forgery (a malicious site could cause a logged-in user's browser to issue requests) • Cross-site scripting (XSS) via injected or unauthorised scripts • Protocol downgrade and SSL-stripping attacks that force a connection back to plain HTTP • MIME-type sniffing that causes a browser to misinterpret and execute an uploaded file's content • Leakage of sensitive URL parameters to third-party sites via the Referrer header • Unauthorised access to browser features (camera, microphone, geolocation) by embedded or malicious scripts • Denial-of-service via oversized payloads, decompression bombs, or memory exhaustion during image processing • Secrets and credentials leaking into logs or error responses, and difficulty tracing an incident back to a specific request • Stack traces exposing file paths, library versions, or internal implementation details to an attacker • Brute-force login attempts, credential stuffing, and request-flooding denial-of-service • Time-of-check-to-time-of-use race conditions, where concurrent requests slip past a check meant to block them • Mass-assignment, where unexpected fields in a request body modify data the client shouldn't be able to touch • Theft, replay, or double redemption of transfer codes from a database breach or concurrent requests • Enumeration of other users' private photos by guessing or incrementing predictable filenames					
Test cases	• Given valid player/match data is submitted, expect CRUD operations to succeed • Given an invalid point submission is made, expect the API to return a clear validation error • Given an unauthenticated request or a request for another coach's data, expect the API to reject it with a 401/403 response					

Client remarks	
For **implemented** backlog items fill in the below fields prior to the demo	
actual time	~1.75 weeks
Link to last git commit	*a0e0f28 · feat(api): require a scoring system when creating a match*
Commit history	*See Appendix C at the end of this document*
Acceptance criteria	Working CRUD endpoints for players and matches, accessible only to the authenticated coach who owns them, alongside photo upload and player transfer management. Batch point ingestion endpoint accepting up to 500 points per request, with idempotent duplicate rejection so retried submissions are safe. Match lifecycle management including capture claims and deletion with tombstones, alongside a point read endpoint for retrieving captured data. Account closure endpoint that cascades deletes across all of a coach's data. Security hardening across all endpoints: CSRF origin checking, security headers (CSP, HSTS), request body size limits, rate limiting, structured logging with request IDs, and generic error responses that never expose internals. Surface filtering across all analytics endpoints. Mean rally length split by point outcome. Last-five-match figures on the squad members endpoint. Player sessions can read their own matches and statistics through scoped read endpoints. Scope adjustment (expanded): analytics endpoints gained surface filtering, rally-length-by-outcome breakdowns, and last-five-match figures. Player sessions can now read their own matches and statistics.
DoD checklist Do not fill in, the tutor will edit these during the final demo	
Compile compiles	*Y/N if no: list any errors here (reviewee has 1 hour to fix non-compiling code, if so attach commit report), compiles after last minute fixes Y/N*
Unit test coverage	*None/Some/All*
Unit tests succeed	*None/Some/All*
Unit tests report	*Add unit test report to appendix T*
Security done?	*y/ n/ N/A (database, web, input, ... cannot be N/A)*
Security actions?	*Describe security fixes implemented or planned*

Backlog item 3						
Name	Accounts And Access					
Description	As a coach, I want my own account and sign-in so that my players and match data are private to me and only reachable by me. As a player I want to sign into my own account and view my own statistics, private to me and only reachable by me.					
Priority	Middle					
Main developer	Danny Gebhard					
Pair programming?	0%					
Time spend	KBLTHO001	40%	BRYLUK001	0%	GBHDAN001	60%
Estimated time	*0.7 weeks*					
Acceptance criteria (DoD):						
Functionality	Must let a coach register, log in, and log out, with passwords hashed, sessions correctly isolating each coach's data, and multi-factor authentication supported to reduce the risk of broken authentication.					
Security issues	• Plaintext password exposure on database compromise • Brute-force cracking of stolen hashes • Credential-stuffing attacks using passwords already known to be compromised elsewhere • Weak, guessable passwords vulnerable to dictionary and pattern-based attacks • Account enumeration, where an attacker maps which emails are registered for phishing or targeted credential stuffing • A timing side-channel that reveals whether an email exists based on response time • Session hijacking via reuse of tokens leaked in a database breach • OTP brute-forcing, OTP reuse from a leaked database, and stale codes remaining usable long after issue • Unauthorised account deletion via a hijacked or unattended session • Residual sensitive data being readable by the next user of a shared or public device					
Test cases	• Given valid registration/login details are submitted, expect the account to be created/logged in successfully; given invalid credentials, expect them to be rejected • Given a password is stored, expect it to be hashed and never stored in plaintext • Given Coach A attempts to access Coach B's data, expect the request to be denied					
Client remarks						
For **implemented** backlog items fill in the below fields prior to the demo						
actual time	1.5 weeks (Danny ~0.75 weeks frontend, Thomas ~0.75 weeks backend)					

Link to last git commit	https://gitlab.cs.uct.ac.za/tldr/matchpoint/-/commit/9478e0d4bd65ffe266e9e71c429343e02d554cb2
Commit history	See Appendix C backlog item 3
Acceptance criteria	• Coach sign-up with email verification (6-digit OTP), sign-in, sign-out, and forgot-password flows, all rate-limited with 429 and Retry-After when limits are exceeded. • Passwords hashed with Argon2id, screened against the HIBP breach list, and validated by a NIST-compliant policy (repetition, sequence, common-word, and context checks). A client-side strength meter blocks weak passwords before submission. • Sessions use opaque tokens stored in HttpOnly cookies with renewal on use, keeping the coach signed in across visits without exposing the token to JavaScript. • Route guards redirect unauthenticated users to sign-in. Account enumeration is defended: all unauthenticated endpoints handle known and unknown identities with identical status codes, bodies, and rate limits (except sign-up OTP request returning a 409 error for already registered emails). • Settings pages for changing password, signing out other sessions, configuring match defaults, toggling appearance, and closing the account (with password re-entry). Profile editing covers display name and profile photo.
DoD checklist Do not fill in, the tutor will edit these during the final demo	
Compile compiles	*Y/N if no: list any errors here (reviewee has 1 hour to fix non-compiling code, if so attach commit report), compiles after last minute fixes Y/N*
Unit test coverage	*None/Some/All*
Unit tests succeed	*None/Some/All*
Unit tests report	*Add unit test report to appendix T*
Security done?	*y/ n/ N/A (database, web, input, ... cannot be N/A)*
Security actions?	*Describe security fixes implemented or planned*

Backlog item 4						
Name	Scoring Engine					
Description	As a coach, I want the app to work out the score from the points I record so that I never track games and sets by hand.					
Priority	High					
Main developer	Luke Bryant					
Pair programming?	0%					
Time spend	KBLTHO001	0%	BRYLUK001	100%	GBHDAN001	0%
Estimated time	*0.5 weeks*					
Acceptance criteria (DoD):						
Functionality	Must correctly calculate game/set/match scores from recorded points, matching standard scoring rules exactly (verified against known scenarios like deuce/tiebreak).					
Security issues	• A client-computer score being trusted as truth • Malformed or hostile data reaching the fold • Cross-match contamination • Injection through free-text carried alongside the score					
Test cases	• Given a standard sequence of points is recorded, expect the game/set/match score to be calculated correctly • Given an edge case scenario (deuce, tiebreak, advantage) occurs, expect the score to resolve correctly per standard rules					
Client remarks						
For **implemented** backlog items fill in the below fields prior to the demo						
actual time	*1.5 week*					
Link to last git commit	*29faae8 · docs: name the API as a second reader of scoring_system*					
Commit history	*See Appendix C at the end of this document*					
Acceptance criteria	• Functionality - All seven scoring systems in docs/scoring-systems.md (traditional, men's Grand Slam, no-ad, match tiebreak, short sets, pro set, and legacy advantage set) are implemented and confirmed with the client. Game, set, and match scores, service rotation, court side, and game/break/set/match point stamps are all derived from recorded points rather than entered manually by the coach. The score is calculated by folding over the match's event log, so a span the coach did not watch is treated as a checkpoint they vouch for rather than as invented points. • Test cases - 71 unit tests pass, covering every DoD edge case, including deuce, no-ad, tiebreaks, short sets, legacy sets, and match-tiebreak scoring. A board snapshot round-trip also confirms the fold and score editor produce the same scoreboard. Three Playwright e2e tests validate the real capture UI: deuce showing AD, a tiebreak deciding a set, and undoing a game-winning point while capture continues.					

	• Security - the scoring rules run on the client so matches can be scored offline. The API treats submitted scores as claims and validates them: values are range-checked, set numbers must be in order, unknown fields return 422 and checkpoints are limited to coach's own match. Invalid, out-of-order, or cross-match events are ignored without crashing the screen. Free text is safely escaped by react and never sent directly to SQL
DoD checklist Do not fill in, the tutor will edit these during the final demo	
Compile compiles	*Y/N if no: list any errors here (reviewee has 1 hour to fix non-compiling code, if so attach commit report), compiles after last minute fixes Y/N*
Unit test coverage	*None/Some/All*
Unit tests succeed	*None/Some/All*
Unit tests report	*Add unit test report to appendix T*
Security done?	*y/ n/ N/A (database, web, input, ... cannot be N/A)*
Security actions?	*Describe security fixes implemented or planned*

Backlog item 5						
Name	Match Data Capture					
Description	As a coach at the court, I want to set up a match in a few taps and then record every point in no more than three taps so that the whole courtside workflow keeps up with live play.					
Priority	Middle					
Main developer	Luke Bryant					
Pair programming?	0%					
Time spend	KBLTHO001	0%	BRYLUK001	100%	GBHDAN001	0%
Estimated time	*0.9 weeks*					
Acceptance criteria (DoD):						
Functionality	Must let a coach set up a match easily and record each point in no more than 3 taps accurately and in real time. Access must be restricted to the authenticated coach's own match					
Security issues	• Data loss through a crash in the scoring path • Unauthorised access to match data • The note field is uncontrolled text input					
Test cases	• Given a coach sets up a new match, expect setup to be completed in ≤3 taps • Given a coach records a point, expect it to be completed in ≤3 taps • Given points are entered rapidly in succession, expect each point to be correctly attributed with no duplicates • The UI should not allow the coach to enter a state that is impossible as per the rules of tennis (e.g. asking for a forced / unforced error classification when the player has not yet served)					
Client remarks						
For **implemented** backlog items fill in the below fields prior to the demo						
actual time	*1 week + 0.5 week after demo*					
Link to last git commit	*af31b11 · fix(web): fade the final shot labels out as the exit zooms*					
Commit history	*See Appendix C at the end of this document*					
Acceptance criteria	• Functionality - when match is started, with saved defaults for surface and scoring system, leaving only opponent selection, with optional event name. Every scoring point is recorded in minimal taps. The client-approved richer detail adds rally length, forced/unforced, wing and shot type as optional taps, while "Didn't see" stores null instead of guessing. Capture is fully offline: taps write synchronously to a local outbox, with access scoped to the authenticated coach and an expiring device claim preventing two devices from recording the same match. • Test cases - 708 tests pass across capture and outbox, with 95.2% statement and 91.4% branch coverage, alongside Playwright e2e tests using the real UI. Duplicate entry is prevented at three tested layers: court taps are blocked during their flash, final-shot dispatch					

	cannot queue twice, and the state machine rejects actions after a point is committed. Invalid states are unreachable by construction, with each screen only exposing valid actions and forced/unforced unavailable until the ball is recorded. • Security - The outbox append never rejects, preventing points from being lost to unhandled failures. Points are stored in IndexedDB before sending, so matches survive reloads, tab crashes and sign-outs, with e2e tests for each and a banner restoring the match at the same score.
DoD checklist Do not fill in, the tutor will edit these during the final demo	
Compile compiles	*Y/N if no: list any errors here (reviewee has 1 hour to fix non-compiling code, if so attach commit report), compiles after last minute fixes Y/N*
Unit test coverage	*None/Some/All*
Unit tests succeed	*None/Some/All*
Unit tests report	*Add unit test report to appendix T*
Security done?	*y/ n/ N/A (database, web, input, ... cannot be N/A)*
Security actions?	*Describe security fixes implemented or planned*

Backlog item 6						
Name	Statistics And Analytics					
Description	As a coach and a player, I want the statistics computed from captured points, shown as a post-match summary and as a dashboard of trends across matches and opponents.					
Priority	Low					
Main developer	Danny Gebhard					
Pair programming?	0%					
Time spend	KBLTHO001	30%	BRYLUK001	0%	GBHDAN001	70%
Estimated time	*0.7 weeks*					
Acceptance criteria (DoD):						
Functionality	Must accurately compute post-match summaries and cross-match trend dashboards from captured data, visible only to the coach who owns that data.					
Security issues	• Unauthorised access to other coaches' player data • Unauthorised creation, modification, or deletion of player data through these endpoints					
Test cases	• Given match data has been captured, expect the post-match summary to match manually calculated values • Given data exists across multiple matches/opponents, expect the dashboard to correctly display and aggregate trends • Given a coach views their statistics, expect them to see only their own data and not another coach's • For each statistic, there should be unit tests to ensure that it is computed correctly, even with edge-case data					
Client remarks						
For **implemented** backlog items fill in the below fields prior to the demo						
actual time	0.75 weeks (Danny ~0.5 weeks frontend, Thomas ~0.25 weeks backend)					
Link to last git commit	https://gitlab.cs.uct.ac.za/tldr/matchpoint/-/commit/b2f65377db87cd99					
Commit history	*See appendix C backlog item 6*					
Acceptance criteria	• Player profile with serve and rally statistics tabs, stat cards with percentage formatting, and various diagrams for visualising categorical distributions (“fault mode split” courts, shot type radars) with preference of minimal clicks (scrolling preferred). • Coach notes accessible per player via a dedicated view. • Squad stats page with a headline card, leaderboard, player insight card scatter plot and trend chart across the roster.					

	• A consistent colour palette differentiating stat categories. • Read-only API endpoints computing analytics from captured points scoped to the authenticated coach's data. • Rally intensity chart on rally tab (representing proportion of rallies longer than 5 shots) and match momentum graphs on matches cards (representing how dominant a win was during a game).
DoD checklist Do not fill in, the tutor will edit these during the final demo	
Compile compiles	*Y/N if no: list any errors here (reviewee has 1 hour to fix non-compiling code, if so attach commit report), compiles after last minute fixes Y/N*
Unit test coverage	*None/Some/All*
Unit tests succeed	*None/Some/All*
Unit tests report	*Add unit test report to appendix T*
Security done?	*y/ n/ N/A (database, web, input, ... cannot be N/A)*
Security actions?	*Describe security fixes implemented or planned*

Backlog item 7						
Name	Offline Capture					
Description	As a coach on a court with no signal, I want capture to keep working and sync when connectivity returns so that I never lose a match's data.					
Priority	Low					
Main developer	Thomas Kabalin					
Pair programming?	0%					
Time spend	KBLTHO001	90%	BRYLUK001	10%	GBHDAN001	0%
Estimated time	*0.7 weeks*					
Acceptance criteria (DoD):						
Functionality	Must keep recording points with no connection, then sync accurately with no data loss once connectivity returns, without exposing cached data to other apps/users.					
Security issues	• Device fingerprinting through client-generated identifiers • Cross-user data exposure on a shared device after sign-out • A malicious client submitting crafted IDs to collide with or predict another user's resources • Unsynced data remaining on a device after its session is revoked for a security reason • Fabricated or tampered point data submitted in bulk during sync					
Test cases	• Given the device has no network connection, expect the app to continue recording points successfully • Given connectivity returns after offline recording, expect queued data to sync correctly with no loss or duplication					
Client remarks						
For **implemented** backlog items fill in the below fields prior to the demo						
actual time	~0.85 weeks					
Link to last git commit	*331acee · docs: describe the corrected-match guard, not the defect it replaced*					
Commit history	*See Appendix C at the end of this document*					
Acceptance criteria	Working offline point recording using a local IndexedDB outbox that persists across page reloads and app restarts, so capture keeps working with no network connection. Automatic sync to the server when connectivity returns, with idempotent ingestion ensuring no data loss or duplication. A capture claim mechanism preventing two devices from recording the same match simultaneously. Tombstone handling so that points queued for a match deleted elsewhere are discarded cleanly on sync. The outbox isolated in its own IndexedDB database, separate from the read cache, so that clearing or evicting cached data can never lose unsynced points. Captured points map onto the full API point contract, including the nullable serve and ending fields.					

	End-to-end offline tests cover the complete point round trip, from capture-screen taps through the outbox to server sync. Cross-device data handling improved, with honest device gating. Scope adjustment (expanded): offline sync now handles the full point payload, which was incomplete at the skateboard stage. End-to-end tests cover the point round trip.
DoD checklist Do not fill in, the tutor will edit these during the final demo	
Compile compiles	*Y/N if no: list any errors here (reviewee has 1 hour to fix non-compiling code, if so attach commit report), compiles after last minute fixes Y/N*
Unit test coverage	*None/Some/All*
Unit tests succeed	*None/Some/All*
Unit tests report	*Add unit test report to appendix T*
Security done?	*y/ n/ N/A (database, web, input, ... cannot be N/A)*
Security actions?	*Describe security fixes implemented or planned*

Backlog item 8						
Name	Home Page & App Navigation Flow					
Description	As a coach, I want a home page that gives me a clear overview when I log in, with easy navigation to my players, matches, and stats, so that I can quickly get to what I need without hunting through menus and As a player I want a home page that gives me a clear overview when I log in, with easy navigation to my matches, and stats, so that I can quickly get to what I need without hunting through menus.					
Priority	Middle					
Main developer	Danny Gebhard					
Pair programming?	0%					
Time spend	KBLTHO001	0%	BRYLUK001	0%	GBHDAN001	100%
Estimated time	*0.5 weeks*					
Acceptance criteria (DoD):						
Functionality	Must present an authenticated coach with a home page summarizing key info (e.g., recent matches, quick actions) and provide clear, consistent navigation to all core app pages (players, matches, stats, account), restricted to that coach's own data only.					
Security issues	• Unauthorised or unauthenticated access to data via the API • Unauthenticated users reaching protected pages or views directly by URL • Enumeration of valid application routes or resources via differing 404 versus authenticated-page behaviour					
Test cases	• Given an authenticated coach logs in, expect the home page to load and display only that coach's own data • Given a coach with recent matches recorded, expect the home page to display a summary of those recent matches • Given a coach taps/clicks a navigation link (e.g., Players, Matches, Stats), expect the app to route to the correct corresponding page • Given an unauthenticated user attempt to access the home page directly (e.g., via URL), expect the app to redirect them to the login page • Given a coach navigates between core pages (Players, Matches, Stats, Settings), expect all links to work with no dead ends or broken routes • Given a coach on a typical mobile connection loads the home page, expect it to fully load within an acceptable time (e.g., ≤2 seconds)					
Client remarks						
For **implemented** backlog items fill in the below fields prior to the demo						
actual time	*1 week*					
Link to last git commit	*https://gitlab.cs.uct.ac.za/tldr/matchpoint/-/commit/2bc9b64236828431faae1021efa2f372c7df4bc6*					

Commit history	See Appendix C backlog item 8
Acceptance criteria	• Home page showing the coach's display name, squad snapshot (player and match counts), a last-viewed player card, a player bubble row with avatars, a player search bar, and an add-player tile. • Bottom navigation with four tabs (Home, Players, Matches, Stats) providing one-tap access to every area of the app. • Player roster page with list, add, edit, and delete actions. Match list page with add and delete actions. • Profile editing for display name and profile photo, with the MatchPoint logo mark in the app header (Which also serves as a navigation element to home page). • A custom 404 page for unmatched routes, and a PWA install guide with platform-specific instructions for iOS, Android, and desktop. • Route guards redirect unauthenticated users to sign-in; all data fetched through authenticated API calls. • Bottom tab navigation with 3 tabs (Me, Matches, Settings) for logged in player. • 'Start match' action available on every screen besides squad stats • Scrolling functionality within a page throughout the app, with minimal drill down design where possible.
DoD checklist Do not fill in, the tutor will edit these during the final demo	
Compile compiles	*Y/N if no: list any errors here (reviewee has 1 hour to fix non-compiling code, if so attach commit report), compiles after last minute fixes Y/N*
Unit test coverage	*None/Some/All*
Unit tests succeed	*None/Some/All*
Unit tests report	*Add unit test report to appendix T*
Security done?	*y/ n/ N/A (database, web, input, ... cannot be N/A)*
Security actions?	*Describe security fixes implemented or planned*

Backlog item 9						
Name	Test data generation					
Description	As a developer, I want realistic seed/test data for coaches, players, matches, and points so that the team can properly test functionality, edge cases, and performance without relying on manual data entry.					
Priority	Middle					
Main developer	Luke Bryant					
Pair programming?	0%					
Time spend	KBLTHO001	20%	BRYLUK001	80%	GBHDAN001	0%
Estimated time	*0.4 weeks*					
Acceptance criteria (DoD):						
Functionality	Create fake data for testing, analytics side of the application. These fake accounts and match data is used during the demo.					
Security issues	• Exposure of a privileged or destructive operation to unauthenticated or unauthorised users via a public API • Accidental or malicious invocation of the test data injection script against a production environment, corrupting or polluting real data					
Test cases	• Given the test data script is run, expect it to populate the database with valid coach, player, match, and point records with no errors • Given generated match data, expect it to include edge case scenarios (e.g. deuce, tiebreak, double faults) so scoring logic can be fully tested • Given the test data script is run in a non-production environment, expect it to have no effect on production data • Given the test data script is run multiple times, expect it to not create duplicate or conflicting records					
Client remarks						
For **implemented** backlog items fill in the below fields prior to the demo						
actual time	*0.1 week+0.1 week post demo Luke, 0.1 Thomas*					
Link to last git commit	*afb5c09 · feat: file dated coach notes on every demo player*					
Commit history	*See Appendix C at the end of this document*					
Acceptance criteria	• Functionality - the generator drives the real capture state machine and scoring fold rather than inventing points. The committed seed contains 3 coaches, 36 players, 360 matches, 48,671 points and 194 dated coach notes, generated deterministically from seed 13 across 12 player archetypes. • Security - no public access, the import and purge cannot be run through the app or API. They require direct access to the server. Production safety, running them requires a manual backup first. The purge only affects the demo coaches, and their password are generated each run and never stored in the code					
DoD checklist Do not fill in, the tutor will edit these during the final demo						

Compile compiles	*Y/N if no: list any errors here (reviewee has 1 hour to fix non-compiling code, if so attach commit report), compiles after last minute fixes Y/N*
Unit test coverage	*None/Some/All*
Unit tests succeed	*None/Some/All*
Unit tests report	*Add unit test report to appendix T*
Security done?	*y/ n/ N/A (database, web, input, ... cannot be N/A)*
Security actions?	*Describe security fixes implemented or planned*

GenAI declaration	
Did you use GenAI	***Yes***
Describe use	*Describe the nature and extent of GenAI use and the specific model used.* ***Briefly detail GenAI use for each implemented backlog item.*** *You are allowed to use GenAI, but you* ***must*** *adhere to the GenAI policy shown on the CSC3003S Amathuba site.* **Thomas:** Various Claude models were used within Claude Code to assist with planning and writing the code for my backlog items (1, 2, and 7), as well as my parts of other backlog items (3, 6, and 9). Planning was an iterative process conducted in collaboration with GenAI (Claude). All code in my sections was written by GenAI (as allowed by the client), under varying levels of autonomy depending on the task. This ranged from Claude implementing a GitLab issue independently from start to finish, to closely supervised collaboration in which I directed each step of the implementation. Every AI-generated change was submitted as a GitLab merge request. I manually reviewed each merge request before it was merged, and no code was merged into the codebase without my review and explicit approval. Below are counts of how many commits each model was used for: Claude Opus 5 249 Claude Fable 5 84 Claude Sonnet 5 41 Claude Fable 5.1 14 Claude Opus 4.6 14 Claude Opus 4.8 2 **Danny:** Claude Sonnet 5 was used to write all the code in my section (Backlog items 3,6 and 8) and Claude Opus was used for planning solutions to tough problems ran into along the way. All code in my sections was written by GenAI (as allowed by the client), under varying levels of autonomy depending on the task. This ranged from Claude implementing a GitLab issue independently from start to finish, to closely supervised collaboration in which I directed each step of the implementation.

	Every AI-generated change was submitted as a GitLab merge request. I manually reviewed each merge request before it was merged, and no code was merged into the codebase without my review and explicit approval. The choice of visuals (graph types and style) and cards were done by myself - Claude was used to implement my own design decisions. Gemini 3.7 Flash was used to generate ideas of types of statistics that may prove interesting for a coach and were mostly implemented in the form of aggression, passive, error prone and steady. **Luke**: Claude Sonnet 5 and Claude Opus 5 were used for assistance in coding and generating all test cases. They were both also used in Claude Design to create initial protypes for the capture UI. Claude Opus 5 was also used to help make the fake session dataset. Claude Fable 5 was used once in the generating of a solution plan for a bug we ran into which was then implemented by Claude Opus 5. In assignment 2, GenAI was used to analyse the codebase and surface actions that have been taken in order to prevent security issues, and list places where the implementation has deviated from the original plan. In assignment 3, GenAI was used to check and ensure all changes made to the codebase form assignment 2 were mentioned. Formatting and sentence structure evaluation was performed with Claude Sonnet 5 on appendix C and T.
Instructions to run the program	Go to https://matchpoint.thomaskabalin.com/ To run the application locally, in a development environment, follow the instructions in the README (https://gitlab.cs.uct.ac.za/tldr/matchpoint/-/blob/main/README.md).
Data needed	The demo data is already in the production database. Demo coaches exist with the following credentials: Emails: • thandi.mokoena@example.com • derek.wallace@example.com • farah.ismail@example.com Password (same for all 3): **statue-canvas-lemon-teapot**
Link to data	https://gitlab.cs.uct.ac.za/tldr/matchpoint/-/tree/main/apps/web/scripts/fake-data/import
Unit tests	Refer to Appendix T for all unit tests, split per backlog item.

Appendix C

Backlog Item 1

Date	Hash	Author	MR	Commit message
2026-08-08	bb8f8fb	Thomas Kabalin	!5	feat(api): add player model and initial migration
2026-08-08	1eb9983	Thomas Kabalin	!5	test(api): add transaction-rollback postgres fixtures
2026-08-08	9e04f42	Thomas Kabalin	!6	feat(api): add match model and migration
2026-08-10	3a8f0cc	Thomas Kabalin	!5	refactor(api): mirror the test tree onto the source layout
2026-08-10	8c4cf6d	Thomas Kabalin	!6	test(api): pin the match model's design decisions
2026-08-10	f9f4974	Thomas Kabalin	!6	docs: correct why str_enum avoids native Postgres enums
2026-08-04	17bc19d	Thomas Kabalin		feat(api): add the sqlalchemy engine and alembic setup
2026-08-11	b46daa9	Thomas Kabalin	!7	feat(api): add coach model and migration
2026-08-11	00b2c79	Thomas Kabalin	!7	test(api): pin the coach model's design decisions
2026-08-11	126f00d	Thomas Kabalin	!7	docs: add an ERD and the schema conventions behind it
2026-08-11	98c6132	Thomas Kabalin	!7	feat(api): store coach emails lowercased
2026-08-11	f71b898	Thomas Kabalin	!6	refactor(api): scope the match format field to the sets axis
2026-08-11	beec14a	Thomas Kabalin	!7	chore(api): drop the migrations versions placeholder
2026-08-11	3dd8a79	Thomas Kabalin	!9	feat(api): add the coach sign-up endpoint
2026-08-11	0be97bf	Thomas Kabalin	!9	test(api): cover coach sign-up
2026-08-11	a669583	Thomas Kabalin	!9	docs: record the password hashing choice
2026-08-11	00ec089	Thomas Kabalin	!10	feat(api): add the coach sessions table
2026-08-11	da724cf	Thomas Kabalin	!10	test(api): cover the coach sessions table
2026-08-11	8854143	Thomas Kabalin	!10	feat(api): add coach login and logout
2026-08-11	45c01be	Thomas Kabalin	!13	feat(api): scope players and matches to the authenticated coach
2026-08-11	8aa0791	Thomas Kabalin	!13	test(api): cover coach scoping on players and matches
2026-08-12	9d781b6	Thomas Kabalin	!24	feat(api): add the email verifications table
2026-08-12	90453eb	Thomas Kabalin	!24	test(api): cover the email verifications table

2026-08-13	bd9cc14	Thomas Kabalin	!28	feat(api): require a verified email to create a coach
2026-08-13	b9307e6	Thomas Kabalin	!32	feat(api): reset a forgotten password with an emailed code
2026-08-13	ed9d01f	Thomas Kabalin	!32	test(api): cover password reset end to end
2026-08-13	8435303	Thomas Kabalin	!36	docs: record the match deletion policy
2026-08-13	cc55be5	Thomas Kabalin	!35	docs: record that list endpoints do not paginate
2026-08-15	89412e7	Thomas Kabalin	!35	docs: record the timestamp source and archiving's reach
2026-08-15	863e1f4	Thomas Kabalin	!62	feat(api): record a match result
2026-08-15	d81c398	Thomas Kabalin	!77	feat(api): store per-coach match-setup defaults
2026-08-17	a35602c	Thomas Kabalin	!102	feat(api): store a scoring system instead of a sets format
2026-08-17	c64ab8b	Thomas Kabalin	!103	feat(api): verify a reset code without spending it
2026-08-17	de5f334	Thomas Kabalin	!111	feat(api): let a coach close their account
2026-08-17	922ec97	Thomas Kabalin	!111	docs: record account closure as the exception to soft deletion
2026-08-18	8881991	Thomas Kabalin	!111	feat(api): refuse an account closure while players are still active
2026-08-18	d1711bd	Thomas Kabalin	!111	docs: narrow the archiving rule to data inside an account
2026-08-18	b29487a	Thomas Kabalin	!131	feat(api): transfer a player to another coach via a transfer code
2026-08-18	577acd2	Thomas Kabalin	!132	feat(api): store which device is capturing a match
2026-08-19	5abefd2	Thomas Kabalin	!131	fix(api): refuse account closure while a player has a live transfer code
2026-08-19	f972828	Thomas Kabalin	!149	refactor(api): delete players outright instead of archiving them
2026-08-19	889d473	Thomas Kabalin	!150	feat(api): report whether a player has a live transfer code
2026-08-19	22f62c9	Thomas Kabalin	!156	feat(api): add photo_key to coaches and players
2026-08-19	5bf5165	Thomas Kabalin	!156	feat(api): upload and clear a player photo
2026-08-19	caebf03	Thomas Kabalin	!160	feat(api): add the derived match_sets table
2026-08-20	5ca0b57	Thomas Kabalin	!164	docs(data-model): record capture claim, transfer re-mint, and coach defaults rationale
2026-08-20	ab73ed4	Thomas Kabalin	!160	docs(api): name the set-closing stamp is_set_end, as #175 settled it

2026-08-20	95cd0f1	Thomas Kabalin	!170	feat(api): add the points table
2026-08-20	d211f74	Thomas Kabalin	!171	feat(api): validate a batch of captured points
2026-08-20	bed36a3	Thomas Kabalin	!173	docs: record the capture-claim refusal in the deletion convention
2026-08-20	0879923	Thomas Kabalin	!175	feat(api): give every player a unique generated username
2026-08-20	0760071	Thomas Kabalin	!175	feat(api): mint a one-time password when a player is created
2026-08-20	ed348c7	Thomas Kabalin	!175	feat(api): player_sessions table, and the deletes that must clear it
2026-08-20	c92951d	Thomas Kabalin	!175	feat(api): player login, logout, and their own /me on a player session
2026-08-21	bc1b942	Thomas Kabalin	!182	feat(api): delete a match along with its points
2026-08-21	30e7c8e	Thomas Kabalin	!182	feat(api): refuse to re-create a deleted match
2026-08-21	bea0def	Thomas Kabalin	!182	docs: record the new match-delete policy and its tombstone
2026-08-21	59b8396	Thomas Kabalin	!182	feat(api): erase a coach's deleted-match tombstones on account closure
2026-08-21	bc7f33b	Thomas Kabalin		docs: shorten data-model prose
2026-08-23	674280d	Thomas Kabalin	!188	refactor(api): drop serve1_target/serve2_target from points
2026-08-24	73b6295	Thomas Kabalin	!199	docs(data-model): what completing a match freezes
2026-08-24	2de8cca	Thomas Kabalin	!205	docs: draw the coach relationship on deleted matches in the ERD
2026-08-24	da8cb84	Thomas Kabalin	!207	refactor(api): split the sample-data import into run_import and main
2026-08-24	4d6d7e9	Thomas Kabalin	!207	fix(api): mint player usernames in the sample-data import
2026-08-24	5a1187d	Thomas Kabalin	!207	fix(api): report a bad matches.csv row by its line number
2026-08-24	707ed2d	Thomas Kabalin	!207	feat(api): make the sample-data import re-runnable
2026-08-24	4dbed59	Thomas Kabalin	!207	chore(api): measure and type-check the scripts directory
2026-08-24	2a8cb96	Thomas Kabalin	!207	fix(api): name the CSV row when a join between files finds nothing
2026-08-31	e85a7ab	Thomas Kabalin	!221	feat(api): store coach notes against a player
2026-08-31	aab06cd	Thomas Kabalin	!218	refactor(api): drop mid from the court depth enum

2026-08-10	25f862e	Thomas Kabalin	!6	docs: add a skill for changing a field on an API model
2026-08-19	c4e0b90	Thomas Kabalin	!149	docs: replace the archiving convention with hard deletion
2026-09-03	2b70f96	Thomas Kabalin	!225	docs(data-model): say when a serve fault mode is set
2026-09-03	b8f5580	Thomas Kabalin	!225	docs(data-model): list every case that leaves a serve box null
2026-09-12	1f57324	Thomas Kabalin	!241	docs: record what the migration walk now catches
2026-09-15	418385c	Thomas Kabalin	!288	docs(data-model): write down the two capture fields the API drops

Backlog Item 2

Date	Hash	Author	MR	Commit message
2026-08-08	55e0c28	Thomas Kabalin	!5	feat(api): add players create/list/get endpoints
2026-08-10	3a8f0cc	Thomas Kabalin	!5	refactor(api): mirror the test tree onto the source layout
2026-08-10	2bc6dfa	Thomas Kabalin	!6	feat(api): add matches create/list/get endpoints
2026-08-10	563c304	Thomas Kabalin	!6	test(api): add unit tests for the match schema rules
2026-08-11	f71b898	Thomas Kabalin	!6	refactor(api): scope the match format field to the sets axis
2026-08-11	3dd8a79	Thomas Kabalin	!9	feat(api): add the coach sign-up endpoint
2026-08-11	7596f58	Thomas Kabalin	!9	test(api): cover 422 password redaction
2026-08-11	ea3d30a	Thomas Kabalin	!9	feat(api): redact submitted passwords from 422 bodies
2026-08-11	24453d3	Thomas Kabalin	!11	feat(api): rate limit coach sign-up
2026-08-11	83c0b3f	Thomas Kabalin	!11	test(api): cover the sign-up rate limit
2026-08-11	45c01be	Thomas Kabalin	!13	feat(api): scope players and matches to the authenticated coach
2026-08-11	8aa0791	Thomas Kabalin	!13	test(api): cover coach scoping on players and matches
2026-08-12	060abc9	Thomas Kabalin	!24	feat(api): add OTP email verification for sign-up

2026-08-13	52809ec	Thomas Kabalin	!27	feat(api): send sign-up codes through Resend
2026-08-13	138893a	Thomas Kabalin	!27	test(api): cover the Resend sender and its 502
2026-08-13	cd94dc5	Thomas Kabalin	!27	docs: record Resend in the tech stack and the VPS key step
2026-08-13	b9307e6	Thomas Kabalin	!32	feat(api): reset a forgotten password with an emailed code
2026-08-13	9a2162a	Thomas Kabalin	!36	feat(api): delete a match
2026-08-13	e21484c	Thomas Kabalin	!35	feat(api): archive and restore a player
2026-08-13	4f85b6f	Thomas Kabalin	!38	feat(api): update players and matches with PATCH
2026-08-14	e4a433a	Thomas Kabalin	!39	feat(api): refuse state-changing requests from a foreign origin
2026-08-15	12fdf95	Thomas Kabalin	!35	refactor(api): take the session first in owned_player
2026-08-15	ec8b3ca	Thomas Kabalin	!35	refactor(api): reuse owned_player in update_player
2026-08-15	5a1a857	Thomas Kabalin	!35	feat(api): refuse a new match against an archived player
2026-08-15	863e1f4	Thomas Kabalin	!62	feat(api): record a match result
2026-08-15	98ef7ff	Thomas Kabalin	!63	feat(api): set security headers on every API response
2026-08-15	d308e2e	Thomas Kabalin	!64	feat(api): bound request body size at 1 MB
2026-08-15	d27a7cc	Thomas Kabalin	!65	feat(api): log every request as a structured line with a request id
2026-08-15	ca4cb4e	Thomas Kabalin	!66	feat(api): bound an authenticated coach's request rate
2026-08-15	dfa604b	Thomas Kabalin	!67	feat(api): refuse unknown keys in request bodies
2026-08-15	0505578	Thomas Kabalin	!67	refactor(api): share the update loop as UpdateModel.apply_to
2026-08-15	0b52471	Thomas Kabalin	!67	refactor(api): reuse match_for_coach in update_match
2026-08-15	62c09fd	Thomas Kabalin	!78	test(api): freeze the limiter clock in bucket-draining tests

2026-08-15	629a7ac	Thomas Kabalin		style: drop the "load-bearing" phrase from comments and docstrings
2026-08-17	a35602c	Thomas Kabalin	!102	feat(api): store a scoring system instead of a sets format
2026-08-17	c64ab8b	Thomas Kabalin	!103	feat(api): verify a reset code without spending it
2026-08-17	4c6f2fa	Thomas Kabalin	!108	fix(api): answer an unhandled exception with a generic JSON body
2026-08-18	e4c6d59	Thomas Kabalin	!130	feat(api): accept a client-supplied match id on create
2026-08-18	b29487a	Thomas Kabalin	!131	feat(api): transfer a player to another coach via a transfer code
2026-08-18	9b297c0	Thomas Kabalin	!132	feat(api): claim a match for capture, and lock it once complete
2026-08-18	cfa0308	Thomas Kabalin	!143	feat(api): let a test run scale the rate-limit buckets up
2026-08-19	f972828	Thomas Kabalin	!149	refactor(api): delete players outright instead of archiving them
2026-08-19	889d473	Thomas Kabalin	!150	feat(api): report whether a player has a live transfer code
2026-08-19	7523acd	Thomas Kabalin	!156	feat(api): add a photo store served at /media
2026-08-19	22f62c9	Thomas Kabalin	!156	feat(api): add photo_key to coaches and players
2026-08-19	5bf5165	Thomas Kabalin	!156	feat(api): upload and clear a player photo
2026-08-19	b31f2e3	Thomas Kabalin	!156	refactor(api): move shared photo-upload helpers out of coaches router
2026-08-19	caebf03	Thomas Kabalin	!160	feat(api): add the derived match_sets table
2026-08-20	14c8adc	Thomas Kabalin	!170	feat(api): refuse deleting a match that has points
2026-08-20	d211f74	Thomas Kabalin	!171	feat(api): validate a batch of captured points
2026-08-20	b9842b9	Thomas Kabalin	!171	feat(api): derive a match's set scores from its points
2026-08-20	8ff1bdc	Thomas Kabalin	!171	feat(api): ingest a batch of captured points
2026-08-20	79221b7	Thomas Kabalin	!172	feat(api): read a match's points

2026-08-20	a7ba608	Thomas Kabalin	!173	feat(api): refuse deleting a match another device is capturing
2026-08-20	0879923	Thomas Kabalin	!175	feat(api): give every player a unique generated username
2026-08-20	0760071	Thomas Kabalin	!175	feat(api): mint a one-time password when a player is created
2026-08-20	ed348c7	Thomas Kabalin	!175	feat(api): player_sessions table, and the deletes that must clear it
2026-08-20	15c3eb3	Thomas Kabalin	!174	feat(api): accept only version-4 client match ids
2026-08-20	c92951d	Thomas Kabalin	!175	feat(api): player login, logout, and their own /me on a player session
2026-08-20	970324d	Thomas Kabalin	!175	feat(api): a coach regenerates a player's password
2026-08-21	bc1b942	Thomas Kabalin	!182	feat(api): delete a match along with its points
2026-08-21	30e7c8e	Thomas Kabalin	!182	feat(api): refuse to re-create a deleted match
2026-08-21	05df2c9	Thomas Kabalin	!174	fix(api): tolerate clock_timestamp() drift between created_at and updated_at
2026-08-23	674280d	Thomas Kabalin	!188	refactor(api): drop serve1_target/serve2_target from points
2026-08-24	8c5933a	Thomas Kabalin	!199	feat(api): metadata edits on a completed match
2026-08-28	d29fa03	Luke Bryant	!208	fix(api): tell a client its queued edit names a deleted match
2026-08-28	a8f0c4a	Thomas Kabalin	!208	fix(api): answer 410 where a match is deleted while a write waits on its row
2026-08-28	126b5fc	Thomas Kabalin	!211	fix(api): remember a deleted player's matches, so a queued flush is discarded
2026-08-28	ede74dd	Thomas Kabalin	!211	docs: say a player delete leaves tombstones too
2026-08-31	7e62292	Thomas Kabalin	!219	docs: record what makes a match in progress
2026-08-31	9b98efa	Thomas Kabalin	!219	feat(api): answer which of a coach's matches capture is under way on
2026-08-31	e85a7ab	Thomas Kabalin	!221	feat(api): store coach notes against a player
2026-08-04	bc1541b	Thomas Kabalin		docs: add api, migrations and api-client rules

2026-08-04	e6fb67e	Thomas Kabalin		feat(api): scaffold the fastapi service
2026-08-06	20f63a1	Thomas Kabalin		docs: fix the pytest coverage command and reflow the query-safety rule
2026-08-06	4f02953	Thomas Kabalin		docs: warn against SQL and HTML injection in Claude rules
2026-08-09	bc6f51e	Thomas Kabalin	!1	docs: document the testing framework
2026-08-10	0677a62	Thomas Kabalin	!5	ci: run api tests against a postgres service
2026-08-10	1b7ccfa	Thomas Kabalin	!5	test(api): cover the migration round-trip
2026-08-10	324e475	Thomas Kabalin	!6	test(api): cover the request-scoped session dependency
2026-08-10	47a8153	Thomas Kabalin	!5	test(api): pin the player model's design decisions
2026-08-10	a3b6f33	Thomas Kabalin	!5	test(api): pin the input boundaries on the players endpoints
2026-08-10	c822060	Thomas Kabalin	!6	test(api): assert the migrated schema matches the models
2026-08-10	f287a8b	Thomas Kabalin	!6	docs: record the 422 and clock_timestamp rules for the API
2026-08-10	f6efec7	Thomas Kabalin	!5	test(api): add unit tests for the player schema rules
2026-08-11	157dec3	Thomas Kabalin	!6	docs: scope the regeneration reminder to the files that cause it
2026-08-11	1d18cd7	Thomas Kabalin	!6	feat(api): publish the OpenAPI spec as a build input
2026-08-11	49f366e	Thomas Kabalin	!6	docs: record the client generation step
2026-08-11	61a1524	Thomas Kabalin	!7	fix(web): fall back to a relative API base URL without a DOM
2026-08-11	9770fac	Thomas Kabalin	!6	docs: write down the model and route conventions
2026-08-11	aff4fe1	Thomas Kabalin	!6	feat: generate a typed API client from the OpenAPI spec
2026-08-12	c969810	Thomas Kabalin	!14	feat(api): report the deployed commit from /health
2026-08-13	0796e38	Thomas Kabalin	!38	docs: record the XUpdate convention

2026-08-13	33d4b4f	Thomas Kabalin	!35	docs: allow an explicit include_archived flag in the archived-rows rule
2026-08-15	2ebe24a	Thomas Kabalin	!34	docs(rules): record who owns the commit, and what route tests cannot see
2026-08-15	9069f27	Thomas Kabalin	!35	docs(rules): record when a response gets declared, not just how
2026-08-15	b6e131c	Thomas Kabalin	!67	docs: prescribe RequestModel and apply_to in the route rules
2026-08-15	f7e0ad9	Thomas Kabalin	!70	build(api-client): generate Zod schemas from the spec
2026-08-17	d80c4f7	Thomas Kabalin	!108	fix(api): validate a pooled connection before handing it to a request
2026-08-19	b50178d	Thomas Kabalin	!156	feat(api): upload and clear a coach photo
2026-09-03	c964e97	Thomas Kabalin	!225	test(api): date the default match in the current year
2026-09-12	8833c8d	Thomas Kabalin	!241	test(api): assert each downgrade reverts the schema its upgrade built
2026-09-12	d27f424	Danny Gebhard	!240	fix: read the coach off the device, gate cross-device deletes honestly, clean up a deleted player's log rows
2026-09-14	2d8e19c	Thomas Kabalin	!268	feat(api): delete a point or a checkpoint the coach undid
2026-09-14	7fca97b	Thomas Kabalin	!268	feat(api): fold the newest score checkpoint into the derived set scores
2026-09-14	e6794c2	Thomas Kabalin	!268	feat(api): store the score checkpoints capture already queues
2026-09-15	541578e	Thomas Kabalin	!288	fix(api): refuse an ace that is not also the point's ending
2026-09-15	a0e0f28	Thomas Kabalin	!285	feat(api): require a scoring system when creating a match
2026-09-15	c52bc12	Thomas Kabalin	!288	feat(api): store the court side a point was played from

Backlog item 3

Date	Hash	Author	MR	Commit message
2026-08-12	fe61a8ae	Danny Gebhard	!15	feat(web): add sign-in/sign-up/home route placeholders
2026-08-12	27d4147e	Danny Gebhard	!15	fix(web): update e2e smoke test for the new sign-in placeholder
2026-08-12	3542815c	Danny Gebhard	!16	feat(web): move the player-roster demo from / to /players
2026-08-12	bb18d3f4	Danny Gebhard	!17	feat(web): add sign-in form UI
2026-08-12	84ae52f8	Danny Gebhard	!18	feat(web): wire the sign-in form to POST /auth/login
2026-08-12	b5e9a8f6	Danny Gebhard	!19	feat(web): add a sign-up link to the sign-in screen
2026-08-12	a26ed1e0	Danny Gebhard	!20	feat(web): add sign-up email step UI
2026-08-12	6d0c3b5e	Danny Gebhard	!21	feat(web): wire sign-up to request-otp and add the OTP entry step
2026-08-12	a6695834	Thomas Kabalin	!9	docs: record the password hashing choice
2026-08-12	7596f583	Thomas Kabalin	!9	test(api): cover 422 password redaction
2026-08-12	ea3d30ab	Thomas Kabalin	!9	feat(api): redact submitted passwords from 422 bodies
2026-08-12	0be97bfe	Thomas Kabalin	!9	test(api): cover coach sign-up
2026-08-12	3dd8a79e	Thomas Kabalin	!9	feat(api): add the coach sign-up endpoint
2026-08-12	422b9f13	Thomas Kabalin	!9	test(api): cover password hashing
2026-08-12	a155c252	Thomas Kabalin	!9	feat(api): hash passwords with argon2id
2026-08-13	892db62b	Danny Gebhard	!22	feat(web): add password + confirm step to sign-up
2026-08-13	86afe4a2	Danny Gebhard	!23	feat(web): wire the password step to POST /coaches
2026-08-13	97e825fa	Thomas Kabalin	!10	style(docs): trim definitions and obvious explanations

Date	Hash	Author	MR	Commit message
2026-08-13	07457674	Thomas Kabalin	!10	style(docs): remove AI writing patterns from docs
2026-08-13	db9a3932	Thomas Kabalin	!10	docs: record the session mechanism
2026-08-13	eabd43cd	Thomas Kabalin	!10	test(api): cover GET /coaches/me
2026-08-13	4cada64f	Thomas Kabalin	!10	feat(api): identify the caller and turn the coach by-id route into /me
2026-08-13	49008237	Thomas Kabalin	!10	test(api): cover login and logout
2026-08-13	88541437	Thomas Kabalin	!10	feat(api): add coach login and logout
2026-08-13	da724cff	Thomas Kabalin	!10	test(api): cover the coach sessions table
2026-08-13	00ec0895	Thomas Kabalin	!10	feat(api): add the coach sessions table
2026-08-13	7d49f111	Thomas Kabalin	!10	fix(scripts): refuse to run dev.sh from a git worktree
2026-08-13	22d6418d	Thomas Kabalin	!11	test(api): pin the login limit's cost claims and forged-cookie logout
2026-08-13	8f0e3b42	Thomas Kabalin	!11	docs: record the rate limiting choice
2026-08-13	dfeff4d4	Thomas Kabalin	!11	test(api): cover the login rate limit
2026-08-13	54462c1c	Thomas Kabalin	!11	feat(api): rate limit coach login
2026-08-13	83c0b3f0	Thomas Kabalin	!11	test(api): cover the sign-up rate limit
2026-08-13	24453d33	Thomas Kabalin	!11	feat(api): rate limit coach sign-up
2026-08-13	90453eb2	Thomas Kabalin	!24	test(api): cover the email verifications table
2026-08-13	9d781b6d	Thomas Kabalin	!24	feat(api): add the email verifications table
2026-08-13	060abc98	Thomas Kabalin	!24	feat(api): add OTP email verification for sign-up
2026-08-13	04691e9d	Thomas Kabalin	!24	docs: shorten Argon2, sessions, and rate-limiting rationale
2026-08-13	4c78e1e3	Thomas Kabalin	!24	chore(claude): add Resend skills

Date	Hash	Author	MR	Commit message
2026-08-13	c6d56346	Thomas Kabalin	!24	test(api): cover requesting and verifying sign-up codes
2026-08-13	23fe5175	Thomas Kabalin	!27	test(api): keep the suite off a real email provider
2026-08-13	ff2252cc	Thomas Kabalin	!27	chore(infra): pass the Resend key to the api container
2026-08-13	1fea21c0	Thomas Kabalin	!27	ci: check a real send on every production deploy
2026-08-13	138893a9	Thomas Kabalin	!27	test(api): cover the Resend sender and its 502
2026-08-13	52809ecf	Thomas Kabalin	!27	feat(api): send sign-up codes through Resend
2026-08-13	cd94dc59	Thomas Kabalin	!27	docs: record Resend in the tech stack and the VPS key step
2026-08-13	72f26696	Thomas Kabalin	!27 !28	docs(api): trim the comments in the coaches router
2026-08-13	a0ee6451	Thomas Kabalin	!27 !28	fix(web): send one sign-up request per click
2026-08-13	bd9cc14f	Thomas Kabalin	!27 !28	feat(api): require a verified email to create a coach
2026-08-13	9de6ca9c	Thomas Kabalin	!27 !28	docs: bound comment length in CLAUDE.md
2026-08-13	45c01bee	Thomas Kabalin	!13	feat(api): scope players and matches to the authenticated coach
2026-08-13	361c7632	Thomas Kabalin	!13	test(web): pin the player fixture to the generated contract
2026-08-13	8aa0791f	Thomas Kabalin	!13	test(api): cover coach scoping on players and matches
2026-08-13	44ee2b10	Danny Gebhard	!30	feat(web): wire the OTP step to POST /auth/verify-otp
2026-08-13	97041e04	Thomas Kabalin	!32	docs: record the account-enumeration policy
2026-08-13	36a21946	Thomas Kabalin	!32	test(api): pin that rate limits answer known and unknown emails alike

Date	Hash	Author	MR	Commit message
2026-08-13	ed9d01ff	Thomas Kabalin	!32	test(api): cover password reset end to end
2026-08-13	b9307e6f	Thomas Kabalin	!32	feat(api): reset a forgotten password with an emailed code
2026-08-14	a21227a9	Danny Gebhard	!41	feat(web): add the forgot-password route and email step UI
2026-08-14	236caadc	Danny Gebhard	!42	feat(web): wire forgot-password to POST /auth/forgot-password, add reset step UI
2026-08-14	36353095	Danny Gebhard	!44	feat(web): add a forgot-password link to the sign-in screen
2026-08-15	48c65fd7	Thomas Kabalin	!46	docs(security): record the re-authentication a password change demands
2026-08-15	ba853d98	Thomas Kabalin	!46	feat(api): change a password with the current one
2026-08-16	6d2b26ce	Danny Gebhard	!80	feat(web): add a "back to sign in" link to sign-up and forgot-password
2026-08-16	646340db	Danny Gebhard	!81	feat(web): make the sign-in failure message more explicit
2026-08-16	22bbcdf0	Danny Gebhard	!82	feat(web): make the sign-up duplicate-email message more explicit
2026-08-16	cbb0ec40	Danny Gebhard	!83	feat(web): add a sign-out control to the settings page
2026-08-16	d3dfbb91	Danny Gebhard	!84	feat(web): show a live countdown on 429 responses
2026-08-16	bfa393ce	Danny Gebhard	!84	feat(web): add recovery paths to the sign-up OTP and password steps
2026-08-16	cd2cebdc	Danny Gebhard	!94	feat(web): add a show/hide toggle to every password field
2026-08-16	54bbd5bd	Danny Gebhard	!95	feat(web): real-time password strength meter with submit-gating
2026-08-16	0ad73b49	Danny Gebhard	!95	test(web): paste the too-long password fixture instead of typing it

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2026-08-16	43a3d8eb	Danny Gebhard	!96	feat(web): build the shared auth-page shell and styled form primitives
2026-08-16	d9414707	Danny Gebhard	!97 !99	feat(web): style the sign-in page per the wireframe
2026-08-16	6a2a72ab	Danny Gebhard	!98	feat(web): style the sign-up page across its three steps
2026-08-16	602c4118	Danny Gebhard	!100	feat(web): style the forgot-password page across its three steps
2026-08-18	1ff4128e	Thomas Kabalin	!104	docs(security): record the password validation policy
2026-08-18	da450878	Thomas Kabalin	!104	feat(api): refuse repeated, sequential, common and self-naming passwords
2026-08-18	ab925ed5	Thomas Kabalin	!107	docs(security): record the breach-list check and its fail-open
2026-08-18	c16c4bb8	Thomas Kabalin	!107	feat(api): screen new passwords against the Pwned Passwords corpus
2026-08-18	131c749b	Danny Gebhard	!134	feat(web): replace sign-up's plain OTP field with the 6-box input
2026-08-18	10a5e1cb	Danny Gebhard	!134	feat(web): split forgot-password's reset step into OTP and password steps
2026-08-19	6413622d	Thomas Kabalin	!145	docs(tech-stack): record the hand-authored email bodies
2026-08-19	8ece27ad	Thomas Kabalin	!145	feat(api): send the code emails as designed HTML with a text alternative
2026-08-20	21a6d730	Danny Gebhard	!165	feat(web): rebuild Settings as a drill-down menu (#178)
2026-08-20	192da1ac	Danny Gebhard	!168	feat(web): close an account and set match defaults (#184, #104)
2026-08-20	6f07577f	Thomas Kabalin	!169	feat(api): check a password without closing the account

Date	Hash	Author	MR	Commit message
2026-08-21	f13cadff	Thomas Kabalin	!180	fix(api): lock the coach row before a credential check is acted on
2026-08-21	59b83966	Thomas Kabalin	!182	feat(api): erase a coach's deleted-match tombstones on account closure
2026-08-20	0879923a	Thomas Kabalin	!175	feat(api): give every player a unique generated username
2026-08-20	07600711	Thomas Kabalin	!175	feat(api): mint a one-time password when a player is created
2026-08-20	ed348c79	Thomas Kabalin	!175	feat(api): player_sessions table, and the deletes that must clear it
2026-08-20	c92951d9	Thomas Kabalin	!175	feat(api): player login, logout, and their own /me on a player session
2026-08-20	7ba06a97	Thomas Kabalin	!175	feat(api): a player changes their own password
2026-08-21	31cd7b79	Thomas Kabalin	!175	fix(api): lock the player row before a session is minted
2026-08-24	5d1526a4	Danny Gebhard	!194	feat(web): visualization/UX batch — error boundary, toggle-only dark mode, dark-mode chart contrast, dismissible attention card
2026-08-24	05e522ad	Danny Gebhard	!197	fix(web): theme.ts applies the stored theme itself instead of relying solely on index.html
2026-08-24	b03d0cc4	Thomas Kabalin	!201	test(web): cover a failed revoke of other sessions
2026-08-27	b66c86e3	Luke Bryant	!208	feat(web): explain each scoring system on Match Defaults
2026-08-28	4ec792bd	Luke Bryant	!208	fix(web): the five defects the branch review turned up
2026-08-28	b36b0e91	Thomas Kabalin	!208	fix(web): stop the theme module blanking every page at boot
2026-09-11	556eb761	Danny Gebhard	!230	fix(web): validate the password inline on the delete-account step (#211)
2026-09-11	3cb10f47	Thomas Kabalin	!237	feat(api): let a player session read its own matches and statistics

Date	Hash	Author	MR	Commit message
2026-09-12	19c84299	Danny Gebhard	!242	fix(web): land on the app, not the sign-in form, with a live session (closes #257)
2026-09-12	56a92973	Danny Gebhard	!242	fix(web): stop / bouncing into an infinite redirect loop on a real 401
2026-09-12	aceac635	Danny Gebhard	!244	feat(web): let a player sign in and see their own scoped view (closes #203)
2026-09-14	111a551a	Luke Bryant	!278	fix(web): warn about unsynced points before signing out
2026-09-15	f98b79f9	Thomas Kabalin	!285	fix(web): group Account & Security under named sections
2026-09-16	9478e0d4	Danny Gebhard	!304	fix(web): deactivate dark mode
2026-08-11	b7ab8d9	Thomas Kabalin	!7	test(api): cover the coach read endpoint
2026-08-11	c43ce7c	Thomas Kabalin	!7	feat(api): add the coach read endpoint
2026-08-13	32a453c	Thomas Kabalin		docs: expand security.md with the full set of security decisions
2026-08-13	3e2bfa8	Thomas Kabalin	!37	docs: stop security.md naming which coach routes exist
2026-08-13	45c615c	Danny Gebhard	!33	refactor(web): route the auth slice through the generated API client
2026-08-13	769a60f	Danny Gebhard	!31	fix(web): stop sign-up crashing on a real 422, collect display_name
2026-08-13	c7341ee	Thomas Kabalin	!37	feat(api): change a coach's display name
2026-08-13	d2fc938	Thomas Kabalin	!34	feat(api): sign out a coach's other sessions
2026-08-13	e5164ac	Thomas Kabalin	!29	feat(api): sweep expired sessions probabilistically
2026-08-14	1ebf60d	Thomas Kabalin	!40	docs: record the API's SSRF posture
2026-08-14	88e0bc7	Thomas Kabalin	!39	docs: record the CSRF analysis beside the enumeration policy
2026-08-15	0237617	Thomas Kabalin		fix(infra): allow inline styles in the app shell CSP

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2026-08-15	04ef951	Thomas Kabalin	!65	docs: record the request log in the tech stack and security policy
2026-08-15	157d556	Thomas Kabalin	!64	docs(security): record the request body ceiling
2026-08-15	4b21748	Thomas Kabalin	!66	docs: record the per-coach rate-limit tier
2026-08-15	56cea10	Thomas Kabalin		docs(security): record the style-src carve-out
2026-08-15	6a6f865	Thomas Kabalin	!63	docs: record the security-header policy
2026-08-15	74b3347	Thomas Kabalin	!63	feat(infra): set the app shell's security headers in Caddy
2026-08-15	92ec74e	Thomas Kabalin	!62	docs(api): stop change_password contrasting DELETE /auth/sessions
2026-08-15	fe6724d	Thomas Kabalin	!70	test(web): check auth request bodies against the generated schemas
2026-08-16	0216a84	Danny Gebhard	!93	fix(web): sign-up password field allows lengths beyond the API's 128-char limit
2026-08-16	27cd59e	Danny Gebhard	!92	docs(api): document the password validation policy
2026-08-16	3d124fd	Danny Gebhard	!86	feat(web): attach response status to generated client errors
2026-08-16	9f0a073	Danny Gebhard	!87	feat(web): show a sign-in prompt instead of a generic error on a 401
2026-08-16	a76b2e9	Danny Gebhard	!79	feat(web): wire the forgot-password reset step to POST /auth/reset-password
2026-08-16	f422b52	Danny Gebhard	!88	feat(web): wire up change password on settings
2026-08-17	212292d	Danny Gebhard	!115	fix(web): sequential strength-meter color, matching input radii, password manager cooperation
2026-08-17	26d4d12	Danny Gebhard	!113	feat(web): replace hand-rolled auth card/heading styles with shadcn Card
2026-08-17	adaa89b	Thomas Kabalin	!103	docs(security): record the answered-code-becomes-a-cookie pattern

Date	Hash	Author	MR	Commit message
2026-08-17	cc560a7	Danny Gebhard	!114	feat(web): replace hand-rolled form/error alert text with shadcn Alert
2026-08-17	d6dc342	Thomas Kabalin	!103	refactor(web): send the reset code and the new password as two calls
2026-08-17	f05b3df	Thomas Kabalin	!108	docs(security): record what an unhandled exception tells the client
2026-08-18	24b365d	Thomas Kabalin	!131	docs(security): record the transfer-code credential and its bounds
2026-08-18	2b02eb3	Thomas Kabalin	!111	docs(security): say what the closure's wrong-guess budget does not promise
2026-08-18	6fd3476	Danny Gebhard	!135	fix(web): reword sign-in's create-account and forgot-password links
2026-08-18	78f88cb	Thomas Kabalin	!142	fix(api): keep a stolen session from starving the routes that end it
2026-08-18	83a26e3	Danny Gebhard	!129	feat(web): swap the auth wordmark for the logo, enlarge and un-bold titles
2026-08-18	8cf8e34	Thomas Kabalin	!142	docs(security): record the recovery budget and the unknown-token charge
2026-08-18	a48eb3c	Thomas Kabalin	!141	docs(security): record what the end of a session leaves on the device
2026-08-18	a4e3a8f	Thomas Kabalin	!143	feat(api): add an email sender that writes to a file
2026-08-18	bbc5ddb	Thomas Kabalin		docs: record per-hash salting in the password storage section
2026-08-18	c6debfa	Thomas Kabalin	!111	style: replace the em dashes in the account-closure comments
2026-08-18	d52e5d6	Thomas Kabalin	!111	test(api): freeze the clock in the closure rate-limit tests
2026-08-19	066f641	Thomas Kabalin	!141	fix(web): let a coach retry a failed local-data purge on sign-out
2026-08-19	888332c	Thomas Kabalin	!156	docs(security): record how profile photos are stored and served

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2026-08-19	cdb7a71	Thomas Kabalin	!131	test(api): pin the waiver against an active player's transfer code too
2026-08-19	d8e17a0	Thomas Kabalin	!160	test(api): cover the vanished-coach 401 on the photo routes
2026-08-19	df351e9	Thomas Kabalin	!150	docs(security): note what the outstanding-code flag discloses
2026-08-20	24d1390	Thomas Kabalin	!175	docs(security): the player principal
2026-08-20	b36e3c7	Thomas Kabalin	!169	docs(security): record the closure's password-check endpoint
2026-08-20	fb81673	Thomas Kabalin	!164	docs(security): record the crash-500 header gap, photo orphan window, and transfer-code closure gate
2026-08-21	b2f682f	Thomas Kabalin	!180	docs(security): the ordering a revocation depends on
2026-08-21	f32bec8	Thomas Kabalin	!175	docs(security): the ordering a revocation depends on
2026-08-21	f9de32e	Thomas Kabalin	!175	style(docs): shorten the player-account additions to security.md
2026-09-11	5e42518	Thomas Kabalin	!237	docs: record what a player session now reaches
2026-09-11	6a5bc72	Danny Gebhard	!238	fix(web): fix the FAQ route's parenting and the legal-page shell (#193, #194)
2026-09-11	9808723	Danny Gebhard	!233	feat(web): add privacy policy and terms of service plumbing (#194)
2026-09-11	cc7bbca	Danny Gebhard	!233	feat(web): show the real app version in Settings (#195)
2026-09-11	efa2264	Danny Gebhard	!233	feat(web): add a static FAQ page for Settings' Help & feedback (#193)
2026-09-14	6c81276	Thomas Kabalin	!272	test(web): pin the outcome, auth and failed-read cases e2e
2026-09-15	59b09da	Thomas Kabalin	!293	fix(web): say what the reset OTP step's escape hatch actually does

Date	Hash	Author	MR	Commit message
2026-09-15	bbc83af	Thomas Kabalin	!293	fix(web): match the sign-up OTP step's escape-hatch copy to the reset one
2026-09-15	be53196	Thomas Kabalin	!293	fix(web): even out the OTP step's footer spacing
2026-09-15	d43b253	Thomas Kabalin	!293	feat(web): let the sign-up OTP step return to sign in

Backlog Item 4

Date	Hash	Author	MR	Commit message
2026-08-12	8d07ad0	Luke Bryant	!25	feat(web): add Point scoring model
2026-08-12	29812e4	Luke Bryant	!25	refactor(web): simplify Point model
2026-08-12	a128079	Luke Bryant	!25	feat(web): rework Point outcome model, exact rally length
2026-08-12	6b8cfb7	Luke Bryant	!25	test(web): add Point unit tests; fix mutable-reference leak
2026-08-14	c302841	Luke Bryant	!69	feat(web): add Match class
2026-08-15	e89ba14	Luke Bryant	!69	feat(web): add createMatch factory with ScoringSystem-driven sets
2026-08-15	bf23554	Luke Bryant	!71	refactor(web): stop pre-populating Match.sets, extract maxSetsFor
2026-08-15	5561dae	Luke Bryant	!71	fix(web): silence no-empty-object-type on the MatchSet placeholder
2026-08-15	f0cd192	Luke Bryant	!72	fix(web): rename Match.timestamp to playedOn, store date-only
2026-08-15	cc12a1f	Luke Bryant	!73	fix(web): add Match.opponentName, generate a stand-in opponentId
2026-08-15	fd63123	Luke Bryant	!74	fix(web): default scoringSystem/surfaceType instead of requiring them
2026-08-15	8e15476	Luke Bryant	!75	fix(web): make Match.firstServer an id, not a role label
2026-08-17	914a6f0	Luke Bryant	!71	feat(web): add MatchSet, a flat record referencing Match by matchId
2026-08-17	302c048	Luke Bryant	!117	feat(web): add gameStateFor, deriving a game's score from its Points
2026-08-17	b10daa0	Luke Bryant	!117	refactor(web): move maxSetsFor to state.ts, add isAdScoringFor
2026-08-17	b5da17c	Luke Bryant	!118	feat(web): add Point.isTiebreakPoint
2026-08-17	1c3233b	Luke Bryant	!118	feat(web): add MatchSet.tiebreak
2026-08-17	559fb55	Luke Bryant	!118	feat(web): add setRulesFor and setStateFor
2026-08-17	d49f987	Luke Bryant	!119	feat(web): add matchRulesFor
2026-08-17	5a86636	Luke Bryant	!119	feat(web): add matchStateFor
2026-08-18	a5fcd6d	Luke Bryant	!120	feat(web): add Match.outcomeOverride for retirement/walkover/default
2026-08-19	3f42970	Luke Bryant	!146	refactor(web): rework Point to the perspective-based capture model
2026-08-19	32a9133	Luke Bryant	!148	feat(web): add the single-point capture flow state machine
2026-08-19	c7f43d1	Luke Bryant	!155	feat(web): close the state.ts gaps between committed points and live match state
2026-08-19	bdaadab	Luke Bryant	!159	feat(web): add the match-level flow layer over capture-flow and state
2026-08-20	2b1212d	Luke Bryant		docs: confirm the four scoring-system values docs/scoring-systems.md left open

2026-08-20	29b1117	Luke Bryant	!166	feat(web): derive which service box a point is served from
2026-08-20	52bfcb8	Luke Bryant	!166	feat(web): stop asking the coach to tap the service box
2026-08-20	4e1bfd8	Luke Bryant		docs: confirm the four scoring-system values docs/scoring-systems.md left open
2026-08-20	1c38750	Luke Bryant	!167	feat(web): drop shot detail on a return error, place it in the outcome itself
2026-08-21	5f2fbdd	Luke Bryant	!179	feat(web): derive the current score as a two-row scoreboard
2026-08-21	8512ef3	Luke Bryant	!183	feat(web): record the deuce/advantage state a point was played under
2026-08-21	19c6859	Luke Bryant	!183	refactor(web): promote courtSide to Point's only deuce/ad field
2026-08-23	02425ac	Luke Bryant	!208	refactor(web): let a landed serve record no placement
2026-08-24	04bc754	Luke Bryant	!208	feat(web): record a serve nobody watched
2026-08-24	c6b63fd	Luke Bryant	!208	feat(web): skip a point from the serve screen
2026-08-24	3a8f9a6	Luke Bryant	!208	refactor(web): skip the shot screen on an unforced error
2026-08-24	151b28b	Luke Bryant	!208	refactor(web): the final ball is four zones, and always one that landed
2026-08-24	5e00035	Luke Bryant	!208	feat(web): synthesise the points behind a skipped-to score
2026-08-26	3a6db6f	Luke Bryant	!208	fix(web): stop a let from carrying the rallied length into the replay
2026-08-26	6f27402	Luke Bryant	!208	feat(web): let the rally screen's two taps land in either order
2026-08-27	cb8d0cf	Luke Bryant	!208	feat(web): fold the score from an event log, so a coach can state it
2026-08-27	b53f8e6	Luke Bryant	!208	refactor(scoring): drop womens_grand_slam as a distinct scoring system
2026-08-27	b66c86e	Luke Bryant	!208	feat(web): explain each scoring system on Match Defaults
2026-08-04	a9e8aab	Thomas Kabalin		feat(web): add the capture feature slice
2026-09-15	019701d	Luke Bryant	!281	test(web): close branch-coverage gaps in capture-flow, match, and point
2026-09-15	29faae8	Thomas Kabalin	!297	docs: name the API as a second reader of scoring_system
2026-09-15	7e796aa	Luke Bryant	!282	test(web): add e2e coverage for score-boundary and undo capture flows

Backlog Item 5

Date	Hash	Author	MR	Commit message
2026-08-22	6a85b0b	Luke Bryant	!186	feat(web): add carpet and unspecified surface options to match setup
2026-08-23	83b0c7d	Luke Bryant	!208	feat(web): court diagram shared by the capture screens
2026-08-23	6398bce	Luke Bryant	!208	feat(web): serve screen
2026-08-24	51ea023	Luke Bryant	!208	feat(web): match header shared by the capture screens
2026-08-24	0b0d9c0	Luke Bryant	!208	feat(web): return screen
2026-08-24	a98745d	Luke Bryant	!208	feat(web): rally screen
2026-08-24	1222b0f	Luke Bryant	!208	feat(web): final shot screen
2026-08-24	be8047a	Luke Bryant	!208	feat(web): final ball screen
2026-08-24	8fcf4fd	Luke Bryant	!208	feat(web): match setup screen
2026-08-24	630f093	Luke Bryant	!208	feat(web): result screen
2026-08-24	12fcd0b	Luke Bryant	!208	feat(web): unusual point end sheet
2026-08-24	ba0393c	Luke Bryant	!208	feat(web): skip ahead sheet
2026-08-24	03d7753	Luke Bryant	!208	feat(web): run the capture flow
2026-08-24	06c4580	Luke Bryant	!208	feat(web): paint the court its own surface
2026-08-25	f9d028a	Luke Bryant	!208	feat(web): back steps out of a match it hasn't started
2026-08-26	07894fc	Luke Bryant	!208	chore: keep design-tool handoff bundles out of the repo
2026-08-26	e0717f9	Luke Bryant	!208	fix(web): queue a match's outcome even when ending it queued nothing else
2026-08-26	80d6b7b	Luke Bryant	!208	fix(web): keep real points when skipping ahead to a score
2026-08-26	19382cc	Luke Bryant	!208	fix(web): never let skip ahead rewrite who served first
2026-08-26	6f27402	Luke Bryant	!208	feat(web): let the rally screen's two taps land in either order
2026-08-26	9507441	Luke Bryant	!208	fix(web): open skip ahead after the hold releases, not mid-press
2026-08-27	f12ef38	Luke Bryant	!208	fix(web): disable serve/return buttons while a court tap's flash is pending
2026-08-27	65911a5	Luke Bryant	!208	fix(web): stop back from reopening setup after the match is queued
2026-08-27	236a7d9	Luke Bryant	!208	feat(web): add a coach note button to the live capture header
2026-08-27	5051cfc	Luke Bryant	!208	feat(web): back button on match setup
2026-08-27	f6c2abe	Luke Bryant	!208	fix(web): stop live-match reading a ref during render, and a flaky select click
2026-08-27	41c073d	Luke Bryant	!208	feat(web): flash the point, game or set a captured point won
2026-08-27	a41f25a	Luke Bryant	!208	fix(web): stop a long-press stranding the next keyboard activation
2026-08-27	cb8d0cf	Luke Bryant	!208	feat(web): fold the score from an event log, so a coach can state it

2026-08-27	9e7b24c	Luke Bryant	!208	feat(web): let a coach say "I didn't see that" without losing the point
2026-08-27	a16639c	Luke Bryant	!208	feat(web): keep the match in a durable log, so a dead phone doesn't lose it
2026-08-28	9f7fab9	Luke Bryant	!208	fix(web): hold the court at one size, so a tap lands where the last one did
2026-08-28	11454ee	Luke Bryant	!208	fix(web): start a match without waiting on the coach, and honour their defaults
2026-08-28	a1e728f	Luke Bryant	!208	fix(web): sit each capture card inside a court region, not near one
2026-08-28	2839951	Luke Bryant	!208	fix(web): let back leave a match queued before anything was captured
2026-08-28	4ec792b	Luke Bryant	!208	fix(web): the five defects the branch review turned up
2026-08-28	d29fa03	Luke Bryant	!208	fix(api): tell a client its queued edit names a deleted match
2026-08-28	68edf7e	Luke Bryant	!208	fix(web): count only the queued rows a drain can actually send
2026-08-28	3bb3f0c	Luke Bryant	!208	fix(web): keep a voice note whole when the speech service ends it
2026-08-28	8d181e6	Luke Bryant	!208	chore: keep the schema-gaps note in docs/scratch, not the source tree
2026-08-28	525c00b	Luke Bryant	!208	feat(web): let a coach hold the note button to type a note
2026-08-28	a00fa04	Thomas Kabalin	!212	test(web): drop user-event's per-action delay across the suite
2026-09-11	072c3fd	Luke Bryant	!236	fix(web): drop the stored code from the unusual point end reasons
2026-09-12	08219a3	Luke Bryant	!243	feat(web): show an ace as a spade, not a bolt
2026-09-12	33055ca	Luke Bryant	!243	fix(web): remove the zone labels from the final ball screen
2026-09-12	3714b60	Luke Bryant	!243	feat(web): infer the point winner from where the final ball finished
2026-09-12	4570cb5	Luke Bryant	!243	feat(web): zoom into the final ball's region for the rest of the point
2026-09-12	61bfb4c	Luke Bryant	!243	feat(web): draw rally length as a basket of balls, not four labels
2026-09-12	63a6228	Luke Bryant	!243	feat(web): zoom past the boundary lines, into plain court colour
2026-09-12	81549c0	Luke Bryant	!246	feat(web): let a coach name the event on match setup
2026-09-12	8463fd2	Luke Bryant	!243	refactor(web): move final-shot wing headings onto the court, above the buttons
2026-09-12	890e29e	Luke Bryant	!243	feat(web): give forced/unforced a real symbol, app wide
2026-09-12	c0b9626	Luke Bryant	!243	feat(web): give the final shot buttons the analytics shot-type icons
2026-09-12	cc52f8e	Luke Bryant	!246	feat(web): suggest a player's previous opponents on match setup

2026-09-13	0a0a50b	Luke Bryant	!252	fix(web): widen the final shot cards' gap past sub-pixel rounding
2026-09-13	1acc05d	Luke Bryant	!246	fix(web): match the Opponent field's real combobox role in e2e
2026-09-13	415c06d	Luke Bryant	!252	feat(web): pair the final shot wing headings with their own icons
2026-09-13	d39c541	Luke Bryant	!252	feat(web): zoom the court back out before leaving the final shot screen
2026-09-13	d92320c	Luke Bryant	!252	fix(web): make the final shot wing headings prominent and stop card bleed
2026-09-14	0a6ab30	Luke Bryant	!280	refactor(web): remove unusualNote now that note covers every point
2026-09-14	0af20f9	Luke Bryant	!264	fix(web): replace Opponent's native datalist with a hand-rolled combobox
2026-09-14	0f90421	Luke Bryant	!273	feat(web): confirm a capture note with a "Note saved" toast
2026-09-14	34e1d20	Luke Bryant	!270	feat(web): long-press capture's back button to go straight home
2026-09-14	384d848	Luke Bryant	!273	feat(web): hint the note button's hold-to-type gesture once
2026-09-14	585d6e6	Luke Bryant	!263	fix(web): zoom via real layout instead of transform, eliminating mid-zoom blur
2026-09-14	59d9852	Luke Bryant	!274	fix(web): match final ball's Let/Didn't see order to serve screen
2026-09-14	93f7cd0	Luke Bryant	!261	feat(web): live-fill voice notes and move note entry to the top
2026-09-14	9bc8d72	Luke Bryant	!263	fix(web): spill the 11+ basket's overflow balls in front, not stacked above
2026-09-14	a4020fc	Luke Bryant	!267	docs(web): fix stale wing/shotType comments to last-successful-shot semantics
2026-09-14	b0a5734	Luke Bryant	!275	fix(web): let a long opponent name truncate instead of covering its Chips sibling
2026-09-14	cf597f4	Thomas Kabalin	!272	test(web): cover the capture journey and the finished match's result
2026-09-14	f6d05c9	Luke Bryant	!273	feat(web): hint back and skip's hold gestures too
2026-09-15	08ccd63	Luke Bryant	!287	feat(web): seat the sync status in each capture screen's heading row
2026-09-15	3e930d1	Luke Bryant	!291	feat(web): label whose half is whose on the serve and return screens
2026-09-15	601084b	Thomas Kabalin	!285	fix(web): save a dictated note after a point boundary takes it
2026-09-15	6471589	Thomas Kabalin	!285	fix(web): fix match setup's required/optional field labels
2026-09-15	70453b5	Thomas Kabalin	!285	fix(web): keep the setup screen's action clear of the fields

2026-09-15	7a095d0	Luke Bryant	!291	feat(web): zoom the court back out when a point ends unforced
2026-09-15	7e6724e	Thomas Kabalin	!285	fix(web): rename result screen's button to Match statistics
2026-09-15	80b2d3a	Thomas Kabalin	!285	fix(web): drop the em dashes from user-facing copy
2026-09-15	80ffffa	Thomas Kabalin	!285	fix(web): put the return screen's Let and Didn't see in the shared order
2026-09-15	8c524ad	Thomas Kabalin	!293	fix(web): lift the note panel over the court's own markers
2026-09-15	c1ac382	Luke Bryant	!291	feat(web): mark which side the server is serving from with a tennis ball
2026-09-15	f354e29	Thomas Kabalin	!285	feat(web): let the note button's Save end a dictation, and carry it across a point boundary
2026-09-15	fd140cd	Luke Bryant	!291	refactor(web): replace the third-party unforced-error icon with ThumbsDown
2026-09-16	66b011b	Thomas Kabalin	!299	test(web): raise the capture-scoring and capture-claim timeouts to 60s
2026-09-16	8b81fdb	Luke Bryant	!300	fix(web): box the final ball screen's Didn't see button
2026-09-16	a23f76f	Luke Bryant	!300	fix(web): keep the court frame still across every capture screen
2026-09-16	af31b11	Luke Bryant	!305	fix(web): fade the final shot labels out as the exit zooms
2026-09-16	cd91c77	Luke Bryant	!300	fix(web): clip the ending cards as the unforced exit zooms out
2026-09-16	d6b5658	Luke Bryant	!300	fix(web): box the serve and return screens' Didn't see button

Backlog item 6

Date	Hash	Author	MR	Commit message
2026-08-18	0cb45c8f	Danny Gebhard	!126	fix(web): address Thomas's review on !126
2026-08-18	75b23c82	Danny Gebhard	!126	feat(web): build the seven player-stats tabs
2026-08-18	035a5cdc	Danny Gebhard	!126	feat(web): build the wireframe's home, roster and match-list screens
2026-08-18	613c4a44	Danny Gebhard	!126	feat(web): add the provisional analytics layer behind the stats screens
2026-08-18	264c2d63	Danny Gebhard	!126	feat(web): add the categorical palette tokens and the shared stat primitives
2026-08-18	3ab1b382	Danny Gebhard	!126	chore(web): install the shadcn components the analytics screens need
2026-08-23	a118d618	Thomas Kabalin	!177	refactor(web): collapse the winner/forced-error split capture never records
2026-08-23	1a4ee0e9	Thomas Kabalin	!177	feat(api): player analytics endpoints scoped to the capture record
2026-08-23	09b2944d	Danny Gebhard	!189	feat(web): redesign the player statistics drill-down
2026-08-23	36442d40	Danny Gebhard	!189	feat(web): restore season record, player details and notes to the stats page
2026-08-31	3131022	Thomas Kabalin	!217	refactor(api): drop the analytics endpoints no screen reads
2026-08-31	aab06cd	Thomas Kabalin	!218	refactor(api): drop mid from the court depth enum
2026-08-31	15b00b7	Thomas Kabalin	!220	feat(api): scope the player analytics tabs to one opponent
2026-08-31	c5b3248f	Danny Gebhard	!216	fix(web): harden Serve/Rally tabs and the court heatmap for real analytics payloads
2026-08-31	ca09ca13	Danny Gebhard	!222	feat(web): wire the squad snapshot and roster to real analytics data
2026-08-31	31487a4e	Danny Gebhard	!223	feat(web): wire the player profile and per-match stats to real analytics data

Date	Hash	Author	MR	Commit message
2026-08-31	f72a0c43	Danny Gebhard	!224	feat(web): wire the opponent axis to real data (MR 4 of the stats-wiring program)
2026-09-11	b0a93698	Danny Gebhard	!228 !227	feat(api): add a surface filter to the analytics endpoints (#225)
2026-09-11	535140ac	Danny Gebhard	!228	feat(web): add surface as a filter dimension on the player stats page (#225)
2026-09-11	ffca5548	Danny Gebhard	!229	feat(web): add a shared surface filter to the squad stats page (#225)
2026-09-11	3cb10f47	Thomas Kabalin	!237	feat(api): let a player session read its own matches and statistics
2026-09-12	4c66715c	Danny Gebhard	!239	feat(web): wire player notes to the real endpoints (closes #250, #252)
2026-09-03	905b850a	Thomas Kabalin	!225	feat(api): take an unwatched serve or ending as null on a point
2026-09-03	fbb83337	Thomas Kabalin	!225	refactor(api): name the squad tally's watched-serve count serves_seen
2026-09-12	32d8bee3	Thomas Kabalin	!241	fix(api): hold a lopsided match at the ratio's ceiling in a stat's series
2026-09-12	88a673b1	Danny Gebhard	!247	feat(web): delete the fixture layer, its last dead export and stale comments (closes #224)
2026-09-13	9af9a9e8	Thomas Kabalin	!248	feat(api): add last-five-match figures to the squad members endpoint
2026-09-13	965fa231	Thomas Kabalin	!249	feat(api): report mean rally length split by point outcome
2026-09-13	0c81487d	Thomas Kabalin	!250	refactor(api): name the window in the squad member stats fields
2026-09-13	3bf4b75f	Thomas Kabalin	!251	fix(api): match rally-length histogram captions to capture's buckets
2026-09-13	d6434da0	Thomas Kabalin	!253	feat(api): count the errors behind the squad forced-to-unforced ratio

Date	Hash	Author	MR	Commit message
2026-09-13	785b4ca4	Danny Gebhard	!255	feat(web): revamp Players tab cards with per-player stats (#262 WIP)
2026-09-13	6dcef113	Danny Gebhard	!255	fix(web): back navigation, real bar heights, raw F:U counts on the player card
2026-09-13	0eaa049b	Danny Gebhard	!255	fix(web): align player-card bar charts, retire header click-through
2026-09-13	d8542af3	Danny Gebhard	!255	fix(web): glanceable stat-card polish, colour-system fixes, shot radar rework
2026-09-13	c04ff84c	Danny Gebhard	!255	fix(web): reword the 1st-serve stat label in plain English
2026-09-14	2cba2df1	Danny Gebhard	!256	fix(web): rework rally length, forced:unforced and last-5 visuals on the player card
2026-09-14	7a039827	Thomas Kabalin	!269	fix(api): anchor the squad trend on the roster's last active week
2026-09-14	35b403dc	Thomas Kabalin	!271	fix(api): divide the error rates by the endings a coach actually saw
2026-09-14	722eaf1c	Danny Gebhard	!279	feat(web): redesign the Squad stats page and add Player Insights
2026-09-15	1b026c84	Thomas Kabalin	!283	fix(api): anchor the squad trend on a decided or captured match
2026-09-15	b7953988	Thomas Kabalin	!288	feat(api): record how a match ended, not just who won
2026-09-15	6c16f71a	Thomas Kabalin	!289	test(web): hold the stubbed fetch until the insights skeleton is seen
2026-09-15	827f6b19	Thomas Kabalin	!289	test(web): state the insights card's poll budgets at the call site
2026-09-15	7a8717fd	Thomas Kabalin	!294	fix(web): treat an unobserved player insights card as pending
2026-09-15	a9a940ba	Thomas Kabalin	!297	feat(api): add a per-match rally-length trend to the rally tab

Date	Hash	Author	MR	Commit message
2026-09-15	3a409f21	Danny Gebhard	!296	feat(web): unify player notes into one dated list, redesign win/loss and surface cards
2026-09-15	a91ff2f2	Danny Gebhard	!296	feat(web): redesign the Serve tab around fault-mode courts, retire the drill-down
2026-09-15	6961bdf3	Danny Gebhard	!296	feat(web): redesign the Rally tab's final-ball court and point-enders visuals
2026-09-16	1da44c04	Thomas Kabalin	!301	feat: count the faults behind the serve tab's percentages
2026-09-16	df643885	Thomas Kabalin	!301	fix(web): weight the opponent card's combined return figures by their returns
2026-09-16	cdac508f	Thomas Kabalin	!301	feat(api): plot the rally trend as the share of points past five shots
2026-09-16	fef04a80	Danny Gebhard	!302	feat(web): rally intensity charts, momentum notes, and stats-page polish
2026-09-16	f6cd47f1	Thomas Kabalin	!303	style(web): drop the baseline claim from the rally intensity caption
2026-09-16	f14bbafb	Thomas Kabalin	!303	style(web): trim the error insight hint to the tap that opens it
2026-09-16	8b75230b	Danny Gebhard	!304	feat(web): replace the Forced:Unforced leaderboard, and surface failed squad-stats queries
2026-09-16	515e98b8	Danny Gebhard	!304	revert(web): drop the Pressure Index/Resilience/Tactical Efficiency leaderboard
2026-09-16	0424506b	Danny Gebhard	!304	fix(web): zoom player insights by spreading avatars apart, not magnifying them
2026-09-16	b2f65377	Danny Gebhard	!304	feat(web): move Wing to its own section, drop Shot type from the rally drill
2026-08-23	25974c9	Danny Gebhard	!192	feat(web): squad stats page — how the squad compares and trends
2026-08-23	cc0da19	Danny Gebhard	!192	fix(e2e): match the smoke test to the real Squad stats page

Date	Hash	Author	MR	Commit message
2026-08-23	dbe48d3	Danny Gebhard	!190	feat(web): squad-scoped needs-attention on coach home
2026-08-24	0439fcb	Thomas Kabalin	!201	test(web): cover the analytics query-option defaults
2026-08-24	06cd448	Thomas Kabalin	!200	feat(api): serve and rally analytics scoped to one match
2026-08-24	32f8038	Thomas Kabalin	!201	fix(web): keep a shot radar's series name out of the CSS property name
2026-08-24	4139c0f	Thomas Kabalin	!202	feat(api): serve outcomes, point counts and rally length on the two tabs
2026-08-24	4caf98c	Thomas Kabalin	!200	feat(api): fault modes and a serve-in trend on the serve tab
2026-08-24	5852149	Thomas Kabalin	!200	feat(api): point-ender styles on the rally tab
2026-08-24	5f69cb1	Danny Gebhard	!194	fix(web): mark W/L on every wins-losses record, stop clipping the roster halo ring
2026-08-24	7676f6a	Thomas Kabalin	!205	docs(web): point the squad placeholders at the endpoints behind them
2026-08-24	8c00fdd	Thomas Kabalin	!198	feat(api): bucket serve and rally analytics by set and phase
2026-08-24	c14d787	Thomas Kabalin	!204	fix(api): floor the points a tracked stat's baseline is measured over
2026-08-24	c401aa3	Thomas Kabalin	!196	fix(web): let charts lay out under jsdom
2026-08-24	fb0a0dc	Thomas Kabalin	!205	feat(api): serve the Squad stats page's leaderboard and trend
2026-08-24	fed37c5	Thomas Kabalin	!203	feat(api): serve the squad snapshot's per-player inputs
2026-09-15	4665588	Danny Gebhard	!296	docs: update color and icon rules for the player stats redesign
2026-09-15	a1fe14c	Thomas Kabalin	!295	docs(web): register the Info icon for stat-explainer popovers

Date	Hash	Author	MR	Commit message
2026-09-16	96c020a	Thomas Kabalin	!303	feat(web): explain the momentum chart behind an info button
2026-09-17	8eacc3c	Thomas Kabalin	!308	refactor: drop the five overview fields nothing renders, and Metric with them
2026-09-17	dab0c9b	Thomas Kabalin	!308	refactor(web): delete BulletBar and StatHero, the components Metric fed
2026-09-17	ee0b8d8	Thomas Kabalin	!308	refactor: stop counting the opponent aces nothing reads
2026-09-17	fc4cd3f	Thomas Kabalin	!307	refactor: drop the analytics fields no screen renders

Backlog Item 7

Date	Hash	Author	MR	Commit message
2026-08-15	70ff86b	Thomas Kabalin	!70	feat(web): map a captured Match onto the API's create payload
2026-08-17	a35602c	Thomas Kabalin	!102	feat(api): store a scoring system instead of a sets format
2026-08-17	f0177fa	Thomas Kabalin	!109	chore(web): add dexie for the offline outbox
2026-08-17	6dac6aa	Thomas Kabalin	!109	feat(web): add the Dexie outbox storage layer
2026-08-17	90e8c5b	Thomas Kabalin	!109	docs(testing): note the fake-indexeddb shim in the setup file
2026-08-17	38c932a	Thomas Kabalin	!109	docs(web): carve out features/outbox from the no-cross-imports rule
2026-08-18	cbc49ff	Thomas Kabalin	!109	docs(web): point the outbox comments at the settled sync decisions
2026-08-18	22255bd	Thomas Kabalin	!140	feat(web): send the client-generated match id when syncing a match
2026-08-18	df72239	Thomas Kabalin	!141	feat(web): purge local data on sign-out and a coach switch
2026-08-19	2248029	Thomas Kabalin	!152	feat(web): mint and persist a device id
2026-08-20	a7ba608	Thomas Kabalin	!173	feat(api): refuse deleting a match another device is capturing
2026-08-20	2526607	Thomas Kabalin	!173	feat(web): name this device when deleting a match
2026-08-20	bed36a3	Thomas Kabalin	!173	docs: record the capture-claim refusal in the deletion convention
2026-08-20	53c2229	Thomas Kabalin	!174	feat(web): record why an outbox row was parked
2026-08-20	1e40e15	Thomas Kabalin	!174	feat(web): drain the outbox into the API
2026-08-20	15c3eb3	Thomas Kabalin	!174	feat(api): accept only version-4 client match ids
2026-08-20	e36cb92	Thomas Kabalin	!174	feat(web): wire the drain to every flush opportunity
2026-08-21	e5e2464	Thomas Kabalin	!176	test(web): e2e offline round trip against a production build
2026-08-21	6516675	Thomas Kabalin	!181	feat(web): discard outbox rows for a deleted match
2026-08-21	12ee9aa	Thomas Kabalin	!181	feat(web): purge one match's rows from the outbox
2026-08-21	05df2c9	Thomas Kabalin	!174	fix(api): tolerate clock_timestamp() drift between created_at and updated_at
2026-08-21	9dc84bb	Thomas Kabalin	!174	fix(web): update sync point fixtures for the courtSide/location split

2026-08-21	eb04542	Thomas Kabalin		docs: add the offline sync mechanism doc
2026-08-21	83f2fbc	Thomas Kabalin	!185	test(web): cover the parts of the offline round trip points don't block
2026-08-21	d4667e9	Thomas Kabalin	!185	docs: note the offline read side in the testing layer split
2026-08-24	a011d8f	Thomas Kabalin	!201	test(web): cover the drain's match-side failure verdicts
2026-08-26	e0717f9	Luke Bryant	!208	fix(web): queue a match's outcome even when ending it queued nothing else
2026-08-27	a16639c	Luke Bryant	!208	feat(web): keep the match in a durable log, so a dead phone doesn't lose it
2026-08-28	68edf7e	Luke Bryant	!208	fix(web): count only the queued rows a drain can actually send
2026-08-28	8d181e6	Luke Bryant	!208	chore: keep the schema-gaps note in docs/scratch, not the source tree
2026-08-28	7eb8880	Thomas Kabalin	!208	fix(web): keep a refused match's undrainable rows out of the failed count
2026-08-28	c75d1fd	Thomas Kabalin	!208	fix(web): forget a deleted match's log when the drain hears it is gone
2026-08-28	2a6ab5f	Thomas Kabalin	!208	docs: drop broken scratch-doc reference from offline-sync
2026-08-11	50c45d9	Thomas Kabalin	!12	test(web): cover the update prompt
2026-08-11	f07a395	Thomas Kabalin	!12	feat(web): precache the app shell as an installable PWA
2026-08-12	7450b58	Thomas Kabalin	!12	refactor(web): move the PWA options beside the prompt they configure
2026-08-12	ca2c81c	Thomas Kabalin	!12	test(web): cover the PWA config and icon assets
2026-08-15	2b6973a	Thomas Kabalin	!76	feat(web): ban raw fetch in feature code
2026-08-15	5bdb9fa	Thomas Kabalin	!70	docs: prescribe the API contract checks in the web test rules
2026-08-18	59a4f15	Thomas Kabalin	!133	feat(web): persist the read cache to IndexedDB
2026-08-20	bce4bf3	Thomas Kabalin	!164	docs(testing): record the offline test-layer split, coverage policy, and mutation checks
2026-09-03	701efdd	Thomas Kabalin	!225	test(web): give the offline Playwright project a timeout its polls fit
2026-09-03	76d0a4f	Thomas Kabalin	!225	test(web): cover the point half of the offline round trip end to end
2026-09-03	7acdb1f	Thomas Kabalin	!225	test(web): seed the offline suite from capture-screen taps

2026-09-03	ff88472	Thomas Kabalin	!225	feat(web): map captured points onto the API's point contract
2026-09-12	426a43a	Thomas Kabalin	!241	test(web): cover the outbox's schema and its version bump
2026-09-12	560920e	Thomas Kabalin	!225	refactor(web): send the point note without the unusual-note fold
2026-09-12	e7b83f5	Danny Gebhard	!240	feat: wire the in-progress banner to a real capture claim (closes #251)
2026-09-14	54003a6	Thomas Kabalin	!268	docs(web): say the checkpoint and delete endpoints now exist
2026-09-14	5b08c1c	Luke Bryant	!267	test(web): pin the note's client-side bound to the API's own contract
2026-09-14	603a3e4	Luke Bryant	!277	fix(web): drain checkpoint and eventDelete rows, stop gating the outcome on them
2026-09-14	cbbf470	Thomas Kabalin	!259	docs: stop tech-stack.md claiming the sync indicator ships
2026-09-14	fba8587	Luke Bryant	!266	feat(web): claim, renew and hand over the single-writer lock from capture
2026-09-15	13d33af	Luke Bryant	!284	feat(web): add the sync status indicator to live capture (closes #115)
2026-09-15	331acee	Thomas Kabalin	!294	docs: describe the corrected-match guard, not the defect it replaced
2026-09-15	53d8853	Thomas Kabalin	!294	refactor(web): drop the outbox's unused failed-row readers
2026-09-15	5503d5e	Thomas Kabalin	!294	fix(web): discard every row of a deleted match, not just its points
2026-09-15	99448b4	Luke Bryant	!287	feat(web): retry failed points straight from the sync badge
2026-09-15	ce51631	Thomas Kabalin	!294	fix(web): catch a failed retry from the sync badge
2026-09-15	e96fab1	Thomas Kabalin	!294	fix(web): stop the sync badge blinking on every capture tap

Backlog item 8

Date	Hash	Author	MR	Commit message
2026-08-15	d3b77207	Danny Gebhard	!47	feat(web): move QueryClientProvider to the root layout, retire the players App scaffold
2026-08-15	b5c8dbff	Danny Gebhard	!48	feat(web): give home its own feature slice
2026-08-15	d56f3814	Danny Gebhard	!68	feat(web): add the bottom nav to the players roster page
2026-08-15	a814c561	Danny Gebhard	!68	feat(web): link players to their profile page, track last-viewed
2026-08-15	cfe4c65b	Danny Gebhard	!68	feat(web): add the player profile route and page
2026-08-15	c048b7de	Danny Gebhard	!68	feat(web): add the bottom nav to home
2026-08-15	c4a0acfa	Danny Gebhard	!68	feat(web): add the squad snapshot box to home
2026-08-15	462f3db3	Danny Gebhard	!68	feat(web): add the last-viewed-player box to home
2026-08-15	1a454dbb	Danny Gebhard	!68	feat(web): add the My Players bubble row to home
2026-08-15	26804b5d	Danny Gebhard	!68	feat(web): add a player search bar to home
2026-08-15	5d57e862	Danny Gebhard	!68	feat(web): show the signed-in coach's name on home
2026-08-15	9a0ce3c5	Danny Gebhard	!68	feat(web): add profile/settings buttons and placeholder routes to home
2026-08-15	4d39408a	Danny Gebhard	!68	feat(web): add the bottom nav component
2026-08-15	c844ad84	Danny Gebhard	!68	feat(web): add the /stats route
2026-08-15	7166d91a	Danny Gebhard	!68	feat(web): add the /matches route
2026-08-15	407952b2	Thomas Kabalin	!68	fix(web): render the player profile at its own route
2026-08-15	0abc65f6	Thomas Kabalin	!68	fix(web): count matches by local calendar date

Date	Hash	Author	MR	Commit message
2026-08-17	f1e290e7	Danny Gebhard	!116	feat(web): player profile tabs and match setup screen
2026-08-18	0cb45c8f	Danny Gebhard	!126	fix(web): address Thomas's review on !126
2026-08-18	75b23c82	Danny Gebhard	!126	feat(web): build the seven player-stats tabs
2026-08-18	035a5cdc	Danny Gebhard	!126	feat(web): build the wireframe's home, roster and match-list screens
2026-08-18	613c4a44	Danny Gebhard	!126	feat(web): add the provisional analytics layer behind the stats screens
2026-08-18	264c2d63	Danny Gebhard	!126	feat(web): add the categorical palette tokens and the shared stat primitives
2026-08-18	3ab1b382	Danny Gebhard	!126	chore(web): install the shadcn components the analytics screens need
2026-08-18	85fb095c	Danny Gebhard	!127	fix(web): restore role="alert" on a failed player load
2026-08-18	1e5a5e1d	Danny Gebhard	!127	fix(web): say "not found" for a player id that isn't the coach's
2026-08-18	1b197c16	Danny Gebhard	!128	feat(web): require date of birth and gender when adding a player
2026-08-18	6eea5a40	Danny Gebhard	!128	fix(web): cap a top/bottom sheet's height to the viewport
2026-08-18	59a559c2	Danny Gebhard	!136	feat(web): add a custom 404 page for unmatched routes
2026-08-18	97427887	Danny Gebhard	!137	fix(web): wire dark mode to the OS preference, fix its brand blue, quiet scrollTo test noise
2026-08-18	251f45e5	Danny Gebhard	!138	feat(web): add a player edit form, fix home's Add tile opening the roster instead
2026-08-19	7e4533dd	Danny Gebhard	!139	feat(web): add a PWA install guide with platform-specific steps
2026-08-19	5a565e64	Danny Gebhard	!153	feat(web): add a delete action for players

Date	Hash	Author	MR	Commit message
2026-08-19	11655ef2	Danny Gebhard	!154	feat(web): add a delete action for matches
2026-08-19	92b64bf6	Danny Gebhard	!161	feat(web): edit display name and profile photo from Profile (#179)
2026-08-19	1e3abdc8	Danny Gebhard	!161	feat(web): nudge a photo-less coach from Home, show their avatar in the header (#179)
2026-08-19	442fe32e	Danny Gebhard	!161	test(web): promote the profile photo e2e check to a real spec
2026-08-20	cd50fd4b	Danny Gebhard	!162	feat(web): add, change, or clear a player photo (#192)
2026-08-20	84495dc3	Danny Gebhard	!163	feat(web): transfer a player to another coach via a one-time code (#185
2026-08-29	e3eeae2	Luke Bryant	!209	feat(web): a fake season, captured the way the app captures one
2026-08-30	1f2cb11	Thomas Kabalin	— (direct push to main)	docs: document the CSV format import_sample_data.py expects
2026-08-30	71412e8	Thomas Kabalin	— (direct push to main)	docs: note the shared demo password in the sample-data README
2026-08-30	9db6a26	Luke Bryant	!215	feat(web): generate the season in the API importer's format too
2026-08-30	83709b2	Thomas Kabalin	!215	docs: point the demo-data docs at where the CSVs actually live
2026-08-31	ca09ca13	Danny Gebhard	!222	feat(web): wire the squad snapshot and roster to real analytics data
2026-08-31	31487a4e	Danny Gebhard	!223	feat(web): wire the player profile and per-match stats to real analytics data
2026-09-12	19c84299	Danny Gebhard	!242	fix(web): land on the app, not the sign-in form, with a live session (closes #257)
2026-09-12	56a92973	Danny Gebhard	!242	fix(web): stop / bouncing into an infinite redirect loop on a real 401

Date	Hash	Author	MR	Commit message
2026-09-12	aceac635	Danny Gebhard	!244	feat(web): let a player sign in and see their own scoped view (closes #203)
2026-09-12	86e0d33d	Luke Bryant	!245	feat(web): add the start-match action to the home screen
2026-09-12	a18f93bc	Luke Bryant	!246	fix(web): give the capture screen exactly one viewport, not one plus the header
2026-09-13	82d0de3a	Luke Bryant	!252	fix(web): give capture's court container query a definite ancestor
2026-09-13	785b4ca4	Danny Gebhard	!255	feat(web): revamp Players tab cards with per-player stats (#262 WIP)
2026-09-14	07d7bd56	Danny Gebhard	!256	feat(web): redesign the Matches tab cards with inline players, set scores and momentum
2026-09-14	82d06229	Danny Gebhard	!276	fix(web): stop Home and Matches clipping card content on mobile
2026-09-15	a9fdda98	Thomas Kabalin	!285	fix(web): keep the floating action clear of the bottom nav
2026-09-15	d58b1d3a	Thomas Kabalin	!285	feat(web): take the coach home when they tap the header mark
2026-09-15	e82c8d85	Danny Gebhard	!292	refactor(web): score Home's squad snapshot by season, not last-5 stats
2026-09-15	76fc1b32	Thomas Kabalin	!293	fix(web): drop the needs-attention caveat from home's stat legend
2026-09-15	53403964	Thomas Kabalin	!295	feat(web): hide Home's squad-stat legend behind per-row info popovers
2026-09-15	cab42962	Thomas Kabalin	!295	style(web): lighten squad-row player names off the label's weight
2026-09-15	b11e0309	Thomas Kabalin	!295	fix(web): thicken squad-roster form band and stop its dots clipping
2026-09-15	29a07db0	Thomas Kabalin	!295	feat(web): scale up Home's roster tally icons as the shown count shrinks

Date	Hash	Author	MR	Commit message
2026-09-16	4df76587	Danny Gebhard	!304	fix(web): flatten player routes so Settings/Matches/Install don't render as Me
2026-09-16	2bc9b642	Danny Gebhard	!304	feat(web): redesign the Matches card, stop Home's per-player queries, and show a match's event
2026-09-16	36be7ff7	Danny Gebhard	!304	style(web): rename Home's serve rows to name what they measure
2026-09-16	9478e0d4	Danny Gebhard	!304	fix(web): deactivate dark mode
2026-08-03	625f5fe	Thomas Kabalin		docs: restructure CLAUDE.md and extract web UI rule
2026-08-04	37480c5	Thomas Kabalin		feat(web): add shadcn/ui with the base-luma button
2026-08-04	ab6f982	Thomas Kabalin		docs: scope web rules to styling, components and capture
2026-08-04	addd00f	Thomas Kabalin		feat(web): scaffold react app with vite and vitest
2026-08-05	c316442	Thomas Kabalin		chore(web): add head metadata and a robots.txt
2026-08-11	1c174c6	Thomas Kabalin	!6	feat(web): read the roster through the generated client
2026-08-11	49f92a5	Thomas Kabalin	!7	test(web): build a fresh Response per stubbed fetch
2026-08-11	9c2a9d4	Danny Gebhard	!8	feat(web): wire up TanStack Router and TanStack Query
2026-08-11	eb6d24f	Thomas Kabalin	!6	test(web): stub fetch in the app shell test
2026-08-11	f7209fe	Thomas Kabalin	!12	feat(web): add the app icons and browser favicon
2026-08-13	c53a1dd	Danny Gebhard	!26	feat(web): finish the home page
2026-08-15	017d3e6	Thomas Kabalin	!76	docs: document the API client and date conventions

Date	Hash	Author	MR	Commit message
2026-08-16	1563e32	Danny Gebhard	!91	docs(web): document the reasoning behind the color palette rule
2026-08-16	6f96a6e	Danny Gebhard	!90	docs(web): add color rule for frontend palette and patterns
2026-08-16	a009441	Danny Gebhard	!88	feat(web): guard protected routes with a session check
2026-08-17	88f2cfe	Danny Gebhard	!106	docs(web): update color rule and palette reasoning for latest corrections
2026-08-18	475b8e6	Danny Gebhard	!129	feat(web): show the MatchPoint mark at the top of every app screen
2026-08-18	6110c78	Danny Gebhard	!121	fix(web): player avatar colors use the current palette, drop blue
2026-08-18	7ffcb77	Danny Gebhard	!124	fix(web): converge every shadcn component on the rounded-sm radius default
2026-08-18	b31969c	Danny Gebhard	!125	docs(rules): add dashboard color roles to the color rule
2026-08-18	c5800b8	Danny Gebhard	!123	docs(rules): add shadcn defaults rule
2026-08-19	11e71c2	Danny Gebhard	!151	docs(web): add the UX-laws design rule for frontend work
2026-08-19	2b03085	Thomas Kabalin	!158	test(web): give the persisted-cache seed the suite's poll budget
2026-08-19	65acd9e	Danny Gebhard	!157	docs: generalize the wireframe skill, refine Fitts's Law guidance
2026-08-19	7b91d6b	Thomas Kabalin	!158	test(web): await the router's load before asserting on the route tree
2026-08-19	b0e6486	Thomas Kabalin	!149	refactor(web): stop branching on a player's archived state
2026-08-19	be7a235	Thomas Kabalin	!147	test(web): cover the auth guard's 429 on the session check
2026-08-23	882737c	Danny Gebhard	!191	fix(web): visual polish and fixes on the stats and home pages

Date	Hash	Author	MR	Commit message
2026-08-24	2d15e01	Thomas Kabalin	!201	test(web): give the sheets that navigate a router to navigate with
2026-08-24	31c0be1	Danny Gebhard	!195	docs(web): canonical icon registry, shadcn skill fixes, sharper color/theme rules
2026-08-24	40ddc47	Thomas Kabalin	!201	test(web): cover the photo hooks' failure paths and the name-edit sheet
2026-08-24	422940b	Thomas Kabalin	!201	test(web): cover the start-match player picker
2026-08-24	4b8794a	Thomas Kabalin	!201	test(web): cover the player sheets' exits and abandoned drafts
2026-08-24	7430e37	Thomas Kabalin	!201	test(web): cover the add-player sheet's failure and reset paths
2026-08-24	9b14a30	Thomas Kabalin	!201	test(web): cover match-list's surface, result and date filters
2026-08-24	d7b9bd5	Danny Gebhard	!193	feat(web): rebuild the matches page with filters, edit/delete and per-match stats
2026-09-11	1db05ad	Danny Gebhard	!235	feat(web): reveal a new player's username and one-time password (#219)
2026-09-11	fced8d3	Danny Gebhard	!235	feat(web): let a coach regenerate a player's password (#220)
2026-09-12	56b0880	Danny Gebhard	!240	test(web): scope the roster e2e assertion to the row, not page-wide text
2026-09-12	615224c	Danny Gebhard	!240	fix(web): pass the owner kind recordOwnerIfUnknown now requires
2026-09-14	3e70fb9	Danny Gebhard	!276	refactor(web): reuse e2e/api.ts's HTTP helpers in overflow.ts
2026-09-14	7124853	Thomas Kabalin	!272	fix(web): keep the agent overlay out of Playwright runs
2026-09-14	87032c7	Danny Gebhard	!260	fix(web): stop the Matches tab card overflowing horizontally on mobile WebKit
2026-09-14	bf291ed	Danny Gebhard	!265	fix(web): share one baseline across a match card's set momentum rows

Date	Hash	Author	MR	Commit message
2026-09-14	e40c5e2	Thomas Kabalin	!272	refactor(web): share the e2e API and capture helpers across specs
2026-09-15	1331290	Thomas Kabalin	!285	fix(web): lock a scrolling sheet or list to the vertical axis
2026-09-15	144cfb7	Thomas Kabalin	!285	fix(web): leave dominant hand unselected by default
2026-09-15	1c4e49d	Thomas Kabalin	!285	fix(web): show a pointer cursor on every button
2026-09-15	3abb8bd	Thomas Kabalin	!293	fix(web): stop the header logo shifting when a tab needs a scrollbar
2026-09-15	3edbaf7	Thomas Kabalin	!285	fix(web): let a select's popup be as wide as its own options
2026-09-15	558cd58	Thomas Kabalin	!293	fix(web): correct what the start-match screen's caption claims
2026-09-15	56e6896	Thomas Kabalin	!285	feat(web): open the edit sheet from a player's photo
2026-09-15	5861ee8	Thomas Kabalin	!293	docs: record that the hover cursor is a base rule
2026-09-15	66901a8	Thomas Kabalin	!293	fix(web): show a pointer cursor on every button
2026-09-15	7893080	Danny Gebhard	!292	fix(web): make Start-a-match and match-stats back buttons context-aware
2026-09-15	7e04213	Thomas Kabalin	!285	fix(web): clamp a long player note to four lines
2026-09-15	8f0d64c	Thomas Kabalin	!285	feat(web): show a player's username on their details card
2026-09-15	a09b3ab	Luke Bryant	!290	fix(web): stop the info toast from intercepting taps underneath it
2026-09-15	b1d82ec	Thomas Kabalin	!293	feat(web): let a picked dominant hand be cleared again

Date	Hash	Author	MR	Commit message
2026-09-15	c04597d	Thomas Kabalin	!293	feat(web): name the tracked player on a match's stats page
2026-09-15	cc6d2f3	Thomas Kabalin	!285	fix(web): give each digit of a code its own box
2026-09-15	fa84767	Thomas Kabalin	!285	feat(web): say which fields a new player or match must have
2026-09-16	0d75346	Thomas Kabalin	!299	feat(web): honour prefers-reduced-motion across the app
2026-09-16	93b76e6	Danny Gebhard	!304	test(web): delete the read-failures pin now that #274 is fixed

Backlog Item 9

Date	Hash	Author	MR	Commit message
2026-08-21	a4b74a3	Thomas Kabalin	!178	feat(api): add a one-off script to import sample coaches/players/matches/points from CSV
2026-08-23	674280d	Thomas Kabalin	!188	refactor(api): drop serve1_target/serve2_target from points
2026-08-24	da8cb84	Thomas Kabalin	!207	refactor(api): split the sample-data import into run_import and main
2026-08-24	4d6d7e9	Thomas Kabalin	!207	fix(api): mint player usernames in the sample-data import
2026-08-24	5a1187d	Thomas Kabalin	!207	fix(api): report a bad matches.csv row by its line number
2026-08-24	707ed2d	Thomas Kabalin	!207	feat(api): make the sample-data import re-runnable
2026-08-24	4dbed59	Thomas Kabalin	!207	chore(api): measure and type-check the scripts directory
2026-08-24	2a8cb96	Thomas Kabalin	!207	fix(api): name the CSV row when a join between files finds nothing

Date	Hash	Author	MR	Commit message
2026-08-28	e3eeae2	Luke Bryant	!209	feat(web): a fake season, captured the way the app captures one
2026-08-30	9db6a26	Luke Bryant	!215	feat(web): generate the season in the API importer's format too
2026-09-13	134bb97	Thomas Kabalin	!251	docs(api): correct three stale field-pairing rules in the import sample-data guide
2026-09-13	93d9911	Luke Bryant	!254	fix(web): record return_outcome IN when a return stays in play
2026-09-13	c5549df	Thomas Kabalin	!251	docs(api): note the rally-length bucket ranges in the import sample-data guide
2026-09-14	24c73f6	Thomas Kabalin	!257	feat(api): add a purge script for replacing a demo season
2026-09-14	2f494c2	Thomas Kabalin	!257	feat(api): mint a demo password that can be read aloud
2026-09-14	afbbd9e	Thomas Kabalin	!257	docs(api): say that re-importing does not replace a regenerated season
2026-09-15	22ab0ea	Luke Bryant	!298	feat: give every demo player and coach a profile photo
2026-09-15	7485096	Luke Bryant	!298	feat(web): deal each demo coach twelve archetyped players
2026-09-15	afb5c09	Luke Bryant	!298	feat: file dated coach notes on every demo player

2026-09-15	d4c1b3f	Luke Bryant	!298	feat: carry each match's result reason through the demo import
2026-09-15	f6f0942	Luke Bryant	!298	feat: generate every gap capture leaves, and import it

Appendix T

Strictly, a unit test checks one function on its own, with no database or network involved. Most of our backend tests instead run real HTTP requests against a real test database, which is technically an integration test.

We chose this deliberately: a database-backed feature (queries, constraints, aggregates) can pass against a fake database and still fail against the real one, so testing against real Postgres gives much stronger confidence than mocking it.

Real unit tests cover pure, self-contained logic: a function that computes or transforms something with no database, network, or UI involved. Anything that has to prove a whole feature actually works uses an integration test instead.

Backlog item 1

Test Results Summary

API (pytest): 216/216 passed across 12 files

API — pytest (216/216)

tests/models/test_coach.py — 29/29

- test_coach_round_trips
- test_created_at_differs_between_rows_in_one_transaction
- test_updated_at_advances_when_a_row_changes
- test_database_requires_every_column [email, display_name, password_hash]
- test_database_rejects_a_duplicate_email
- test_database_rejects_an_email_that_is_not_lowercased
- test_the_orm_lowercases_an_email_on_the_way_in
- test_database_rejects_an_email_over_the_column_width
- test_database_rejects_a_display_name_over_the_column_width
- test_database_rejects_default_surface_outside_the_enum
- test_database_rejects_default_scoring_system_outside_the_enum
- test_database_accepts_every_member_of_the_reused_enums (12 cases)
- test_database_accepts_a_coach_with_no_preferences
- test_database_accepts_a_password_hash_of_any_length
- test_a_coach_without_a_photo_has_no_photo_url
- test_a_stored_photo_key_becomes_a_url

tests/models/test_coach_session.py — 8/8

- test_coach_session_round_trips
- test_database_requires_every_column [coach_id, token_hash, expires_at]
- test_database_rejects_a_duplicate_token_hash
- test_database_rejects_a_token_hash_over_the_column_width
- test_database_requires_the_coach_to_exist
- test_a_coach_with_sessions_cannot_be_hard_deleted

tests/models/test_email_verification.py — 13/13

- test_email_verification_round_trips
- test_database_requires_every_column [email, attempts, expires_at]
- test_database_rejects_a_second_row_for_the_same_address
- test_database_rejects_a_duplicate_token_hash
- test_database_permits_many_rows_at_the_code_stage
- test_database_rejects_a_hash_over_the_column_width [otp_hash, token_hash]
- test_database_rejects_an_email_over_the_column_width
- test_database_rejects_an_email_that_is_not_lowercased
- test_database_rejects_a_row_holding_both_secrets
- test_database_rejects_a_row_holding_neither_secret

tests/models/test_password_reset.py — 13/13

- test_password_reset_round_trips
- test_database_requires_every_column [coach_id, attempts, expires_at]
- test_database_rejects_a_second_row_for_the_same_coach
- test_database_rejects_an_unknown_coach
- test_database_refuses_to_delete_a_coach_with_a_pending_reset
- test_database_rejects_a_hash_over_the_column_width [otp_hash, token_hash]
- test_database_rejects_a_duplicate_token_hash
- test_database_permits_many_rows_at_the_code_stage
- test_database_rejects_a_row_holding_both_secrets
- test_database_rejects_a_row_holding_neither_secret

tests/models/test_player.py — 23/23

- test_player_round_trips
- test_committed_rows_do_not_leak_between_tests_a
- test_committed_rows_do_not_leak_between_tests_b
- test_only_the_name_parts_and_the_owner_are_required
- test_created_at_differs_between_rows_in_one_transaction
- test_updated_at_advances_when_a_row_changes
- test_has_transfer_code_follows_the_code_row
- test_has_transfer_code_answers_for_one_player_at_a_time
- test_database_requires_both_name_parts [first_name, last_name]
- test_database_rejects_dominant_hand_outside_the_enum
- test_database_rejects_gender_outside_the_enum
- test_database_rejects_a_name_over_the_column_width [first_name, last_name]
- test_database_requires_a_username
- test_database_rejects_a_duplicate_username
- test_database_rejects_a_username_over_the_column_width
- test_database_requires_an_owner
- test_database_rejects_an_unknown_coach_id
- test_database_refuses_to_delete_a_coach_who_still_has_players
- test_coach_id_is_indexed
- test_a_player_without_a_photo_has_no_photo_url

- test_a_stored_photo_key_becomes_a_url

tests/models/test_player_session.py — 8/8

- test_player_session_round_trips
- test_database_requires_every_column [player_id, token_hash, expires_at]
- test_database_rejects_a_duplicate_token_hash
- test_database_rejects_a_token_hash_over_the_column_width
- test_database_requires_the_player_to_exist
- test_a_player_with_sessions_cannot_be_hard_deleted

tests/models/test_player_transfer_code.py — 7/7

- test_transfer_code_round_trips
- test_database_requires_every_column [player_id, token_hash]
- test_database_rejects_a_second_code_for_the_same_player
- test_database_rejects_a_reused_token_hash
- test_database_refuses_a_code_for_a_player_that_does_not_exist
- test_database_bounds_the_token_hash_at_the_hash_width

tests/models/test_match.py — 23/23

- test_match_round_trips_with_player
- test_only_the_player_and_date_are_required
- test_match_requires_a_played_on_date
- test_match_requires_an_existing_player
- test_a_player_with_matches_cannot_be_deleted
- test_database_rejects_surface_outside_the_enum
- test_database_rejects_scoring_system_outside_the_enum
- test_database_accepts_every_scoring_system (8 cases)
- test_database_rejects_result_outside_the_enum
- test_database_rejects_free_text_over_the_column_width
- test_database_rejects_half_a_capture_claim [half0, half1]
- test_database_accepts_a_whole_capture_claim
- test_player_id_is_indexed
- test_created_at_differs_between_rows_in_one_transaction
- test_updated_at_advances_when_a_row_changes

tests/models/test_match_set.py — 12/12

- test_match_set_round_trips_with_match
- test_a_set_in_flight_has_a_tally_and_no_winner
- test_match_set_requires_an_existing_match
- test_deleting_a_match_takes_its_derived_sets_with_it
- test_a_set_number_is_unique_within_a_match
- test_the_same_set_number_is_free_under_another_match
- test_database_rejects_a_winner_outside_the_enum
- test_database_accepts_every_side [tracked, opponent]
- test_match_id_leads_an_index
- test_created_at_differs_between_rows_in_one_transaction

- test_updated_at_advances_when_a_row_changes

tests/models/test_point.py — 67/67

- test_point_round_trips_with_the_minimal_shape
- test_point_round_trips_with_a_full_two_serve_opponent_serve_point
- test_point_requires_an_existing_match
- test_deleting_a_match_with_points_is_refused
- test_sequence_is_unique_within_a_match
- test_the_same_sequence_is_free_under_another_match
- test_match_id_leads_an_index
- test_nullable_columns_accept_null
- test_database_rejects_a_half_set_pair [half0, half1, half2, half3]
- test_database_rejects_end_reason_outside_the_enum
- test_database_accepts_every_enum_member (52 cases)
- test_created_at_differs_between_rows_in_one_transaction
- test_updated_at_advances_when_a_row_changes
- test_database_rejects_a_mid_ball_depth

apps/api/tests/models/test_player_note.py — 25/25

- test_player_note_round_trips
- test_a_note_filed_under_nothing_stores_an_empty_list
- test_the_note_date_is_independent_of_when_the_row_was_written
- test_deleting_a_player_with_notes_is_refused
- test_database_requires_every_column
- test_database_refuses_a_note_for_a_player_that_does_not_exist
- test_database_refuses_a_focus_area_outside_the_four
- test_database_accepts_no_focus_areas_at_all
- test_database_bounds_a_focus_area_at_the_widest_member
- test_only_the_body_and_the_date_are_required
- test_rejects_a_missing_required_field
- test_rejects_a_key_the_schema_does_not_carry
- test_strips_surrounding_whitespace_from_the_body
- test_rejects_a_whitespace_only_body
- test_accepts_a_body_at_the_length_limit
- test_rejects_an_overlong_body
- test_accepts_a_note_dated_today
- test_rejects_a_note_dated_in_the_future
- test_accepts_a_note_about_a_session_seasons_ago
- test_rejects_an_implausibly_old_note_date
- test_keeps_focus_areas_in_declaration_order
- test_files_a_note_under_each_area_once
- test_rejects_a_focus_area_outside_the_four
- test_rejects_a_null_focus_areas_list
- test_an_update_may_carry_nothing
- test_an_update_carries_each_mutable_field
- test_an_update_clears_the_focus_areas_with_an_empty_list

- test_an_update_rejects_a_null_over_a_non_nullable_column
- test_an_update_applies_the_same_bounds_as_a_create

tests/models/test_score_checkpoint.py — 7/7

- test_score_checkpoint_round_trips_its_snapshot
- test_a_sequence_is_unique_within_a_match
- test_the_same_sequence_in_another_match_is_fine
- test_serves_next_is_constrained_to_the_sides
- test_a_checkpoint_needs_an_existing_match
- test_a_match_with_checkpoints_refuses_to_be_deleted_underneath_them
- test_match_id_leads_an_index

tests/test_migrations.py — 6/6

- test_migrations_upgrade_downgrade_and_upgrade_again
- test_requiring_a_player_owner_refuses_orphans_rather_than_deleting_them
- test_the_username_backfill_numbers_existing_players_deterministically
- test_the_schema_at_head_matches_the_models
- test_dropping_mid_clears_the_placement_rather_than_the_point
- test_every_downgrade_leaves_the_schema_the_upgrade_found
- test_the_court_side_carries_across_the_boxes_and_back

Backlog item 2

Test Results Summary

API (pytest): 596/596 passed across 17 files

API — pytest (596/596)

tests/routers/test_players.py — 118/118

- test_create_player_persists_and_returns_row
- test_create_player_belongs_to_the_caller
- test_create_player_with_every_optional_field
- test_create_player_without_optional_fields
- test_create_player_cannot_set_coach_id
- test_create_player_mints_the_username_and_numbers_namesakes
- test_create_player_numbers_past_usernames_minted_before_this_request
- test_create_player_returns_a_one_time_password_and_stores_only_its_hash
- test_the_password_never_appears_on_a_read
- test_the_minted_password_never_meets_the_policy
- test_creates_beyond_the_minting_burst_are_refused
- test_create_player_retries_when_a_concurrent_create_takes_the_username
- test_create_player_gives_up_after_three_username_collisions
- test_regenerating_gives_a_backfilled_player_a_password_that_signs_them_in
- test_regenerating_stores_only_a_hash_and_replaces_the_old_password
- test_regenerating_ends_the_player_s_live_sessions_and_no_one_else_s
- test_regenerating_another_coach_s_player_is_the_404_an_unknown_id_gets
- test_a_404_from_the_password_route_spends_no_minting_token
- test_creation_and_regeneration_spend_the_same_minting_budget
- test_create_player_re_raises_an_integrity_error_that_is_not_the_username
- test_create_player_rejects_future_date_of_birth
- test_create_player_rejects_implausibly_old_date_of_birth
- test_create_player_accepts_a_plausibly_old_date_of_birth
- test_create_player_accepts_date_of_birth_of_today
- test_create_player_accepts_an_explicit_null_date_of_birth
- test_create_player_rejects_missing_first_name
- test_create_player_rejects_missing_last_name
- test_create_player_rejects_invalid_dominant_hand
- test_create_player_rejects_invalid_gender
- test_create_player_rejects_overlong_name
- test_create_player_accepts_name_at_the_length_limit
- test_create_player_rejects_overlong_notes
- test_create_player_accepts_notes_at_the_length_limit
- test_create_player_rejects_whitespace_only_name
- test_create_player_strips_surrounding_whitespace
- test_list_players_returns_all_sorted_by_last_then_first_name
- test_list_players_omits_another_coachs_players

- test_get_player_returns_one
- test_get_player_of_another_coach_is_404
- test_list_players_with_none_stored_is_an_empty_list
- test_get_player_unknown_id_is_404
- test_get_player_malformed_id_is_422
- test_update_player_changes_only_the_fields_sent
- test_update_player_clears_a_nullable_field_with_an_explicit_null
- test_update_player_rejects_a_null_name_part [first_name, last_name]
- test_update_player_with_an_empty_body_changes_nothing
- test_update_player_bumps_updated_at
- test_update_player_rejects_coach_id
- test_update_player_rejects_a_future_date_of_birth
- test_update_player_of_another_coach_is_404
- test_update_player_unknown_id_is_404
- test_delete_player_takes_their_matches_with_them
- test_delete_player_takes_their_matches_points_with_them
- test_delete_player_takes_a_match_being_captured_with_it
- test_delete_player_leaves_the_rest_of_the_roster_alone
- test_delete_player_is_refused_while_a_transfer_code_is_outstanding
- test_cancelling_the_code_clears_the_way_for_the_delete
- test_delete_player_leaves_another_coachs_roster_alone
- test_delete_player_unknown_id_is_404
- test_player_routes_require_a_session (8 cases)
- test_a_transfer_code_moves_the_player_and_their_matches
- test_a_pasted_code_may_carry_whitespace
- test_minting_again_replaces_the_previous_code
- test_a_redeemed_code_cannot_be_used_again
- test_redeeming_an_unknown_code_is_400
- test_redeeming_your_own_code_is_409_and_leaves_it_usable
- test_deleting_a_transfer_code_stops_it_working
- test_deleting_a_transfer_code_that_was_never_minted_is_204
- test_transfer_code_for_another_coachs_player_is_404 [post, delete]
- test_a_freshly_created_player_has_no_transfer_code
- test_the_roster_flags_only_the_players_holding_a_code
- test_the_roster_reports_every_flag_in_one_statement
- test_minting_a_code_shows_up_on_the_player
- test_cancelling_a_code_clears_the_flag
- test_redeeming_a_code_clears_the_flag_on_the_response
- test_the_flag_and_the_delete_refusal_agree
- test_uploading_a_photo_answers_with_the_player
- test_an_uploaded_photo_is_served_at_the_url_it_reports
- test_replacing_a_photo_removes_the_one_it_replaced
- test_clearing_a_photo_removes_the_file
- test_clearing_a_photo_that_is_not_there_is_a_no_op
- test_another_coachs_player_is_a_404_and_stores_nothing
- test_a_file_that_is_not_an_image_is_refused

- test_deleting_a_player_removes_their_photo
- test_a_transferred_player_keeps_their_photo
- test_delete_player_records_the_ids_of_the_matches_it_freed
- test_delete_player_with_no_matches_records_nothing
- test_a_note_survives_the_write_and_comes_back_on_the_next_read
- test_create_note_files_it_under_the_areas_sent
- test_create_note_defaults_to_no_focus_areas
- test_create_note_cannot_name_a_player_in_the_body
- test_create_note_against_another_coachs_player_is_404
- test_list_notes_is_newest_session_first
- test_list_notes_breaks_a_tie_on_when_the_note_was_written
- test_list_notes_carries_only_this_players_notes
- test_list_notes_with_none_written_is_an_empty_list
- test_list_notes_of_another_coachs_player_is_404
- test_update_note_changes_only_the_fields_sent
- test_update_note_clears_the_focus_areas_with_an_empty_list
- test_update_note_of_another_coachs_player_is_404
- test_a_note_reached_through_the_wrong_player_is_404
- test_delete_note_removes_it_and_leaves_the_rest
- test_delete_note_unknown_id_is_404
- test_delete_player_takes_their_notes_with_them

tests/routers/test_matches.py — 175/175

- test_create_match_persists_and_returns_row
- test_create_match_with_only_the_required_fields
- test_create_match_rejects_unknown_player
- test_create_match_against_another_coachs_player_is_404
- test_create_match_for_a_transferred_player_is_404
- test_create_match_rejects_missing_player_id
- test_create_match_rejects_a_malformed_player_id
- test_create_match_rejects_missing_played_on
- test_create_match_rejects_a_malformed_played_on
- test_create_match_rejects_a_future_played_on
- test_create_match_accepts_a_played_on_of_today
- test_create_match_rejects_an_implausibly_old_played_on
- test_create_match_rejects_an_invalid_surface
- test_create_match_rejects_an_invalid_scoring_system
- test_create_match_rejects_an_invalid_result
- test_create_match_rejects_a_whitespace_only_opponent_name
- test_create_match_rejects_an_overlong_opponent_name
- test_create_match_accepts_free_text_at_the_length_limit
- test_create_match_strips_surrounding_whitespace
- test_create_match_stores_a_client_supplied_id
- test_create_match_mints_a_fresh_id_each_time_none_is_sent
- test_create_match_repeated_with_the_same_id_returns_the_stored_match
- test_create_match_repeated_under_another_player_returns_the_stored_match

- test_create_match_refuses_an_id_the_coach_deleted
- test_create_match_allows_an_id_whose_delete_has_aged_out
- test_create_match_with_another_coachs_match_id_is_404
- test_create_match_rejects_a_malformed_id
- test_create_match_rejects_an_id_the_client_could_not_have_minted [nil, v1, v5]
- test_list_matches_newest_first
- test_list_matches_breaks_same_day_ties_by_creation_order
- test_list_matches_spans_every_player_of_the_caller
- test_list_matches_omits_another_coachs_matches
- test_list_matches_filtered_by_player
- test_list_matches_filtered_by_an_unknown_player_is_empty
- test_list_matches_filtered_by_another_coachs_player_is_empty
- test_list_matches_rejects_a_malformed_player_id
- test_list_matches_with_none_stored_is_an_empty_list
- test_get_match_returns_one
- test_get_match_of_another_coach_is_404
- test_get_match_unknown_id_is_404
- test_get_match_malformed_id_is_422
- test_update_match_changes_only_the_fields_sent
- test_update_match_clears_a_nullable_field_with_an_explicit_null
- test_update_match_records_a_result
- test_update_match_edits_the_metadata_of_a_completed_match
- test_update_match_with_an_empty_body_leaves_a_completed_match_alone
- test_update_match_cannot_change_a_recorded_result
- test_update_match_cannot_change_a_completed_matchs_scoring_system
- test_update_match_cannot_clear_a_completed_matchs_scoring_system
- test_update_match_fills_in_a_missing_scoring_system_after_completion
- test_update_match_resending_the_stored_values_is_a_no_op
- test_update_match_rejects_a_null_played_on
- test_update_match_with_an_empty_body_changes_nothing
- test_update_match_bumps_updated_at
- test_update_match_cannot_move_the_match_to_another_player
- test_update_match_rejects_a_future_played_on
- test_update_match_of_another_coach_is_404
- test_update_match_unknown_id_is_404
- test_delete_match_removes_it
- test_delete_match_takes_its_derived_sets_with_it
- test_delete_match_takes_its_points_with_it
- test_delete_match_leaves_another_matchs_points_alone
- test_delete_match_records_the_id_it_freed
- test_deleting_a_reused_id_starts_its_window_over
- test_delete_match_twice_is_404
- test_delete_match_of_another_coach_is_404
- test_delete_match_unknown_id_is_404
- test_delete_match_malformed_id_is_422
- test_delete_match_is_allowed_on_a_completed_match

- test_delete_match_under_another_devices_claim_is_refused [False, True]
- test_the_capturing_device_may_delete_its_own_match
- test_a_lapsed_claim_does_not_block_the_delete
- test_releasing_the_claim_clears_the_way_for_the_delete
- test_the_capture_refusal_still_stands_over_a_match_with_points
- test_claim_capture_takes_a_free_match
- test_claim_capture_renews_the_claim_of_the_device_holding_it
- test_claim_capture_refuses_a_second_device
- test_claim_capture_takes_over_a_lapsed_claim
- test_claim_capture_on_a_completed_match_is_409
- test_claim_capture_of_another_coachs_match_is_404
- test_claim_capture_unknown_id_is_404
- test_claim_capture_requires_a_device_id
- test_release_capture_frees_the_match_for_another_device
- test_release_capture_on_an_unclaimed_match_is_a_no_op
- test_release_capture_of_another_coachs_match_is_404
- test_recording_the_result_releases_the_capture_claim
- test_ingest_points_stores_the_batch_and_answers_with_the_derived_sets
- test_ingest_stores_every_field_the_client_stamped
- test_ingesting_the_same_batch_again_lands_on_the_same_answer
- test_a_repeated_point_keeps_the_stored_version
- test_an_overlapping_batch_converges
- test_ingest_recomputes_the_sets_a_late_batch_changes
- test_ingest_into_a_completed_match_is_refused
- test_a_sequence_already_held_by_a_different_point_is_refused
- test_a_point_already_stored_against_another_match_is_refused
- test_one_invalid_point_rejects_the_whole_batch
- test_an_empty_batch_is_refused
- test_a_flush_from_the_claim_holder_renews_the_claim
- test_a_flush_from_another_device_is_still_ingested
- test_a_flush_does_not_claim_a_match_nobody_holds
- test_a_flush_does_not_revive_the_lapsed_claim_it_once_held
- test_ingest_into_another_coachs_match_is_404
- test_ingest_into_an_unknown_match_is_404
- test_ingest_into_a_deleted_match_is_410
- test_ingest_into_a_match_whose_delete_has_aged_out_is_404
- test_list_points_orders_by_sequence_regardless_of_insertion_order
- test_list_points_of_a_match_with_none_recorded_is_an_empty_list
- test_list_points_of_another_coachs_match_is_404
- test_list_points_of_an_unknown_match_is_404
- test_list_points_round_trips_every_field_the_ingest_route_stored
- test_list_points_carries_no_ingestion_timestamps
- test_an_integrity_error_that_is_not_a_conflict_is_not_absorbed
- test_the_routes_that_read_the_claim_lock_the_match_row (5 cases)
- test_match_routes_require_a_session (9 cases)
- test_list_in_progress_matches_returns_the_claimed_match

- test_in_progress_does_not_care_which_device_holds_the_claim
- test_in_progress_puts_the_freshest_claim_first
- test_a_match_nobody_is_capturing_is_not_in_progress
- test_a_lapsed_claim_is_not_in_progress
- test_a_completed_match_is_not_in_progress
- test_in_progress_omits_another_coachs_match
- test_create_match_refuses_an_id_a_player_delete_freed
- test_update_a_deleted_match_is_410
- test_update_a_match_a_player_delete_took_is_410
- test_ingest_into_a_match_a_player_delete_took_is_410
- test_create_match_rejects_a_missing_scoring_system
- test_a_flush_claims_a_match_nobody_holds
- test_a_flush_takes_over_a_lapsed_claim
- test_a_flush_that_claims_a_match_puts_it_in_progress
- test_a_claim_taken_by_a_flush_still_blocks_another_devices_delete
- test_releasing_a_flush_claim_clears_the_way_for_another_devices_delete
- test_ingest_score_checkpoints_stores_the_batch_as_sent
- test_ingesting_the_same_checkpoints_again_is_a_no_op
- test_a_sequence_already_held_by_a_different_checkpoint_is_refused
- test_a_checkpoint_sequence_may_equal_a_points
- test_a_checkpoint_already_stored_against_another_match_is_refused
- test_one_invalid_checkpoint_rejects_the_whole_batch
- test_a_checkpoint_flush_from_the_claim_holder_renews_the_claim
- test_a_checkpoint_flush_from_another_device_is_still_ingested
- test_ingest_checkpoints_into_a_completed_match_is_refused
- test_ingest_checkpoints_into_another_coachs_match_is_404
- test_ingest_checkpoints_into_a_deleted_match_is_410
- test_list_score_checkpoints_orders_by_sequence
- test_list_score_checkpoints_round_trips_what_the_ingest_route_stored
- test_list_score_checkpoints_of_another_coachs_match_is_404
- test_delete_point_erases_the_point_and_rederives_the_sets
- test_delete_point_of_an_absent_id_is_still_204
- test_delete_point_leaves_a_point_of_another_match_alone
- test_delete_point_in_a_completed_match_is_refused
- test_delete_point_in_another_coachs_match_is_404
- test_delete_point_in_a_deleted_match_is_410
- test_delete_score_checkpoint_erases_the_row_and_rederives_the_sets
- test_delete_score_checkpoint_of_an_absent_id_is_still_204
- test_delete_score_checkpoint_leaves_another_matchs_alone
- test_delete_score_checkpoint_in_a_completed_match_is_refused
- test_delete_score_checkpoint_in_another_coachs_match_is_404
- test_delete_score_checkpoint_in_a_deleted_match_is_410
- test_delete_match_takes_its_checkpoints_with_it
- test_a_checkpoint_integrity_error_that_is_not_a_conflict_is_not_absorbed

tests/schemas/test_player.py — 48/48

- test_only_the_name_parts_are_required
- test_both_name_parts_are_required [first_name, last_name]
- test_rejects_a_blank_name_part (6 cases)
- test_accepts_a_single_character_name
- test_strips_surrounding_whitespace_from_a_name
- test_strips_a_name_before_measuring_its_length
- test_rejects_a_name_over_the_length_limit
- test_accepts_a_date_of_birth_of_today
- test_rejects_a_date_of_birth_of_tomorrow
- test_accepts_a_date_of_birth_just_inside_the_age_limit
- test_rejects_a_date_of_birth_just_outside_the_age_limit
- test_coerces_a_dominant_hand_string_to_the_enum
- test_rejects_a_dominant_hand_outside_the_enum
- test_coerces_a_gender_string_to_the_enum
- test_rejects_a_gender_outside_the_enum
- test_accepts_notes_at_the_length_limit
- test_rejects_notes_over_the_length_limit
- test_strips_surrounding_whitespace_from_notes
- test_rejects_blank_notes (2 cases)
- test_accepts_an_explicit_null_for_notes
- test_an_empty_update_carries_no_fields
- test_an_update_carries_only_the_fields_sent
- test_an_update_distinguishes_an_absent_field_from_a_null_one
- test_an_update_rejects_a_null_name_part [first_name, last_name]
- test_an_update_accepts_a_null_for_every_nullable_field [date_of_birth, dominant_hand, gender, notes]
- test_an_update_applies_the_name_bounds
- test_an_update_rejects_a_date_of_birth_of_tomorrow
- test_an_update_rejects_a_date_of_birth_outside_the_age_limit
- test_an_update_accepts_a_date_of_birth_inside_the_bounds
- test_a_transfer_code_is_bounded
- test_a_transfer_code_is_a_secret
- test_a_read_carries_the_photo_url_derived_from_the_key
- test_a_created_player_carries_the_password_in_the_clear
- test_neither_create_nor_update_accepts_a_username [PlayerCreate, PlayerUpdate]
- test_neither_create_nor_update_accepts_a_photo_url [PlayerCreate, PlayerUpdate]

tests/schemas/test_match.py — 54/54

- test_only_the_player_and_date_are_required
- test_coerces_an_id_string_to_a_uuid
- test_rejects_a_malformed_id
- test_treats_an_explicitly_null_id_as_an_absent_one
- test_coerces_a_player_id_string_to_a_uuid
- test_rejects_a_malformed_player_id
- test_rejects_a_missing_player_id
- test_rejects_a_missing_played_on

- test_rejects_a_played_on_that_is_not_an_iso_date (4 cases)
- test_accepts_a_played_on_of_today
- test_rejects_a_played_on_of_tomorrow
- test_accepts_a_played_on_just_inside_the_age_limit
- test_rejects_a_played_on_just_outside_the_age_limit
- test_coerces_a_surface_string_to_the_enum
- test_rejects_a_surface_outside_the_enum
- test_coerces_a_scoring_system_string_to_the_enum
- test_rejects_a_scoring_system_outside_the_enum
- test_coerces_a_result_string_to_the_enum
- test_rejects_a_result_outside_the_enum
- test_accepts_an_omitted_or_null_text_field [opponent_name, event]
- test_rejects_a_blank_text_field (6 cases)
- test_accepts_a_single_character_text_field [opponent_name, event]
- test_strips_surrounding_whitespace_from_a_text_field [opponent_name, event]
- test_strips_a_text_field_before_measuring_its_length [opponent_name, event]
- test_rejects_a_text_field_over_the_length_limit [opponent_name, event]
- test_an_empty_update_carries_no_fields
- test_an_update_carries_only_the_fields_sent
- test_an_update_rejects_a_player_id
- test_an_update_rejects_a_null_played_on
- test_an_update_accepts_a_null_for_every_nullable_field [opponent_name, surface, event, scoring_system, result]
- test_an_update_applies_the_text_bounds
- test_an_update_rejects_a_played_on_of_tomorrow
- test_an_update_rejects_a_played_on_outside_the_past_limit
- test_an_update_accepts_a_played_on_inside_the_bounds
- test_only_the_player_date_and_scoring_system_are_required
- test_rejects_a_missing_scoring_system
- test_rejects_a_null_scoring_system

tests/schemas/test_point.py — 71/71

- test_the_default_shape_round_trips
- test_shapes_the_contract_allows (15 cases)
- test_shapes_the_contract_refuses (34 cases)
- test_a_missing_required_field_is_refused [sequence, server, won, serve1_outcome, end_reason]
- test_every_stamp_is_required (7 cases)
- test_the_note_is_stripped_and_bounded
- test_the_full_enum_membership_reaches_the_columns
- test_a_batch_carries_a_device_and_its_points
- test_a_batch_without_a_device_is_refused
- test_an_empty_batch_is_refused
- test_a_batch_at_the_ceiling_is_accepted
- test_a_batch_over_the_ceiling_is_refused
- test_a_batch_cannot_repeat_a_point_id

- test_a_batch_cannot_repeat_a_sequence

tests/schemas/test_score_checkpoint.py — 25/25

- test_a_full_checkpoint_validates_and_keeps_its_enums
- test_the_tiebreak_fields_are_paired
- test_sets_are_numbered_from_one_in_order [starts at two, skips a set, out of order, repeats a set]
- test_a_snapshot_needs_at_least_one_set
- test_a_board_holds_at_most_the_sets_a_match_can_play
- test_a_count_cannot_be_negative [games_tracked, games_opponent, tiebreak_tracked, tiebreak_opponent]
- test_a_count_is_bounded_by_the_integer_column
- test_a_winner_must_be_a_side
- test_a_sequence_cannot_be_negative
- test_every_field_is_required [id, sequence, serves_next, order_changed, after]
- test_an_unknown_field_is_refused
- test_a_batch_cannot_repeat_an_id
- test_a_batch_cannot_repeat_a_sequence
- test_a_batch_is_bounded_like_a_point_batch
- test_the_read_schema_parses_the_stored_document

tests/test_usernames.py — 13/13

- test_the_surname_prefers_consonants_and_the_first_name_reads_plainly
- test_skipped_vowels_fill_in_when_the_consonants_run_short
- test_accents_fold_to_their_bare_letters
- test_punctuation_and_spaces_drop_out_before_folding
- test_a_short_name_is_padded_to_keep_the_prefix_fixed_width
- test_a_name_with_no_latin_letters_folds_to_the_placeholder
- test_casefold_handles_one_to_many_letters
- test_the_prefix_is_always_six_characters
- test_the_first_holder_of_a_name_is_numbered_001
- test_a_namesake_takes_the_next_number
- test_numbering_resumes_past_the_highest_ever_minted_not_the_count
- test_another_name_s_numbers_do_not_interfere
- test_past_999_the_number_widens_rather_than_wrapping

tests/test_photos.py — 23/23

- test_a_jpeg_is_stored_as_a_square_webp
- test_every_accepted_format_round_trips [JPEG, PNG, WEBP]
- test_two_uploads_are_stored_under_different_keys
- test_a_file_that_is_not_an_image_is_refused
- test_an_image_in_an_unlisted_format_is_refused
- test_an_image_claiming_too_many_pixels_is_refused
- test_an_image_at_the_pixel_bound_is_accepted
- test_the_stored_photo_carries_no_exif
- test_a_rotated_photo_is_uprighted_before_it_is_cropped

- test_a_wide_photo_is_centre_cropped_rather_than_squashed
- test_transparency_is_flattened_onto_white
- test_a_palette_image_is_stored_as_colour
- test_an_image_pillow_refuses_to_open_at_all_is_refused
- test_a_truncated_image_is_refused
- test_nothing_is_written_for_an_image_that_is_refused
- test_deleting_a_stored_photo_removes_it
- test_deleting_a_photo_that_is_already_gone_is_not_an_error
- test_deleting_nothing_is_not_an_error
- test_a_key_that_could_escape_the_media_directory_is_refused
- test_a_url_is_built_from_a_key
- test_no_key_is_no_url

tests/test_match_sets.py — 18/18

- test_a_set_in_flight_has_no_winner
- test_the_closing_point_names_the_set_winner
- test_a_set_won_by_the_tracked_player
- test_each_set_gets_its_own_row
- test_points_of_another_match_are_not_counted
- test_a_second_recompute_overwrites_the_row_and_stamps_updated_at
- test_a_set_row_the_points_no_longer_support_is_deleted
- test_a_match_with_no_points_keeps_no_set_rows
- test_an_unknown_match_derives_nothing
- test_a_checkpoint_alone_is_the_board
- test_points_after_a_checkpoint_add_to_its_board
- test_points_before_the_checkpoint_are_superseded_by_it
- test_the_newest_checkpoint_wins
- test_a_tail_point_stamped_into_a_closed_set_is_ignored
- test_the_checkpoints_declared_winner_stands_over_the_tail
- test_a_backwards_correction_shrinks_the_set_list
- test_a_deleted_point_takes_its_set_row_with_it
- test_the_fold_lands_on_the_same_answer_whatever_arrived_first

tests/test_csrf.py — 18/18

- test_a_write_from_our_own_origin_is_allowed [POST, PUT, PATCH, DELETE]
- test_a_write_from_another_origin_is_refused [POST, PUT, PATCH, DELETE]
- test_a_write_from_a_sibling_host_of_our_own_apex_is_refused
- test_a_write_with_no_origin_at_all_is_refused
- test_a_write_with_an_unusable_origin_is_refused (3 cases)
- test_a_read_from_another_origin_is_allowed [GET, HEAD, OPTIONS]
- test_another_port_on_our_own_host_is_another_origin
- test_the_scheme_is_deliberately_not_compared

tests/test_headers.py — 9/9

- test_a_stored_photo_carries_them_too
- test_json_responses_carry_the_strict_headers

- test_error_responses_carry_them_too
- test_a_refused_origin_gets_them_too
- test_an_unhandled_exception_gets_them_too
- test_docs_page_gets_its_scoped_csp
- test_the_docs_csp_survives_the_deployed_root_path
- test_docs_relaxation_does_not_reach_the_spec_route
- test_the_html_pages_nobody_uses_are_gone

tests/test_body_limit.py — 7/7

- test_oversized_declared_body_is_refused
- test_oversized_chunked_body_is_refused
- test_body_at_the_limit_reaches_the_route
- test_overflow_outside_an_exception_handler_is_still_a_413
- test_overflow_after_a_started_response_propagates
- test_unreadable_declared_length_is_left_to_the_count
- test_non_body_messages_are_not_counted

tests/test_rate_limit.py — 17/17

- test_a_key_is_admitted_up_to_the_burst_and_refused_beyond_it
- test_the_scale_setting_widens_a_bucket
- test_the_refusal_says_when_to_retry
- test_tokens_refill_at_the_sustained_rate
- test_an_idle_key_accrues_no_more_than_the_burst
- test_keys_do_not_share_a_bucket
- test_a_refused_request_spends_nothing
- test_a_refunded_request_leaves_the_bucket_as_it_found_it
- test_a_refund_cannot_bank_more_than_the_burst
- test_refunding_a_key_that_never_spent_is_harmless
- test_taking_from_several_buckets_admits_only_when_all_of_them_can_pay
- test_a_bucket_refused_by_another_is_not_charged_for_it
- test_a_slower_per_key_tier_leaves_the_shared_one_to_other_keys
- test_the_refusal_quotes_the_bucket_furthest_from_a_token
- test_the_furthest_bucket_is_quoted_even_when_another_is_refused_first
- test_a_new_key_evicts_buckets_that_have_refilled_to_full
- test_a_new_key_keeps_buckets_still_refilling

tests/test_logs.py — 16/16

- test_every_response_carries_a_request_id
- test_a_well_formed_inbound_id_is_kept
- test_a_malformed_inbound_id_is_replaced (4 cases)
- test_each_request_logs_one_line_with_the_documented_fields
- test_an_authenticated_request_logs_the_coach_id
- test_an_authenticated_player_request_logs_the_player_id
- test_a_response_no_route_saw_is_logged_too
- test_a_crashed_handler_still_gets_its_line
- test_a_line_logged_inside_a_route_inherits_the_request_id

- test_no_secret_from_the_auth_flows_reaches_the_log
- test_secret_named_fields_are_redacted_at_any_depth
- test_a_tty_gets_console_lines_and_a_pipe_gets_json
- test_uvicorns_loggers_are_rerouted_and_its_access_log_dropped

tests/test_errors.py — 11/11

- test_a_secret_field_is_the_only_thing_redacted
- test_a_password_alongside_another_fields_error_is_redacted
- test_a_password_nested_below_the_reported_field_is_redacted
- test_an_unknown_keys_value_is_redacted
- test_an_error_carrying_no_input_is_passed_through
- test_a_field_that_is_not_secret_keeps_its_input
- test_every_secret_field_in_the_schemas_is_named_in_secret_fields
- test_an_unhandled_exception_answers_a_readable_json_body
- test_an_unhandled_exception_tells_the_client_nothing_about_itself
- test_the_500_carries_the_request_id_that_logged_the_traceback
- test_a_failure_above_the_logging_middleware_is_answered_too

apps/api/tests/schemas/test_player_note.py — 25/25

- test_only_the_body_and_the_date_are_required
- test_rejects_a_missing_required_field
- test_rejects_a_key_the_schema_does_not_carry
- test_strips_surrounding_whitespace_from_the_body
- test_rejects_a_whitespace_only_body
- test_accepts_a_body_at_the_length_limit
- test_rejects_an_overlong_body
- test_accepts_a_note_dated_today
- test_rejects_a_note_dated_in_the_future
- test_accepts_a_note_about_a_session_seasons_ago
- test_rejects_an_implausibly_old_note_date
- test_keeps_focus_areas_in_declaration_order
- test_files_a_note_under_each_area_once
- test_rejects_a_focus_area_outside_the_four
- test_rejects_a_null_focus_areas_list
- test_an_update_may_carry_nothing
- test_an_update_carries_each_mutable_field
- test_an_update_clears_the_focus_areas_with_an_empty_list
- test_an_update_rejects_a_null_over_a_non_nullable_column
- test_an_update_applies_the_same_bounds_as_a_create

tests/test_health.py — 3/3

- test_health_returns_ok
- test_health_reports_dev_when_no_commit_is_set
- test_health_reports_the_deployed_commit

tests/test_openapi.py — 6/6

- test_the_committed_spec_matches_the_routes
- test_operation_ids_are_unique
- test_every_request_body_forbids_unknown_keys
- test_an_upload_body_is_typed_as_a_file
- test_main_writes_the_spec_file
- test_the_media_mount_is_not_in_the_spec

Backlog item 3

Test Results Summary

API (pytest): 856/856 passed across 27 files
Web (vitest): 268/268 passed across 35 files
Total: 1124/1124 passed

API — pytest (856/856)

tests/models/test_coach.py — 29/29

- test_coach_round_trips
- test_created_at_differs_between_rows_in_one_transaction
- test_updated_at_advances_when_a_row_changes
- test_database_requires_every_column [email]
- test_database_requires_every_column [display_name]
- test_database_requires_every_column [password_hash]
- test_database_rejects_a_duplicate_email
- test_database_rejects_an_email_that_is_not_lowercased
- test_the_orm_lowercases_an_email_on_the_way_in
- test_database_rejects_an_email_over_the_column_width
- test_database_rejects_a_display_name_over_the_column_width
- test_database_rejects_default_surface_outside_the_enum
- test_database_rejects_default_scoring_system_outside_the_enum
- test_database_accepts_every_member_of_the_reused_enums [default_surface: hard, clay, grass, carpet]
- test_database_accepts_every_member_of_the_reused_enums [default_scoring_system: traditional, mens_grand_slam, womens_grand_slam, no_ad_scoring, match_tiebreak_10_points, short_sets, pro_set, legacy]
- test_database_accepts_a_coach_with_no_preferences
- test_database_accepts_a_password_hash_of_any_length
- test_a_coach_without_a_photo_has_no_photo_url
- test_a_stored_photo_key_becomes_a_url

tests/models/test_coach_session.py — 8/8

- test_coach_session_round_trips
- test_database_requires_every_column [coach_id, token_hash, expires_at]

- test_database_rejects_a_duplicate_token_hash
- test_database_rejects_a_token_hash_over_the_column_width
- test_database_requires_the_coach_to_exist
- test_a_coach_with_sessions_cannot_be_hard_deleted

tests/models/test_email_verification.py — 13/13

- test_email_verification_round_trips
- test_database_requires_every_column [email, attempts, expires_at]
- test_database_rejects_a_second_row_for_the_same_address
- test_database_rejects_a_duplicate_token_hash
- test_database_permits_many_rows_at_the_code_stage
- test_database_rejects_a_hash_over_the_column_width [otp_hash, token_hash]
- test_database_rejects_an_email_over_the_column_width
- test_database_rejects_an_email_that_is_not_lowercased
- test_database_rejects_a_row_holding_both_secrets
- test_database_rejects_a_row_holding_neither_secret

tests/models/test_password_reset.py — 13/13

- test_password_reset_round_trips
- test_database_requires_every_column [coach_id, attempts, expires_at]
- test_database_rejects_a_second_row_for_the_same_coach
- test_database_rejects_an_unknown_coach
- test_database_refuses_to_delete_a_coach_with_a_pending_reset
- test_database_rejects_a_hash_over_the_column_width [otp_hash, token_hash]
- test_database_rejects_a_duplicate_token_hash
- test_database_permits_many_rows_at_the_code_stage
- test_database_rejects_a_row_holding_both_secrets
- test_database_rejects_a_row_holding_neither_secret

tests/models/test_player.py — 23/23

- test_player_round_trips
- test_committed_rows_do_not_leak_between_tests_a
- test_committed_rows_do_not_leak_between_tests_b
- test_only_the_name_parts_and_the_owner_are_required

- test_created_at_differs_between_rows_in_one_transaction
- test_updated_at_advances_when_a_row_changes
- test_has_transfer_code_follows_the_code_row
- test_has_transfer_code_answers_for_one_player_at_a_time
- test_database_requires_both_name_parts [first_name, last_name]
- test_database_rejects_dominant_hand_outside_the_enum
- test_database_rejects_gender_outside_the_enum
- test_database_rejects_a_name_over_the_column_width [first_name, last_name]
- test_database_requires_a_username
- test_database_rejects_a_duplicate_username
- test_database_rejects_a_username_over_the_column_width
- test_database_requires_an_owner
- test_database_rejects_an_unknown_coach_id
- test_database_refuses_to_delete_a_coach_who_still_has_players
- test_coach_id_is_indexed
- test_a_player_without_a_photo_has_no_photo_url
- test_a_stored_photo_key_becomes_a_url

tests/routers/test_auth.py — 131/131

- test_logging_in_returns_the_coach_and_sets_the_session_cookie
- test_the_cookie_is_locked_down
- test_logging_in_stores_the_hash_of_the_token_and_never_the_token
- test_a_login_expires_a_lifetime_from_now
- test_each_login_is_its_own_session
- test_logging_in_clears_the_coach_s_expired_sessions
- test_logging_in_leaves_other_coaches_expired_sessions_alone
- test_a_wrong_password_is_rejected_without_a_session
- test_an_unknown_email_answers_exactly_like_a_wrong_password
- test_an_unknown_email_still_burns_a_verification
- test_a_mixed_case_email_finds_the_account
- test_a_password_below_the_sign_up_floor_is_a_401_not_a_422
- test_a_stale_hash_is_upgraded_on_login

- test_a_current_hash_is_left_alone_on_login
- test_a_login_that_verified_a_retired_password_mints_no_session
- test_a_login_locks_the_coach_row_before_minting_a_session
- test_login_attempts_beyond_the_account_burst_are_refused
- test_the_login_429_answers_a_known_and_an_unknown_address_alike
- test_a_flooded_account_does_not_lock_out_the_rest
- test_the_account_bucket_folds_case
- test_cycling_accounts_does_not_escape_the_address_burst
- test_login_has_a_total_bound_no_address_escapes
- test_a_body_the_schema_rejects_is_still_counted
- test_an_arrival_beyond_the_address_burst_spends_no_verification
- test_a_correct_password_is_never_refused_by_the_account_tier
- test_logging_out_revokes_the_session_and_clears_the_cookie
- test_logging_out_only_ends_the_presented_session
- test_logging_out_signed_out_is_still_a_204
- test_logging_out_with_a_forged_cookie_revokes_nothing
- test_a_rejected_login_does_not_echo_the_password
- test_revoking_ends_the_coach_s_other_sessions
- test_the_token_the_revocation_was_made_with_stops_working
- test_a_revocation_from_a_session_due_for_renewal_hands_back_the_new_token
- test_revoking_with_nothing_else_to_end_is_still_a_204
- test_revoking_needs_a_session_of_its_own
- test_requesting_a_code_emails_it_to_the_address
- test_a_failed_send_answers_502
- test_a_failed_send_gives_the_address_its_code_budget_back
- test_the_code_is_stored_hashed_and_expiring
- test_a_registered_address_is_refused_with_no_code
- test_a_mixed_case_address_meets_the_409
- test_the_stored_address_is_lowercased
- test_requesting_again_replaces_the_code
- test_requesting_a_code_undoes_a_finished_verification

- test_requesting_sweeps_expired_rows_and_leaves_live_ones
- test_a_fourth_code_for_one_address_is_refused
- test_a_registered_address_meets_the_same_429_as_a_fresh_one
- test_the_code_budget_folds_case
- test_cycling_addresses_meets_the_address_burst
- test_request_otp_has_a_total_bound_no_address_escapes
- test_answering_the_code_sets_the_verification_cookie
- test_the_verification_cookie_is_locked_down
- test_answering_the_code_retires_it_for_the_token
- test_the_whole_flow_ends_in_an_account
- test_a_wrong_code_is_refused_and_charged
- test_an_unknown_address_answers_exactly_like_a_wrong_code
- test_an_expired_code_is_refused
- test_an_answered_code_cannot_be_answered_again
- test_verification_folds_the_address_case
- test_the_attempt_budget_kills_the_code
- test_a_guess_short_of_the_budget_does_not_cost_the_code
- test_guess_arrivals_beyond_the_address_burst_are_refused
- test_a_rejected_verification_does_not_echo_the_code
- test_the_flow_runs_on_the_default_logging_sender
- test_forgetting_a_password_emails_a_code
- test_an_unknown_address_answers_the_same_202_with_no_mail_and_no_row
- test_the_reset_code_is_stored_hashed_and_expiring
- test_a_mixed_case_address_finds_its_account
- test_forgetting_again_replaces_the_code
- test_forgetting_undoes_a_finished_verification
- test_forgetting_sweeps_expired_rows_and_leaves_live_ones
- test_a_fourth_code_for_one_account_is_refused
- test_the_forgot_password_429_answers_a_known_and_an_unknown_address_alike
- test_forgot_password_cycling_addresses_meets_the_address_burst
- test_forgot_password_has_a_total_bound_no_address_escapes

- test_a_failed_send_answers_502_and_keeps_the_row
- test_a_failed_reset_send_gives_the_address_its_code_budget_back
- test_answering_the_reset_code_sets_the_reset_cookie
- test_the_reset_cookie_is_locked_down
- test_answering_the_reset_code_retires_it_for_the_token
- test_a_correct_code_resets_the_password
- test_a_reset_to_a_password_the_policy_refuses_is_a_400
- test_a_reset_to_a_breached_password_is_a_400
- test_a_reset_survives_a_corpus_that_cannot_answer
- test_a_reset_revokes_every_live_session
- test_the_reset_row_is_consumed
- test_a_wrong_reset_code_is_refused_and_charged
- test_an_unknown_address_answers_exactly_like_a_wrong_reset_code
- test_an_expired_reset_code_is_refused
- test_an_answered_reset_code_cannot_be_answered_again
- test_the_attempt_budget_kills_the_reset
- test_a_guess_short_of_the_budget_does_not_cost_the_reset
- test_the_reset_flow_folds_the_address_case
- test_reset_guess_arrivals_beyond_the_address_burst_are_refused
- test_a_rejected_reset_verification_does_not_echo_the_code
- test_a_reset_without_the_cookie_is_refused
- test_a_reset_cookie_proving_one_address_cannot_reset_another
- test_an_unknown_address_answers_exactly_like_a_dead_reset_cookie
- test_an_expired_reset_proof_is_refused
- test_reset_arrivals_beyond_the_address_burst_are_refused
- test_a_rejected_reset_does_not_echo_the_new_password
- test_the_current_password_buys_a_new_one
- test_the_token_the_change_was_made_with_stops_working
- test_a_change_from_a_session_due_for_renewal_hands_back_the_new_token
- test_the_new_password_is_what_signs_in_afterwards
- test_a_wrong_current_password_changes_nothing

- test_a_change_to_a_password_the_policy_refuses_is_a_400
- test_a_change_to_a_breached_password_is_a_400
- test_the_policy_takes_the_coach_s_own_words_from_their_row
- test_a_wrong_current_password_answers_before_the_policy_does
- test_changing_a_password_needs_a_session
- test_the_change_ends_the_coach_s_other_sessions
- test_a_pending_reset_code_dies_with_a_deliberate_change
- test_wrong_guesses_beyond_the_coach_s_budget_are_refused
- test_a_correct_password_is_never_refused_by_the_coach_tier
- test_change_arrivals_beyond_the_address_burst_are_refused
- test_an_unauthenticated_flood_spends_none_of_the_change_budget
- test_change_password_has_a_total_bound_no_address_escapes
- test_a_rejected_change_echoes_neither_password
- test_a_change_verified_before_another_write_overwrites_neither_hash_nor_revocation
- test_a_change_locks_the_coach_row_before_writing_the_new_hash
- test_a_coach_beyond_the_per_coach_burst_is_refused
- test_one_coach_s_spending_does_not_charge_another
- test_a_dead_cookie_s_401_does_not_charge_the_coach
- test_a_drained_shared_budget_still_admits_a_revocation
- test_a_drained_shared_budget_still_admits_a_password_change
- test_the_recovery_budget_still_bounds_the_recovery_routes
- test_a_drained_recovery_budget_leaves_the_ordinary_routes_alone
- test_a_flood_of_unknown_tokens_is_refused
- test_a_drained_unknown_token_tier_still_admits_a_live_session
- test_a_request_carrying_no_cookie_is_never_charged

tests/routers/test_coaches.py — 99/99

- test_signing_up_returns_the_new_coach
- test_signing_up_never_echoes_the_password_or_its_hash
- test_signing_up_stores_a_hash_the_password_verifies_against
- test_two_coaches_sharing_a_password_get_different_hashes
- test_a_mixed_case_email_is_stored_and_read_back_lowercased

- test_signing_up_twice_with_one_email_is_a_conflict
- test_an_email_differing_only_in_case_is_a_conflict
- test_a_transferred_player_keeps_their_notes
- test_an_integrity_error_that_is_not_a_duplicate_email_is_not_a_conflict
- test_signing_up_without_a_password_is_rejected
- test_signing_up_with_too_short_a_password_is_rejected
- test_a_rejected_sign_up_does_not_echo_the_password
- test_signing_up_with_a_password_the_policy_refuses_is_a_400 [mostly one repeated character / consecutive characters / too common / contains name or email]
- test_a_refused_password_leaves_the_verification_to_try_again_with
- test_signing_up_with_a_breached_password_is_a_400
- test_a_breached_password_also_leaves_the_verification_standing
- test_signing_up_survives_a_corpus_that_cannot_answer
- test_a_password_refused_by_policy_is_not_echoed
- test_signing_up_without_a_verification_cookie_is_a_401
- test_signing_up_with_a_cookie_for_another_email_is_a_401
- test_signing_up_with_an_expired_verification_is_a_401
- test_signing_up_with_a_token_verify_otp_never_minted_is_a_401
- test_signing_up_consumes_the_verification
- test_a_spent_verification_cookie_cannot_sign_up_again
- test_a_conflict_does_not_burn_the_verification
- test_sign_up_arrivals_beyond_the_burst_are_refused
- test_sign_up_has_a_total_bound_no_address_escapes
- test_a_sign_up_refused_by_the_shared_tier_spends_no_per_address_token
- test_me_returns_the_signed_in_coach_and_nobody_else
- test_me_without_a_cookie_is_a_401
- test_me_with_a_token_no_login_minted_is_a_401
- test_me_with_an_expired_session_is_a_401_and_discards_it
- test_me_renews_a_session_past_its_half_life
- test_a_renewed_session_re_sends_the_cookie
- test_a_renewal_reaches_the_browser_even_on_an_error_response

- test_a_renewal_reaches_the_browser_even_when_the_route_crashes
- test_me_leaves_a_session_in_its_first_half_alone
- test_an_authenticated_request_may_sweep_all_expired_sessions
- test_a_missed_gc_roll_leaves_expired_sessions_alone
- test_me_never_returns_the_password_hash
- test_updating_the_display_name_stores_the_new_one
- test_updating_the_preferences_stores_them
- test_updating_a_preference_to_null_clears_it
- test_updating_to_a_preference_outside_its_enum_is_rejected [default_surface: sand / default_scoring_system: best_of_3]
- test_me_returns_null_preferences_until_they_are_set
- test_an_update_sending_no_fields_changes_nothing
- test_updating_the_display_name_to_null_is_rejected
- test_updating_to_a_display_name_outside_the_bounds_is_rejected [blank / too long]
- test_an_update_refuses_a_field_the_schema_leaves_out [email / password / password_hash]
- test_updating_without_a_cookie_is_a_401
- test_uploading_a_photo_answers_with_the_coach
- test_an_uploaded_photo_is_served_at_the_url_it_reports
- test_replacing_a_photo_removes_the_one_it_replaced
- test_clearing_a_photo_removes_the_file
- test_clearing_a_photo_that_is_not_there_is_a_no_op
- test_a_file_that_is_not_an_image_is_refused
- test_an_image_with_too_many_pixels_is_refused
- test_an_upload_with_no_file_part_is_a_422
- test_uploading_a_photo_needs_a_session
- test_clearing_a_photo_needs_a_session
- test_clearing_a_photo_answers_401_when_the_coach_is_already_gone
- test_uploads_are_rate_limited_per_coach
- test_closing_an_account_erases_the_coach
- test_closing_an_account_takes_every_photo_under_it
- test_closing_an_account_is_refused_while_a_player_is_active

- test_a_refusal_spends_none_of_the_wrong_guess_budget
- test_closing_an_account_may_take_the_roster_along
- test_closing_an_account_takes_a_match_being_captured_with_it
- test_closing_an_account_ends_every_session_and_clears_the_cookie
- test_a_closure_from_a_session_due_for_renewal_sends_no_renewed_cookie
- test_closing_an_account_deletes_a_live_reset_code
- test_closing_an_account_deletes_a_deleted_match_tombstone
- test_a_transfer_code_survives_the_delete_players_waiver
- test_the_email_can_sign_up_again_afterwards
- test_closing_an_account_leaves_every_other_coach_untouched
- test_a_wrong_password_erases_nothing
- test_a_closure_verified_before_a_password_write_erases_nothing
- test_closing_an_account_needs_a_session
- test_a_rejected_closure_echoes_no_password
- test_wrong_guesses_beyond_the_coach_s_budget_are_refused
- test_a_correct_password_is_never_refused_by_the_coach_tier
- test_closure_arrivals_beyond_the_address_burst_are_refused
- test_an_unauthenticated_flood_spends_none_of_the_closure_budget
- test_closing_an_account_has_a_total_bound_no_address_escapes
- test_the_right_password_verifies_and_leaves_the_account_alone
- test_a_wrong_password_is_refused
- test_verifying_needs_a_session
- test_a_rejected_verification_echoes_no_password
- test_wrong_guesses_here_spend_the_closure_s_budget
- test_a_correct_password_still_verifies_while_someone_guesses
- test_verifications_and_closures_share_one_arrival_budget
- test_an_unauthenticated_flood_spends_none_of_that_budget
- test_coaches_are_not_listable

tests/routers/test_matches.py — 150/150

- test_list_in_progress_matches_returns_the_claimed_match
- test_in_progress_does_not_care_which_device_holds_the_claim
- test_in_progress_puts_the_freshest_claim_first

- test_a_match_nobody_is_capturing_is_not_in_progress
- test_a_lapsed_claim_is_not_in_progress
- test_a_completed_match_is_not_in_progress
- test_in_progress_omits_another_coachs_match
- test_create_match_refuses_an_id_a_player_delete_freed
- test_update_a_deleted_match_is_410
- test_update_a_match_a_player_delete_took_is_410
- test_ingest_into_a_match_a_player_delete_took_is_410
- test_create_match_persists_and_returns_row
- test_create_match_with_only_the_required_fields
- test_create_match_rejects_unknown_player
- test_create_match_against_another_coachs_player_is_404
- test_create_match_for_a_transferred_player_is_404
- test_create_match_rejects_missing_player_id
- test_create_match_rejects_a_malformed_player_id
- test_create_match_rejects_missing_played_on
- test_create_match_rejects_a_malformed_played_on
- test_create_match_rejects_a_future_played_on
- test_create_match_accepts_a_played_on_of_today
- test_create_match_rejects_an_implausibly_old_played_on
- test_create_match_rejects_an_invalid_surface
- test_create_match_rejects_an_invalid_scoring_system
- test_create_match_rejects_an_invalid_result
- test_create_match_rejects_a_whitespace_only_opponent_name
- test_create_match_rejects_an_overlong_opponent_name
- test_create_match_accepts_free_text_at_the_length_limit
- test_create_match_strips_surrounding_whitespace
- test_create_match_stores_a_client_supplied_id
- test_create_match_mints_a_fresh_id_each_time_none_is_sent
- test_create_match_repeated_with_the_same_id_returns_the_stored_match
- test_create_match_repeated_under_another_player_returns_the_stored_match
- test_create_match_refuses_an_id_the_coach_deleted
- test_create_match_allows_an_id_whose_delete_has_aged_out

- test_create_match_with_another_coachs_match_id_is_404
- test_create_match_rejects_a_malformed_id
- test_create_match_rejects_an_id_the_client_could_not_have_minted [nil / v1 / v5]
- test_list_matches_newest_first
- test_list_matches_breaks_same_day_ties_by_creation_order
- test_list_matches_spans_every_player_of_the_caller
- test_list_matches_omits_another_coachs_matches
- test_list_matches_filtered_by_player
- test_list_matches_filtered_by_an_unknown_player_is_empty
- test_list_matches_filtered_by_another_coachs_player_is_empty
- test_list_matches_rejects_a_malformed_player_id
- test_list_matches_with_none_stored_is_an_empty_list
- test_get_match_returns_one
- test_get_match_of_another_coach_is_404
- test_get_match_unknown_id_is_404
- test_get_match_malformed_id_is_422
- test_update_match_changes_only_the_fields_sent
- test_update_match_clears_a_nullable_field_with_an_explicit_null
- test_update_match_records_a_result
- test_update_match_on_a_completed_match_is_409
- test_update_match_cannot_change_a_recorded_result
- test_update_match_resending_the_stored_values_is_a_no_op
- test_update_match_rejects_a_null_played_on
- test_update_match_with_an_empty_body_changes_nothing
- test_update_match_bumps_updated_at
- test_update_match_cannot_move_the_match_to_another_player
- test_update_match_rejects_a_future_played_on
- test_update_match_of_another_coach_is_404
- test_update_match_unknown_id_is_404
- test_delete_match_removes_it
- test_delete_match_takes_its_derived_sets_with_it

- test_delete_match_takes_its_points_with_it
- test_delete_match_leaves_another_matchs_points_alone
- test_delete_match_records_the_id_it_freed
- test_deleting_a_reused_id_starts_its_window_over
- test_delete_match_twice_is_404
- test_delete_match_of_another_coach_is_404
- test_delete_match_unknown_id_is_404
- test_delete_match_malformed_id_is_422
- test_delete_match_is_allowed_on_a_completed_match
- test_delete_match_under_another_devices_claim_is_refused [False / True]
- test_the_capturing_device_may_delete_its_own_match
- test_a_lapsed_claim_does_not_block_the_delete
- test_releasing_the_claim_clears_the_way_for_the_delete
- test_the_capture_refusal_still_stands_over_a_match_with_points
- test_claim_capture_takes_a_free_match
- test_claim_capture_renews_the_claim_of_the_device_holding_it
- test_claim_capture_refuses_a_second_device
- test_claim_capture_takes_over_a_lapsed_claim
- test_claim_capture_on_a_completed_match_is_409
- test_claim_capture_of_another_coachs_match_is_404
- test_claim_capture_unknown_id_is_404
- test_claim_capture_requires_a_device_id
- test_release_capture_frees_the_match_for_another_device
- test_release_capture_on_an_unclaimed_match_is_a_no_op
- test_release_capture_of_another_coachs_match_is_404
- test_recording_the_result_releases_the_capture_claim
- test_ingest_points_stores_the_batch_and_answers_with_the_derived_sets
- test_ingest_stores_every_field_the_client_stamped
- test_ingesting_the_same_batch_again_lands_on_the_same_answer
- test_a_repeated_point_keeps_the_stored_version
- test_an_overlapping_batch_converges

- test_ingest_recomputes_the_sets_a_late_batch_changes
- test_ingest_into_a_completed_match_is_refused
- test_a_sequence_already_held_by_a_different_point_is_refused
- test_a_point_already_stored_against_another_match_is_refused
- test_one_invalid_point_rejects_the_whole_batch
- test_an_empty_batch_is_refused
- test_a_flush_from_the_claim_holder_renews_the_claim
- test_a_flush_from_another_device_is_still_ingested
- test_a_flush_does_not_claim_a_match_nobody_holds
- test_a_flush_does_not_revive_the_lapsed_claim_it_once_held
- test_ingest_into_another_coachs_match_is_404
- test_ingest_into_an_unknown_match_is_404
- test_ingest_into_a_deleted_match_is_410
- test_ingest_into_a_match_whose_delete_has_aged_out_is_404
- test_list_points_orders_by_sequence_regardless_of_insertion_order
- test_list_points_of_a_match_with_none_recorded_is_an_empty_list
- test_list_points_of_another_coachs_match_is_404
- test_list_points_of_an_unknown_match_is_404
- test_list_points_round_trips_every_field_the_ingest_route_stored
- test_list_points_carries_no_ingestion_timestamps
- test_an_integrity_error_that_is_not_a_conflict_is_not_absorbed
- test_the_routes_that_read_the_claim_lock_the_match_row [capture / capture-release / patch / points / delete]
- test_match_routes_require_a_session [post /matches, get /matches, get /matches/{id}, patch /matches/{id}, delete /matches/{id}, post /matches/{id}/capture, post /matches/{id}/capture/release, post /matches/{id}/points, get /matches/{id}/points]
- test_list_matches_for_a_player_is_their_own
- test_list_matches_filtered_by_another_player_is_empty_for_a_player
- test_get_match_answers_a_player_about_their_own
- test_get_match_hides_a_stablemates_match_from_a_player
- test_match_routes_a_player_cannot_reach [post /matches, get /matches/in-progress, patch /matches/{id}, delete /matches/{id}, post /matches/{id}/capture, post /matches/{id}/capture/release, post /matches/{id}/points, get /matches/{id}/points, post /matches/{id}/score-checkpoints, get /matches/{id}/score-checkpoints, delete /matches/{id}/points/{id}, delete /matches/{id}/score-checkpoints/{id}]

- test_a_coach_cookie_decides_when_both_are_in_the_jar
- test_a_coach_cookie_that_authenticates_nothing_steps_aside
- test_a_coach_cookie_that_authenticates_nothing_is_no_session_of_its_own
- test_a_drained_coach_budget_refuses_rather_than_narrowing

tests/routers/test_players.py — 121/121

- test_delete_player_records_the_ids_of_the_matches_it_freed
- test_delete_player_with_no_matches_records_nothing
- test_a_note_survives_the_write_and_comes_back_on_the_next_read
- test_create_note_files_it_under_the_areas_sent
- test_create_note_defaults_to_no_focus_areas
- test_create_note_cannot_name_a_player_in_the_body
- test_create_note_against_another_coachs_player_is_404
- test_list_notes_is_newest_session_first
- test_list_notes_breaks_a_tie_on_when_the_note_was_written
- test_list_notes_carries_only_this_players_notes
- test_list_notes_with_none_written_is_an_empty_list
- test_list_notes_of_another_coachs_player_is_404
- test_update_note_changes_only_the_fields_sent
- test_update_note_clears_the_focus_areas_with_an_empty_list
- test_update_note_of_another_coachs_player_is_404
- test_a_note_reached_through_the_wrong_player_is_404
- test_delete_note_removes_it_and_leaves_the_rest
- test_delete_note_unknown_id_is_404
- test_delete_player_takes_their_notes_with_them
- test_create_player_persists_and_returns_row
- test_create_player_belongs_to_the_caller
- test_create_player_with_every_optional_field
- test_create_player_without_optional_fields
- test_create_player_cannot_set_coach_id
- test_create_player_mints_the_username_and_numbers_namesakes
- test_create_player_numbers_past_usernames_minted_before_this_request
- test_create_player_returns_a_one_time_password_and_stores_only_its_hash
- test_the_password_never_appears_on_a_read
- test_the_minted_password_never_meets_the_policy
- test_creates_beyond_the_minting_burst_are_refused
- test_create_player_retries_when_a_concurrent_create_takes_the_username
- test_create_player_gives_up_after_three_username_collisions
- test_regenerating_gives_a_backfilled_player_a_password_that_signs_them_in

- test_regenerating_stores_only_a_hash_and_replaces_the_old_password
- test_regenerating_ends_the_player_s_live_sessions_and_no_one_else_s
- test_regenerating_another_coach_s_player_is_the_404_an_unknown_id_gets
- test_a_404_from_the_password_route_spends_no_minting_token
- test_creation_and_regeneration_spend_the_same_minting_budget
- test_create_player_re_raises_an_integrity_error_that_is_not_the_username
- test_create_player_rejects_future_date_of_birth
- test_create_player_rejects_implausibly_old_date_of_birth
- test_create_player_accepts_a_plausibly_old_date_of_birth
- test_create_player_accepts_date_of_birth_of_today
- test_create_player_accepts_an_explicit_null_date_of_birth
- test_create_player_rejects_missing_first_name
- test_create_player_rejects_missing_last_name
- test_create_player_rejects_invalid_dominant_hand
- test_create_player_rejects_invalid_gender
- test_create_player_rejects_overlong_name
- test_create_player_accepts_name_at_the_length_limit
- test_create_player_rejects_overlong_notes
- test_create_player_accepts_notes_at_the_length_limit
- test_create_player_rejects_whitespace_only_name
- test_create_player_strips_surrounding_whitespace
- test_list_players_returns_all_sorted_by_last_then_first_name
- test_list_players_omits_another_coachs_players
- test_get_player_returns_one
- test_get_player_of_another_coach_is_404
- test_list_players_with_none_stored_is_an_empty_list
- test_get_player_unknown_id_is_404
- test_get_player_malformed_id_is_422
- test_update_player_changes_only_the_fields_sent
- test_update_player_clears_a_nullable_field_with_an_explicit_null
- test_update_player_rejects_a_null_name_part [first_name, last_name]

- test_update_player_with_an_empty_body_changes_nothing
- test_update_player_bumps_updated_at
- test_update_player_rejects_coach_id
- test_update_player_rejects_a_future_date_of_birth
- test_update_player_of_another_coach_is_404
- test_update_player_unknown_id_is_404
- test_delete_player_takes_their_matches_with_them
- test_delete_player_takes_their_matches_points_with_them
- test_delete_player_takes_a_match_being_captured_with_it
- test_delete_player_leaves_the_rest_of_the_roster_alone
- test_delete_player_is_refused_while_a_transfer_code_is_outstanding
- test_cancelling_the_code_clears_the_way_for_the_delete
- test_delete_player_leaves_another_coachs_roster_alone
- test_delete_player_unknown_id_is_404
- test_player_routes_require_a_session [post /players, get /players, get /players/{id}, patch /players/{id}, delete /players/{id}, post /players/{id}/transfer-code, delete /players/{id}/transfer-code, post /players/redeem-transfer-code]
- test_a_transfer_code_moves_the_player_and_their_matches
- test_a_pasted_code_may_carry_whitespace
- test_minting_again_replaces_the_previous_code
- test_a_redeemed_code_cannot_be_used_again
- test_redeeming_an_unknown_code_is_400
- test_redeeming_your_own_code_is_409_and_leaves_it_usable
- test_deleting_a_transfer_code_stops_it_working
- test_deleting_a_transfer_code_that_was_never_minted_is_204
- test_transfer_code_for_another_coachs_player_is_404 [post / delete]
- test_a_freshly_created_player_has_no_transfer_code
- test_the_roster_flags_only_the_players_holding_a_code
- test_the_roster_reports_every_flag_in_one_statement
- test_minting_a_code_shows_up_on_the_player
- test_cancelling_a_code_clears_the_flag
- test_redeeming_a_code_clears_the_flag_on_the_response

- test_the_flag_and_the_delete_refusal_agree
- test_uploading_a_photo_answers_with_the_player
- test_an_uploaded_photo_is_served_at_the_url_it_reports
- test_replacing_a_photo_removes_the_one_it_replaced
- test_clearing_a_photo_removes_the_file
- test_clearing_a_photo_that_is_not_there_is_a_no_op
- test_another_coachs_player_is_a_404_and_stores_nothing
- test_a_file_that_is_not_an_image_is_refused
- test_deleting_a_player_removes_their_photo
- test_a_transferred_player_keeps_their_photo
- test_a_player_session_cannot_read_its_own_player_row
- test_a_player_session_cannot_read_its_own_player_row [/notes]
- test_a_player_session_cannot_read_the_roster

tests/schemas/test_auth.py — 33/33

- test_a_valid_login_parses
- test_the_email_is_stripped_and_validated
- test_a_password_below_the_sign_up_floor_is_accepted
- test_a_password_over_the_ceiling_is_rejected
- test_the_password_is_not_stripped
- test_the_repr_never_contains_the_password
- test_a_valid_otp_request_parses
- test_the_otp_request_email_is_stripped_and_validated
- test_a_valid_verification_parses
- test_a_code_of_the_wrong_length_is_rejected [12345 / 1234567]
- test_a_non_digit_code_is_accepted_by_the_schema
- test_the_repr_never_contains_the_code
- test_a_valid_forgot_password_request_parses
- test_the_forgot_password_email_is_stripped_and_validated
- test_a_valid_reset_code_verification_parses
- test_a_reset_code_of_the_wrong_length_is_rejected [12345 / 1234567]
- test_the_reset_code_verification_repr_never_contains_the_code
- test_a_valid_reset_parses

- test_a_reset_carrying_a_code_is_rejected
- test_a_new_password_below_the_floor_is_rejected
- test_a_new_password_over_the_ceiling_is_rejected
- test_the_new_password_is_not_stripped
- test_the_repr_never_contains_the_new_password
- test_a_valid_change_parses
- test_only_the_new_password_takes_the_sign_up_floor
- test_a_valid_player_login_parses
- test_the_username_is_stripped_and_bounded
- test_the_player_password_has_no_floor_and_keeps_the_ceiling
- test_the_player_login_repr_never_contains_the_password
- test_a_valid_player_change_parses
- test_only_the_player_change_new_password_takes_the_sign_up_floor

tests/schemas/test_coach.py — 33/33

- test_a_valid_sign_up_parses
- test_the_email_is_stripped_and_its_domain_lowercased
- test_a_malformed_email_is_rejected
- test_an_email_over_the_length_bound_is_rejected
- test_the_bounds_reach_the_json_schema
- test_a_password_outside_the_bounds_is_rejected [7 / 129]
- test_a_password_keeps_its_whitespace
- test_a_blank_display_name_is_rejected [empty / whitespace]
- test_coach_create_has_no_password_hash_field
- test_the_password_is_not_in_the_repr
- test_an_update_omitting_the_display_name_dumps_nothing
- test_an_update_carrying_the_display_name_dumps_it_stripped
- test_an_invalid_display_name_is_rejected_on_an_update [empty / whitespace / too long / None]
- test_coach_update_carries_the_name_and_the_preferences_alone
- test_an_update_carrying_a_preference_dumps_it
- test_an_update_may_clear_a_preference_with_null [default_surface, default_scoring_system]

- test_a_preference_outside_its_enum_is_rejected [default_surface: sand / default_scoring_system: best_of_3]
- test_a_closure_carries_the_password_and_the_roster_waiver
- test_a_closure_leaves_the_roster_alone_unless_asked
- test_a_closure_takes_the_ceiling_without_the_floor
- test_verifying_a_password_carries_nothing_else
- test_a_closure_password_is_not_in_the_repr
- test_coach_read_has_no_password_hash_field
- test_reading_a_coach_row_drops_the_password_hash
- test_a_read_carries_the_photo_url_derived_from_the_key
- test_an_update_refuses_a_photo_url

tests/test_auth.py — 7/7

- test_each_token_is_fresh
- test_a_token_is_long_enough_to_be_unguessable
- test_hashing_a_token_is_deterministic
- test_the_hash_is_not_the_token_and_fits_the_column
- test_each_player_password_is_fresh
- test_a_player_password_reads_aloud_without_ambiguity
- test_the_lifetime_is_long_lived

tests/test_breach.py — 19/19

- test_the_vector_is_the_hash_the_api_would_be_asked_for
- test_the_default_checker_is_the_real_one
- test_the_suite_runs_with_the_corpus_switched_off
- test_the_setting_selects_the_real_checker
- test_the_offline_checker_refuses_nothing
- test_only_the_hash_prefix_is_sent
- test_the_answer_is_padded
- test_a_listed_password_is_breached
- test_a_password_the_bucket_does_not_hold_is_not_breached
- test_a_padding_row_is_not_a_sighting
- test_an_answer_of_nothing_but_padding_is_not_a_sighting

- test_the_suffix_comparison_ignores_case
- test_junk_lines_beside_real_rows_are_skipped
- test_a_200_that_is_not_corpus_rows_says_the_corpus_went_unasked
- test_a_refused_lookup_says_the_corpus_went_unasked [429 / 500 / 503]
- test_a_lookup_that_never_answers_says_the_corpus_went_unasked
- test_a_failed_lookup_logs_the_status_and_nothing_of_the_password

tests/test_db.py — 3/3

- test_the_pool_validates_a_connection_before_handing_it_over
- test_get_db_yields_a_usable_session_and_releases_it
- test_the_engine_runs_at_the_isolation_the_otp_attempt_cap_assumes

tests/test_email_templates.py — 25/25

- test_both_parts_carry_the_code [signup_code, password_reset_code]
- test_the_code_is_six_unbroken_digits [signup_code, password_reset_code]
- test_both_parts_say_when_the_code_dies [signup_code, password_reset_code]
- test_the_subject_names_the_app_and_the_code [signup_code, password_reset_code]
- test_the_two_emails_are_told_apart_by_their_subjects
- test_an_unasked_for_code_can_be_ignored [signup_code, password_reset_code]
- test_the_html_is_a_document_a_client_can_read [signup_code, password_reset_code]
- test_every_table_is_marked_presentational [signup_code, password_reset_code]
- test_the_message_has_exactly_one_heading [signup_code, password_reset_code]
- test_the_html_fetches_nothing [signup_code, password_reset_code]
- test_the_brand_blue_dresses_the_message_and_the_data_colors_stay_out [signup_code, password_reset_code]
- test_the_html_answers_a_dark_mode_client [signup_code, password_reset_code]
- test_an_interpolated_value_cannot_carry_markup_into_the_html [signup_code, password_reset_code]

tests/test_errors.py — 11/11

- test_a_secret_field_is_the_only_thing_redacted
- test_a_password_alongside_another_fields_error_is_redacted
- test_a_password_nested_below_the_reported_field_is_redacted
- test_an_unknown_keys_value_is_redacted
- test_an_error_carrying_no_input_is_passed_through

- test_a_field_that_is_not_secret_keeps_its_input
- test_every_secret_field_in_the_schemas_is_named_in_secret_fields
- test_an_unhandled_exception_answers_a_readable_json_body
- test_an_unhandled_exception_tells_the_client_nothing_about_itself
- test_the_500_carries_the_request_id_that_logged_the_traceback
- test_a_failure_above_the_logging_middleware_is_answered_too

tests/test_mail.py — 18/18

- test_the_default_sender_cannot_send
- test_the_selected_sender_is_the_logging_one
- test_the_logging_sender_writes_the_message_to_the_log
- test_the_logging_sender_leaves_the_markup_out
- test_the_resend_sender_posts_the_message_to_the_api
- test_the_resend_sender_logs_the_provider_id
- test_a_refused_send_raises_email_send_error
- test_a_stalled_send_raises_email_send_error
- test_a_rate_limited_send_is_retried_and_succeeds
- test_a_send_rate_limited_twice_still_fails
- test_an_unreadable_success_body_is_still_a_send
- test_a_refused_send_logs_the_provider_s_reason
- test_a_failed_send_keeps_the_code_out_of_the_log
- test_the_file_sender_appends_every_message_as_a_line_of_json
- test_the_setting_selects_the_file_sender
- test_selecting_the_file_sender_without_a_path_fails_loudly
- test_the_setting_selects_the_resend_sender
- test_selecting_resend_without_a_key_fails_loudly

tests/test_migrations.py — 5/5

- test_migrations_upgrade_downgrade_and_upgrade_again
- test_requiring_a_player_owner_refuses_orphans_rather_than_deleting_them
- test_the_username_backfill_numbers_existing_players_deterministically
- test_the_schema_at_head_matches_the_models
- test_dropping_mid_clears_the_placement_rather_than_the_point

tests/test_password_policy.py — 37/37

- test_a_password_that_is_mostly_one_character_is_refused [aaaaaaaa / aaaaa123 / AAAAaaaa / a1a2a3a4a]
- test_a_password_running_through_consecutive_characters_is_refused [12345678 / abcdefgh / 87654321 / hgfedcba / wimbledon12345 / fullwidth digits]
- test_a_common_password_is_refused [password / PASSWORD / matchpoint / password! / Password123! / monkey123 / tennis12345]
- test_a_password_carrying_the_coach_s_own_words_is_refused [the-mike-special / coach-of-the-year / smith-and-wesson / MikeSmith-forever]
- test_an_ordinary_password_is_accepted [a-good-password / correct-horse-battery-st4ple! / wimbledon-in-the-rain / secret-garden-path]
- test_a_player_password_carrying_their_own_words_is_refused
- test_no_email_contributes_no_words
- test_a_short_name_is_not_matched_as_a_substring
- test_the_first_matching_rule_is_the_one_reported
- test_the_refusal_is_a_400_carrying_the_reason
- test_an_accepted_password_raises_nothing
- test_a_password_in_the_corpus_is_refused
- test_a_password_the_corpus_does_not_hold_is_accepted
- test_a_corpus_that_cannot_answer_lets_the_password_through
- test_a_checker_that_raises_anything_else_still_lets_the_password_through
- test_a_locally_refused_password_never_reaches_the_corpus
- test_the_corpus_is_asked_about_the_password_that_gets_hashed

tests/test_rate_limit.py — 17/17

- test_a_key_is_admitted_up_to_the_burst_and_refused_beyond_it
- test_the_scale_setting_widens_a_bucket
- test_the_refusal_says_when_to_retry
- test_tokens_refill_at_the_sustained_rate
- test_an_idle_key_accrues_no_more_than_the_burst
- test_keys_do_not_share_a_bucket
- test_a_refused_request_spends_nothing
- test_a_refunded_request_leaves_the_bucket_as_it_found_it
- test_a_refund_cannot_bank_more_than_the_burst

- test_refunding_a_key_that_never_spent_is_harmless
- test_taking_from_several_buckets_admits_only_when_all_of_them_can_pay
- test_a_bucket_refused_by_another_is_not_charged_for_it
- test_a_slower_per_key_tier_leaves_the_shared_one_to_other_keys
- test_the_refusal_quotes_the_bucket_furthest_from_a_token
- test_the_furthest_bucket_is_quoted_even_when_another_is_refused_first
- test_a_new_key_evicts_buckets_that_have_refilled_to_full
- test_a_new_key_keeps_buckets_still_refilling

tests/test_security.py — 13/13

- test_hashing_the_same_password_twice_gives_different_hashes
- test_the_hash_names_argon2id
- test_the_hash_records_the_chosen_parameters
- test_the_column_storing_the_hash_cannot_truncate_it
- test_verifying_the_right_password_succeeds
- test_verifying_a_wrong_password_fails
- test_verifying_is_case_sensitive
- test_the_same_password_verifies_across_two_encodings_of_it
- test_the_dummy_hash_costs_what_a_real_one_costs
- test_nothing_verifies_against_the_dummy_hash
- test_a_current_hash_does_not_need_a_rehash
- test_a_hash_with_stale_parameters_needs_a_rehash
- test_verifying_against_a_stored_value_that_is_not_a_hash_raises

tests/schemas/test_player.py — 3/3

- test_neither_create_nor_update_accepts_a_username [PlayerCreate]
- test_neither_create_nor_update_accepts_a_username [PlayerUpdate]
- test_a_created_player_carries_the_password_in_the_clear

tests/test_usernames.py — 13/13

- test_the_surname_prefers_consonants_and_the_first_name_reads_plainly
- test_skipped_vowels_fill_in_when_the_consonants_run_short
- test_accents_fold_to_their_bare_letters
- test_punctuation_and_spaces_drop_out_before_folding
- test_a_short_name_is_padded_to_keep_the_prefix_fixed_width
- test_a_name_with_no_latin_letters_folds_to_the_placeholder
- test_casefold_handles_one_to_many_letters

- test_the_prefix_is_always_six_characters
- test_the_first_holder_of_a_name_is_numbered_001
- test_a_namesake_takes_the_next_number
- test_numbering_resumes_past_the_highest_ever_minted_not_the_count
- test_another_name_s_numbers_do_not_interfere
- test_past_999_the_number_widens_rather_than_wrapping

tests/models/test_player_session.py — 8/8

- test_player_session_round_trips
- test_database_requires_every_column [player_id]
- test_database_requires_every_column [token_hash]
- test_database_requires_every_column [expires_at]
- test_database_rejects_a_duplicate_token_hash
- test_database_rejects_a_token_hash_over_the_column_width
- test_database_requires_the_player_to_exist
- test_a_player_with_sessions_cannot_be_hard_deleted

tests/routers/test_player_auth.py — 50/50

- test_a_player_login_returns_their_own_profile_and_sets_the_player_cookie
- test_the_login_response_is_the_trimmed_self_view
- test_the_player_cookie_is_locked_down
- test_a_player_login_stores_the_hash_of_the_token_and_never_the_token
- test_a_wrong_password_is_rejected_without_a_session
- test_an_unknown_username_answers_exactly_like_a_wrong_password
- test_a_player_with_no_password_yet_answers_exactly_like_a_wrong_password
- test_a_miss_still_burns_a_verification [ghost01]
- test_a_miss_still_burns_a_verification [ndlraf001]
- test_a_mixed_case_username_finds_the_account
- test_a_stale_hash_is_upgraded_on_player_login
- test_logging_in_clears_only_the_player_s_own_expired_sessions
- test_wrong_guesses_beyond_the_account_burst_are_refused
- test_the_account_bucket_folds_case
- test_a_correct_password_is_never_refused_by_the_wrong_guess_budget
- test_cycling_usernames_does_not_escape_the_address_burst
- test_player_login_has_a_total_bound_no_address_escapes
- test_a_player_login_flood_spends_none_of_the_coach_login_budget
- test_me_returns_the_trimmed_profile
- test_me_needs_a_player_session
- test_a_coach_session_does_not_authenticate_a_player_route
- test_a_forged_player_cookie_is_refused
- test_an_expired_player_session_is_refused_and_deleted
- test_a_session_past_halfway_is_renewed_with_a_fresh_cookie
- test_a_fresh_session_is_not_renewed
- test_a_gc_roll_hit_sweeps_expired_player_sessions
- test_a_missed_gc_roll_leaves_expired_player_sessions_alone
- test_requests_beyond_the_per_player_budget_are_refused
- test_logging_out_revokes_the_session_and_clears_the_player_cookie
- test_logging_out_only_ends_the_presented_session

- test_logging_out_signed_out_is_still_a_204
- test_logging_out_with_a_forged_cookie_revokes_nothing
- test_a_player_logout_leaves_a_coach_session_on_the_same_browser_alone
- test_the_current_password_buys_a_new_one
- test_the_change_ends_every_other_session_and_keeps_the_caller_signed_in
- test_a_change_from_a_session_due_for_renewal_hands_back_the_new_token
- test_a_wrong_current_password_is_a_403_that_changes_nothing
- test_a_session_whose_player_has_no_hash_hears_the_same_403
- test_wrong_guesses_beyond_the_player_s_budget_are_refused
- test_a_correct_password_is_never_refused_by_the_wrong_guess_budget_here_either
- test_a_change_to_a_breached_password_is_a_400
- test_the_policy_takes_the_player_s_own_words
- test_a_wrong_current_password_answers_before_the_policy_does
- test_an_unauthenticated_flood_spends_none_of_the_change_budget
- test_a_stolen_session_flood_cannot_429_the_change
- test_a_rejected_change_echoes_neither_password
- test_a_login_that_verified_a_regenerated_password_mints_no_session
- test_a_login_locks_the_player_row_before_minting_a_session
- test_a_change_verified_before_a_regenerate_overwrites_neither_hash_nor_revocation
- test_a_change_locks_the_player_row_before_writing_the_new_hash

tests/test_logs.py — 1/1

- test_an_authenticated_player_request_logs_the_player_id

tests/routers/test_analytics.py — 14/14

- test_routes_answer_a_player_about_itself [overview]
- test_routes_answer_a_player_about_itself [serve]
- test_routes_answer_a_player_about_itself [rally]
- test_routes_answer_a_player_about_itself [opponents]
- test_routes_hide_a_stablemate_from_a_player [overview]
- test_routes_hide_a_stablemate_from_a_player [serve]
- test_routes_hide_a_stablemate_from_a_player [rally]
- test_routes_hide_a_stablemate_from_a_player [opponents]
- test_match_routes_answer_a_player_about_their_own_match [serve]
- test_match_routes_answer_a_player_about_their_own_match [rally]
- test_match_routes_hide_a_stablemates_match_from_a_player [serve]
- test_match_routes_hide_a_stablemates_match_from_a_player [rally]
- test_squad_routes_refuse_a_player [/squad/analytics/members]
- test_squad_routes_refuse_a_player [/squad/analytics/trend]

Web — vitest (268/268)

src/components/password-input.test.tsx — 3/3

- starts masked
- toggling reveals the password, and toggling again re-masks it

- forwards autoComplete and name so password managers can recognize the field

src/components/password-strength-meter.test.tsx — 4/4

- labels a score of 0 as very weak
- labels a score of 4 as strong
- fills more of the bar for a higher score
- fill color progresses from vermillion to green as score increases

src/features/auth/auth-shell.test.tsx — 1/1

- renders the MatchPoint mark and its children

src/features/auth/forgot-password.test.tsx — 35/35

- renders the email field
- updates the field as the user types
- blocks submit until the email is filled
- links back to sign in
- sends the email to /api/auth/forgot-password
- moves to the OTP step on a 202 response
- shows an inline error, the retry wait, and stays on the email step on a 429 response
- shows an inline error and stays on the email step on a 502 response
- shows a generic error when the request fails outright
- sends one forgot-password request however impatiently Continue is clicked
- renders the OTP field once the OTP step is reached
- updates the OTP field as the user types
- blocks the OTP submit until a code is entered
- blocks submit while the code is the wrong length, with a visible hint
- sends the email and code to /api/auth/verify-reset-code
- shows the server's error and stays on the OTP step on a 400 response
- shows the retry wait and disables submit on a verify-reset-code 429 response
- shows a generic error when the verify-reset-code request fails outright
- sends one verify-reset-code however impatiently Verify is clicked
- requesting a new code from the OTP step returns to the email step with the email preserved
- renders the new password and confirm fields once the password step is reached

- blocks the password submit until matching passwords are entered
- blocks submit while the new password is too short, with a visible hint
- blocks submit while the passwords don't match, with a visible hint
- sends the email and new password to /api/auth/reset-password, no code field
- shows the server's error and stays on the password step on a 400 response
- shows the server's error and stays on the password step on a 401 response
- shows the server's error, the retry wait, and stays on the password step on a 429 response
- retries the password submit alone on failure, without re-verifying the code
- shows a generic error when the reset-password request fails outright
- sends one reset-password request however impatiently Reset Password is clicked
- reaches the confirmation view once both steps succeed, and its link signs back in
- starting over from the password step returns to the email step with the email preserved
- starting over from the OTP step returns to the email step with the email preserved
- links back to sign in from the OTP step

src/features/auth/sign-in.test.tsx — 28/28

- renders the email and password fields
- updates the fields as the user types
- blocks submit until both fields are filled
- navigates to /home on a successful sign-in
- sends the credentials to /api/auth/login
- shows the server's error message on a failed sign-in
- falls back to an explicit message when the response has no detail to show
- shows the retry wait and disables submit on a 429 response
- shows a generic error when the request fails outright
- links to the sign-up page
- links to the forgot-password page
- wipes what the previous coach left when a different one signs in
- keeps the queue when the same coach signs in again
- refuses a 200 that names no coach rather than signing in as nobody
- keeps the coach on the form when this device cannot be cleared
- renders the identifier and password fields

- navigates to /home on a successful coach sign-in
- sends the credentials to /api/auth/login when the identifier has an @
- shows the server's error message on a failed coach sign-in
- falls back to an explicit message when the coach response has no detail to show
- wipes what the previous owner left when a different coach signs in
- refuses a coach 200 that names no coach rather than signing in as nobody
- navigates to /me on a successful player sign-in
- sends the credentials to /api/auth/player/login when the identifier has no @
- shows the server's error message on a failed player sign-in
- falls back to a username-specific message when the player response has no detail
- refuses a player 200 that names no player rather than signing in as nobody
- wipes what a previous coach left when a player signs in on the same device

src/features/auth/sign-up.test.tsx — 39/39

- renders the email field
- updates the field as the user types
- blocks submit until the email is filled
- links back to sign in
- sends the email to /api/auth/request-otp
- moves to the OTP step on a 202 response
- shows an inline error and stays on the email step on a 409 response
- falls back to an explicit message when the response has no detail to show
- shows the retry wait and disables submit on a 429 response
- shows a generic error when the request fails outright
- sends one request-otp however impatiently Continue is clicked
- renders the OTP field once the OTP step is reached
- updates the OTP field as the user types
- blocks the OTP submit until a code is entered
- blocks submit while the code is the wrong length, with a visible hint
- sends the email and code to /api/auth/verify-otp
- shows the server's error and stays on the OTP step on a 400 response
- shows the retry wait and disables submit on a verify-otp 429 response
- shows a generic error when the verify-otp request fails outright
- sends one verify-otp however impatiently Verify is clicked
- requesting a new code from the OTP step returns to the email step with the email preserved

- renders the display name, password and confirm fields once the password step is reached
- blocks submit until a display name is entered
- blocks submit while the password is too short, with a visible error
- blocks submit while the password is too long, with a visible error
- blocks submit while the passwords differ, with a visible error
- allows submit once the display name is set and the passwords match
- sends the email, password and display name to POST /coaches, no token field
- shows a usable message instead of crashing on a real FastAPI 422 body
- navigates to / on a successful sign-up
- shows the server's error and stays on the password step when verification is missing
- shows the retry wait and disables submit on a sign-up 429 response
- sends one POST /coaches however impatiently Sign Up is clicked
- shows a generic error when the complete-sign-up request fails outright
- starting over from the password step returns to the email step with the email preserved
- links to the Terms of Service
- links to the Privacy Policy
- starting over from the OTP step returns to the email step with the email preserved
- links back to sign in from the OTP step

src/features/players/player-list.test.tsx — 9/9

- renders the players the API returns
- flags a player whose transfer code is still out
- the pending-transfer flag is part of the row, not a control of its own
- requests the roster from the API's own origin
- groups players under their surname initial
- filters the roster by the search term
- reports a failed load instead of rendering an empty roster
- prompts to sign in again on a 401, instead of the generic error
- tapping a player hands it to the caller instead of navigating

src/features/settings/delete-account-sheet.test.tsx — 12/12

- names an outstanding transfer code on the first warning
- says nothing about transfer codes when the roster has none outstanding

- Cancel on the first step closes the sheet
- Back on the password step returns to the roster warning
- the full happy path deletes with delete_players true, purges local data, and navigates home
- a wrong password returns to the password step with the error shown there
- an outstanding transfer code shows its own message on the final step
- a 429 shows a retry countdown and disables the delete button
- says so when the device could not be cleared, and Retry clears the message once storage recovers
- a wrong password on the password step shows the error there and never advances
- a 429 on the password step shows a retry countdown and disables Continue
- a correct password advances straight to the final warning

src/features/settings/settings-account.test.tsx — 12/12

- links back to settings
- change password asks for confirmation before leaving Settings
- confirming the change-password dialog navigates to the forgot-password flow
- cancelling the change-password dialog makes no navigation happen
- offers Sign Out, grouped with the other session actions
- signing out of all other sessions shows a confirm then a success dialog
- cancelling the sign-out-all-sessions confirm makes no request
- opens the delete-account flow
- dismissing the success dialog leaves the settings page as it was
- a failed revoke closes the dialog and says why, in the server's own words
- a failure with nothing to quote still explains itself
- groups every action under a named section

src/features/settings/settings-app.test.tsx — 5/5

- links back to settings
- Install MatchPoint links to the real install guide
- Notifications is still a stub, not interactive
- Appearance toggles between light and dark, and persists the choice
- has no Appearance row

src/features/settings/settings-match-defaults.test.tsx — 7/7

- links back to settings

- prefills 'No preference' when the coach has no defaults set
- prefills from the coach's current defaults
- choosing a surface PATCHes it straight away
- choosing a scoring system PATCHes it straight away
- choosing 'No preference' PATCHes null
- shows a short rule explanation for the selected scoring system

src/features/settings/settings-support.test.tsx — 7/7

- links back to settings
- all three rows are stubs, not interactive
- Help & feedback links to the FAQ page
- Privacy policy links to its own page
- Terms of Service links to its own page
- no stub rows remain on the Support page
- shows the real app version, not a stub

src/features/settings/settings.test.tsx — 6/6

- links back to home
- the Match Defaults row links to its category page
- the Account & Security row links to its category page
- the App row links to its category page
- the Support row links to its category page
- offers Sign Out at the root, for fast single-tap access

src/features/settings/sign-out-action.test.tsx — 9/9

- opens a confirmation dialog when clicked
- Cancel closes the dialog without signing out
- confirming signs out, clears the query cache, and navigates to sign-in directly
- confirming also deletes the points that never synced
- says so with a toast when the device could not be cleared, but still signs out
- warns with a count before confirming when a point hasn't synced
- pluralizes the warning for more than one unsynced row
- counts a parked row toward the warning too
- shows no warning when nothing is queued

src/lib/parse-error.test.ts — 10/10

- returns a string detail as-is

- joins a 422 validation array's messages
- falls back when detail is missing
- falls back when the validation array is empty or unrecognisable
- falls back when there is no error at all
- reports a network failure distinctly from an API error
- reads Retry-After off a 429 response
- returns null when there is no Retry-After header
- returns null when the response itself is undefined
- returns null for a non-numeric or non-positive Retry-After

src/lib/password-strength.test.ts — 4/4

- scores an empty password as 0
- scores an obviously weak password below the submit-gating threshold
- scores a long, random passphrase above the submit-gating threshold
- penalizes a password for containing a supplied user input

src/lib/use-countdown.test.ts — 5/5

- returns null when given null
- returns the initial value immediately
- ticks down by one every second
- stops at zero instead of going negative
- restarts from a new value when given one

src/components/root-error-banner.test.tsx — 1/1

- shows a reload banner instead of a blank screen when a route throws

src/features/home/home.test.tsx — 3/3

- ranks best and worst form, and shows nothing at all for a legitimate empty needs-attention state
- shows a dismissible card when a player is flagged
- dismissing the attention card hides it

src/features/players/players-page.test.tsx — 1/1

- the labelled row opens the redeem-transfer-code sheet

src/lib/attention-flag-dismissed.test.ts — 5/5

- a flag is not dismissed until dismissAttentionFlag is called
- dismissing a flag marks that exact player/tier pair as dismissed
- a different tier for the same player is not considered dismissed

- the same tier for a different player is not considered dismissed
- only the most recently dismissed pair is remembered

src/features/capture/note-button.test.tsx — 1/1

- an errored session never auto-adds what it heard, since the sheet already holds it

src/features/capture/score-sheet.test.tsx — 1/1

- a fractional tiebreak score is refused the same way

src/features/matches/match-list.test.tsx — 1/1

- the edit icon is still offered once the match has a result

src/features/matches/matches-page.test.tsx — 2/2

- the edit icon is still offered once the match has a result
- a completed match edits its metadata, with the scoring system locked

src/features/sync/drain.test.ts — 3/3

- holds the outcome back when the match's points are refused
- sends the outcome after the match's points, never before them
- holds the outcome back when a point batch was only transiently refused

src/lib/theme.test.ts — 6/6

- dark sets the class and both meta tags the browser paints from
- light puts all three back
- survives the tags being absent
- carries the choice through to the meta tags, not just the class
- applies the stored theme with the tags already in place
- resolves light even with a stale stored 'dark' value

src/route-tree.test.tsx — 14/14

- / redirects straight to /home with a live session, rather than showing the form
- / shows the sign-in form with no session
- / goes to /home for a recorded owner with no connection, not the form
- / shows the form, not /home, with no connection and no owner recorded
- / shows the form on a confirmed 401, even with an owner recorded from before
- /me renders the player's own view
- redirects a player session to sign-in on a confirmed 401
- / redirects straight to /home with a live coach session, rather than showing the form
- / redirects straight to /me with a live player session, rather than showing the form
- / shows the sign-in form with no session at all
- / shows the form on a confirmed 401 from both checks, even with an owner recorded from before
- / goes to /home for a recorded coach with no connection, not the form
- / goes to /me for a recorded player with no connection, not the form
- / shows the form, not the app, with no connection and no owner recorded

src/components/bottom-nav.test.tsx — 2/2

- renders all four coach tabs by default
- renders the player's three tabs instead, when given PLAYER_TABS

src/features/matches/player-match-list.test.tsx — 3/3

- renders the player's own matches, with no Edit or Delete controls
- shows an empty state with no matches yet
- reports a failed load instead of rendering an empty list

src/features/players/player-details-card.test.tsx — 1/1

- shows a player's own details with no Edit button and no Notes row

src/features/players/player-self-view.test.tsx — 4/4

- renders the player's own name and avatar
- shows the record and details cards, with no Edit button
- has no Start match button, no Notes button, and no coach-only affordance
- opens on the Serve tab, and switches to Rally

src/features/players/player-settings.test.tsx — 7/7

- renders the change-password fields and the sign-out action
- blocks submit until the new password and its confirmation match
- shows a mismatch error and blocks submit when the confirmation differs
- changes the password and shows a success message
- shows the server's message when the current password is wrong
- shows the server's message when the new password fails its policy
- signs out through playerLogout, then lands on sign-in

src/lib/local-data.test.ts — 16/16

- no owner kind is recorded until one signs in or a guard confirms one
- records the owner's kind alongside its id on sign-in
- records the owner's kind on a route guard's first confirmed session, same as the id
- never overwrites a recorded kind any more than it overwrites the id
- purging forgets the owner's kind along with everything else
- clearing cached reads empties both copies of the cache
- clearing cached reads leaves queued points alone
- purging deletes the queued points as well as the cache
- leaves the outbox usable once it has been purged
- purges when the coach signing in is not the one the device's data belongs to
- keeps the queue when the same coach signs in again
- a cleared read cache still knows whose device this is
- purging forgets the player the device was last showing
- purges a device that cannot say whose data it holds
- a confirmed session claims a device nothing else has claimed
- a confirmed session never re-claims a device that has an owner

src/features/outbox/outbox.test.ts — 1/1

- agrees with watchCounts' snapshot, without staying subscribed

Backlog Item 4

Test Results Summary

Web (vitest): 402/402 passed across 8 files

Web — vitest (402/402)

src/features/capture/point.test.ts — 105/105

• Point.create structural validation > accepts a minimal valid point

• Point.create structural validation > rejects non-object input

• Point.create structural validation > rejects null input

• Point.create structural validation > rejects an empty id

• Point.create structural validation > rejects an empty matchId

• Point.create structural validation > rejects a non-finite timestamp

• Point.create structural validation > rejects a sequence of -1

• Point.create structural validation > rejects a sequence of 1.5

• Point.create structural validation > rejects an unknown server

• Point.create structural validation > courtSide > accepts DEUCE

• Point.create structural validation > courtSide > accepts AD

• Point.create structural validation > courtSide > rejects an unknown courtSide

• Point.create structural validation > courtSide > rejects a null courtSide — unlike deuceOrAd, every point has one

• Point.create structural validation > serves > rejects a serves array of length 3

• Point.create structural validation > serves > rejects a serves array of length 4

• Point.create structural validation > serves > rejects a serve with the wrong number for its position

• Point.create structural validation > serves > rejects an unknown serve outcome

• Point.create structural validation > serves > rejects a non-null location on an OPPONENT serve

• Point.create structural validation > serves > accepts a null location on an OPPONENT serve

• Point.create structural validation > serves > rejects a location on a TRACKED serve that went in — a landed serve has none

• Point.create structural validation > serves > accepts a null location on a TRACKED serve that went in

• Point.create structural validation > serves > accepts a point with no serve at all — the coach missed it

• Point.create structural validation > serves > accepts a lone faulted serve — the second one went unwatched

• Point.create structural validation > serves > rejects a second attempt that doesn't follow a fault

• Point.create structural validation > serves > rejects a non-FailureMode location on a TRACKED serve that faulted

• Point.create structural validation > serves > rejects a location on an ace — a landed serve carries none

• Point.create structural validation > serves > an ace and its endReason require each other

• Point.create structural validation > serves > rejects a double fault whose endReason isn't DOUBLE_FAULT

• Point.create structural validation > serves > accepts a double fault with endReason DOUBLE_FAULT

• Point.create structural validation > serves > rejects a second serve attempt when the first serve didn't fault

• Point.create structural validation > serves > rejects endReason DOUBLE_FAULT when the serves weren't both faulted

• Point.create structural validation > returnOutcome > rejects a non-null returnOutcome when server is TRACKED

• Point.create structural validation > returnOutcome > accepts a null returnOutcome when server is OPPONENT

• Point.create structural validation > returnOutcome > rejects an unknown returnOutcome

• Point.create structural validation > returnOutcome > accepts a NET returnOutcome — the point ends on the return, no contact fields

• Point.create structural validation > returnOutcome > accepts a LONG returnOutcome — the point ends on the return, no contact fields

• Point.create structural validation > returnOutcome > accepts a WIDE returnOutcome — the point ends on the return, no contact fields

• Point.create structural validation > rejects a non-boolean won

• Point.create structural validation > rejects an unknown endReason

• Point.create structural validation > unusualReason / unusualNote > rejects a non-null unusualReason when endReason isn't UNUSUAL

• Point.create structural validation > unusualReason / unusualNote > rejects a non-null unusualNote when endReason isn't UNUSUAL

• Point.create structural validation > unusualReason / unusualNote > accepts an UNUSUAL point with no reason or note — the escape hatch stays optional

• Point.create structural validation > unusualReason / unusualNote > accepts an UNUSUAL point with a reason and note

• Point.create structural validation > unusualReason / unusualNote > rejects an unknown unusualReason

• Point.create structural validation > note > defaults to null

• Point.create structural validation > note > accepts a coach note on an ordinary (non-UNUSUAL) point

• Point.create structural validation > note > rejects an empty string — null is how 'no note' is spelled

• Point.create structural validation > note > truncates a note past what the API accepts rather than losing the point

• Point.create structural validation > wing / shotType > accepts both set together

• Point.create structural validation > wing / shotType > rejects wing set without the other

• Point.create structural validation > wing / shotType > rejects shotType set without the other

• Point.create structural validation > wing / shotType > rejects a non-null wing/shotType on a ACE point

• Point.create structural validation > wing / shotType > rejects a non-null wing/shotType on a DOUBLE_FAULT point

• Point.create structural validation > wing / shotType > rejects a non-null wing/shotType on a UNUSUAL point

• Point.create structural validation > wing / shotType > rejects a non-null wing/shotType on a return-error point

• Point.create structural validation > ballIn / ballLocation > accepts a CourtZone location when ballIn is true

• Point.create structural validation > ballIn / ballLocation > accepts a FailureMode location when ballIn is false

• Point.create structural validation > ballIn / ballLocation > rejects a FailureMode location when ballIn is true

• Point.create structural validation > ballIn / ballLocation > rejects a CourtZone location when ballIn is false

• Point.create structural validation > ballIn / ballLocation > accepts a null ballIn on a FORCED/MISTAKE point — the coach missed where it landed

• Point.create structural validation > ballIn / ballLocation > rejects a ballLocation with no ballIn beside it

• Point.create structural validation > ballIn / ballLocation > rejects a non-null ballIn/ballLocation on a ACE point

• Point.create structural validation > ballIn / ballLocation > rejects a non-null ballIn/ballLocation on a DOUBLE_FAULT point

• Point.create structural validation > ballIn / ballLocation > rejects a non-null ballIn/ballLocation on a UNUSUAL point

• Point.create structural validation > ballIn / ballLocation > rejects a non-null ballIn/ballLocation on a return-error point

• Point.create structural validation > rallyLengthBucket > accepts a valid bucket

• Point.create structural validation > rallyLengthBucket > accepts null — never silently defaulted

• Point.create structural validation > rallyLengthBucket > does not override a provided bucket even for an ACE

• Point.create structural validation > rallyLengthBucket > rejects an unknown bucket

• Point.fromJSON / toJSON > round-trips through JSON

• immutability > mutating the input serves array after create() does not affect the Point

• immutability > mutating toJSON()'s output does not affect the Point

• immutability > mutating toJSON()'s ballLocation object does not affect the Point

• isAce > true when endReason is ACE

• isAce > false otherwise

• isDoubleFault > true when both serves faulted

• isDoubleFault > false on a single serve

• isDoubleFault > isAce and isDoubleFault are mutually exclusive

• context > is required — a point always knows where in the match it sat

• context > rejects a non-positive setNumber

• context > rejects a non-positive gameNumber

• context > rejects a non-boolean inTiebreak

• context > rejects a non-boolean isBreakPoint

• context > rejects a non-boolean isGamePoint

• context > rejects a non-boolean isGameEnd

• context > rejects a non-boolean isSetPoint

• context > rejects a non-boolean isSetEnd

• context > rejects a non-boolean isMatchPoint

• context > is copied out by toJSON rather than shared

• firstServeIn > true with a single serve

• firstServeIn > false once a second serve was needed

• firstServeIn > null when the serve was never recorded — not the same as it having missed

• firstServeIn > false when the first serve faulted and the second went unwatched

• lastShotPlayer > TRACKED when the tracked player won

• lastShotPlayer > OPPONENT when the tracked player lost

• Point.create structural validation > serves > rejects a serve entry that isn't even an object

• Point.create structural validation > serves > rejects a serve entry whose number is outside 1 or 2

• Point.create structural validation > serves > rejects a location on a TRACKED serve that went in — a landed serve has none

• Point.create structural validation > returnOutcome > rejects a returnOutcome with no observed serve to attach it to

• Point.create structural validation > unusualReason > accepts an UNUSUAL point with no reason — the escape hatch stays optional

• Point.create structural validation > unusualReason > accepts an UNUSUAL point with a reason

• Point.create structural validation > wing / shotType > rejects an unknown wing

• Point.create structural validation > wing / shotType > rejects an unknown shotType

• Point.create structural validation > ballIn / ballLocation > accepts a null ballIn on a FORCED/MISTAKE point — the coach missed where it landed

• Point.create structural validation > ballIn / ballLocation > rejects a ballIn that isn't a boolean or null

src/features/capture/match.test.ts — 37/37

• createMatch > builds a Match from the given fields

• createMatch > generates a fresh matchId on every call

• createMatch > still runs Match.create's validation, e.g. rejects opponentId equal to trackedPlayerId

• createMatch > rejects an unknown surfaceType

• createMatch > rejects a datetime string, not just a date — playedOn must match the backend's date-only played_on

• createMatch > rejects an unknown scoringSystem

• createMatch > defaults > scoringSystem defaults to traditional when not given

• createMatch > defaults > surfaceType defaults to unspecified when not given

• createMatch > un-rostered opponent > given an opponentId, opponentName is null — the name lives on that Player row

• createMatch > un-rostered opponent > given an opponentName instead, generates a stand-in opponentId

• createMatch > un-rostered opponent > generates a fresh stand-in opponentId on every call

• Match.create opponentName validation > rejects an empty opponentName

• Match.create firstServer validation > accepts trackedPlayerId

• Match.create firstServer validation > accepts opponentId — the opponent can serve first

• Match.create firstServer validation > rejects a firstServer that isn't a participant, e.g. the old "player"/"opponent" role label

• Match.create surfaceType validation > accepts unspecified as a real surfaceType, not just hard/clay/grass/carpet

• outcomeOverride > defaults to null — a newly-started match hasn't ended

• outcomeOverride > createMatch accepts one — a back-filled match already known to have ended early

• outcomeOverride > Match.create accepts retirement as a reason

• outcomeOverride > Match.create accepts walkover as a reason

• outcomeOverride > Match.create accepts default as a reason

• outcomeOverride > Match.create rejects an unknown reason

• outcomeOverride > Match.create rejects a winner that isn't a participant

• Match.fromJSON / toJSON > round-trips through JSON

• Match.fromJSON / toJSON > reads a womens_grand_slam payload back as traditional, the way the API's migration did

• createMatch > event > defaults to null when not given

• createMatch > event > is passed through unchanged when given

• Match.create input validation > rejects input that is not an object, not an object

• Match.create input validation > rejects input that is null, not an object

• Match.create matchId validation > rejects a matchId of null

• Match.create matchId validation > rejects a matchId of

• Match.create trackedPlayerId / opponentId validation > rejects a trackedPlayerId of null

• Match.create trackedPlayerId / opponentId validation > rejects a trackedPlayerId of

• Match.create trackedPlayerId / opponentId validation > rejects a opponentId of null

• Match.create trackedPlayerId / opponentId validation > rejects a opponentId of

• Match.create event validation > rejects an empty event

• outcomeOverride > Match.create rejects an outcomeOverride that isn't an object, e.g. a bare string

src/features/capture/match-set.test.ts — 26/26

• MatchSet.create structural validation > accepts a minimal valid set

• MatchSet.create structural validation > rejects non-object input

• MatchSet.create structural validation > rejects null input

• MatchSet.create structural validation > rejects a setNumber of 0

• MatchSet.create structural validation > rejects a setNumber of -1

• MatchSet.create structural validation > rejects a setNumber of 1.5

• MatchSet.create structural validation > rejects an empty matchId

• MatchSet.create structural validation > rejects an empty trackedPlayerId

• MatchSet.create structural validation > rejects an empty opponentId

• MatchSet.create structural validation > rejects trackedPlayerId equal to opponentId

• MatchSet.create structural validation > rejects a negative trackedPlayerGames

• MatchSet.create structural validation > rejects a negative opponentGames

• MatchSet.create structural validation > rejects a non-integer trackedPlayerGames

• MatchSet.create structural validation > rejects a non-integer opponentGames

• MatchSet.create structural validation > accepts a null winner — the set is still in progress

• MatchSet.create structural validation > rejects a winner that isn't a participant

• MatchSet.create tiebreak validation > accepts a null tiebreak — the set wasn't decided by a breaker

• MatchSet.create tiebreak validation > accepts a valid tiebreak score

• MatchSet.create tiebreak validation > accepts the match_tiebreak_10_points degenerate case: 0/0 games, tiebreak decides it

• MatchSet.create tiebreak validation > rejects a non-object, non-null tiebreak

• MatchSet.create tiebreak validation > rejects a negative tiebreak.trackedPlayerPoints

• MatchSet.create tiebreak validation > rejects a negative tiebreak.opponentPoints

• MatchSet.create tiebreak validation > rejects a tiebreak.target of 0

• MatchSet.create tiebreak validation > rejects a tiebreak.target of -1

• MatchSet.create tiebreak validation > rejects a tiebreak.target of 1.5

• MatchSet.fromJSON / toJSON > round-trips through JSON

src/features/capture/state.test.ts — 56/56

• maxSetsFor > traditional's max sets is 3

• maxSetsFor > mens_grand_slam's max sets is 5

• maxSetsFor > no_ad_scoring's max sets is 3

• maxSetsFor > match_tiebreak_10_points's max sets is 3

• maxSetsFor > short_sets's max sets is 3

• maxSetsFor > pro_set's max sets is 1

• maxSetsFor > legacy's max sets is 3

• matchRulesFor > traditional needs 2 sets to win, out of 3 playable

• matchRulesFor > mens_grand_slam needs 3 sets to win, out of 5 playable

• matchRulesFor > no_ad_scoring needs 2 sets to win, out of 3 playable

• matchRulesFor > match_tiebreak_10_points needs 2 sets to win, out of 3 playable

• matchRulesFor > short_sets needs 2 sets to win, out of 3 playable

• matchRulesFor > pro_set needs 1 sets to win, out of 1 playable

• matchRulesFor > legacy needs 2 sets to win, out of 3 playable

• isAdScoringFor > traditional is ad scoring: true

• isAdScoringFor > mens_grand_slam is ad scoring: true

• isAdScoringFor > no_ad_scoring is ad scoring: false

• isAdScoringFor > match_tiebreak_10_points is ad scoring: true

• isAdScoringFor > short_sets is ad scoring: false

• isAdScoringFor > pro_set is ad scoring: true

• isAdScoringFor > legacy is ad scoring: true

• setRulesFor > traditional's non-deciding set: 6 games, tiebreak at 6, target 7

• setRulesFor > no_ad_scoring's non-deciding set: 6 games, tiebreak at 6, target 7

• setRulesFor > short_sets's non-deciding set: 4 games, tiebreak at 3, target 5

• setRulesFor > pro_set's non-deciding set: 8 games, tiebreak at 8, target 7

• setRulesFor > mens_grand_slam's regular set uses a 7-point breaker

• setRulesFor > mens_grand_slam's decider uses a 10-point breaker

• setRulesFor > set 3 of a best-of-5 mens_grand_slam match is NOT the decider

• setRulesFor > legacy's regular set tiebreaks normally at 6-6

• setRulesFor > legacy's decider has no tiebreak at all

• setRulesFor > match_tiebreak_10_points' regular set plays normally

• setRulesFor > match_tiebreak_10_points' decider is a single breaker — no games phase

• setRulesFor > validation > rejects a setNumber of 0

• setRulesFor > validation > rejects a setNumber of -1

• setRulesFor > validation > rejects a setNumber of 1.5

• setRulesFor > validation > rejects a maxSets of 0

• setRulesFor > validation > rejects a maxSets of -1

• setRulesFor > validation > rejects a maxSets of 1.5

• setRulesFor > validation > rejects setNumber greater than maxSets

• matchStateFor > no sets is 0-0 with no winner

• matchStateFor > counts sets won by each side

• matchStateFor > an in-progress set doesn't count toward either side

• matchStateFor > traditional (best-of-3, setsToWin: 2) > 2-0 is a win

• matchStateFor > traditional (best-of-3, setsToWin: 2) > 2-1 is a win

• matchStateFor > traditional (best-of-3, setsToWin: 2) > 1-1 is not decided yet

• matchStateFor > traditional (best-of-3, setsToWin: 2) > the opponent can win too

• matchStateFor > mens_grand_slam (best-of-5, setsToWin: 3) > 3-1 is a win

• matchStateFor > mens_grand_slam (best-of-5, setsToWin: 3) > 2-2 is not decided yet — traditional's setsToWin (2) would say otherwise

• matchStateFor > pro_set (setsToWin: 1) is decided by a single set

• matchStateFor > consistency checks > rejects sets that don't match the given participants

• finalWinnerFor > retirement override wins regardless of matchSets

• finalWinnerFor > walkover override wins regardless of matchSets

• finalWinnerFor > default override wins regardless of matchSets

• finalWinnerFor > no override, decided by sets, matches matchStateFor's own winner

• finalWinnerFor > no override, not yet decided -> null

• finalWinnerFor > uses the match's own scoringSystem rules, not a hardcoded best-of-3 assumption

src/features/capture/score.test.ts — 71/71

• folding points into a game > a new match is love-all in set 1, nothing observed yet

• folding points into a game > four straight points win a game, and the game resets to love-all

• folding points into a game > ad scoring needs two clear points from deuce

• folding points into a game > no-ad scoring ends the game on the first point past 40-40

• folding games into a set > six games to love takes the set and opens the next one

• folding games into a set > 5-5 is not a set — two clear games are needed

• folding games into a set > games-all starts the breaker, and the breaker's points are not games

• folding games into a set > a breaker won 7-0 records the set as 7-6

• folding games into a set > a breaker needs two clear points, same as a game

• folding games into a set > short sets run to four games with a breaker at 3-3, first to five

• folding games into a set > legacy's decider has no breaker and plays on past 6-6

• folding games into a set > the match-tiebreak decider is a breaker from its first point and credits no game

• folding sets into a match > two sets decide a best-of-three, and the board stops there

• folding sets into a match > a pro set is one set, so winning it wins the match

• folding sets into a match > three sets do not decide a best-of-five

• folding sets into a match > a point past a decided match leaves the board alone

• rotation and court side > serve alternates every game from whoever served first

• rotation and court side > the opponent serving first flips every answer

• rotation and court side > rotation carries across a set boundary rather than restarting

• rotation and court side > a breaker takes one slot, and serve inside it alternates every two points

• rotation and court side > court side alternates by points played, in a game and in a breaker alike

• rotation and court side > a fresh game is always the deuce side

• interpretSnapshot > reads completed sets, the live set and the game

• interpretSnapshot > accepts a completed traditional set of 6-0

• interpretSnapshot > accepts a completed traditional set of 6-4

• interpretSnapshot > accepts a completed traditional set of 7-5

• interpretSnapshot > accepts a completed traditional set of 7-6

• interpretSnapshot > refuses 7-4 as a completed set

• interpretSnapshot > refuses 8-6 as a completed set

• interpretSnapshot > refuses 6-6 as a completed set

• interpretSnapshot > names the set that is wrong

• interpretSnapshot > a completed last set that doesn't decide the match opens the next one

• interpretSnapshot > a decided match has no current set and no game

• interpretSnapshot > refuses a set played after the match was already won

• interpretSnapshot > refuses a set still in progress ahead of a later one

• interpretSnapshot > refuses a game score that can't happen, and one past its format

• interpretSnapshot > refuses a game alongside a breaker, and one after the match is over

• interpretSnapshot > a set at games-all carries its breaker score instead

• interpretSnapshot > refuses a tiebreak score on a set that never had one, and a decided one still running

• interpretSnapshot > refuses a tiebreak score that disagrees with who won the set

• interpretSnapshot > the match-tiebreak decider plays no games

• interpretSnapshot > refuses more sets than the format plays

• interpretSnapshot > a board round-trips through its own snapshot

• checkpoints > applying one moves the score to what the coach declared

• checkpoints > declared games are never counted as observed

• checkpoints > games already watched stay watched when a later checkpoint keeps them

• checkpoints > a correction that shrinks the score can't leave more observed than played

• checkpoints > the serving answer is cross-checked against the score

• checkpoints > on an empty log the answer defines who served first instead

• checkpoints > an explicit order change re-anchors the rotation rather than refusing

• checkpoints > a checkpoint that changes nothing is refused

• the fold over a whole log > points and checkpoints fold together in sequence order

• the fold over a whole log > an event out of order, or from another match, is ignored rather than thrown

• the fold over a whole log > points past a decided match are ignored, not thrown

• the fold over a whole log > a checkpoint the fold can't read is ignored, and the board survives

• stamps > a point mid-game is none of them

• stamps > 40-0 on serve is a game point, not a break point

• stamps > 0-40 on serve is a break point, not a game point

• stamps > no-ad deuce is both at once

• stamps > ad scoring at deuce is neither — two points are still needed

• stamps > serving for the set stamps a set point, and winning it stamps the end

• stamps > a set point for the receiver is still a set point

• stamps > the last point of a match stamps a match point and the set end

• stamps > inside a breaker the stamps answer for the breaker as its own game

• scoreboardFor > a brand-new match is love-all: one games column, one call column

• scoreboardFor > the call column reads deuce and advantage

• scoreboardFor > a completed set keeps its own column and the live set gains one

• scoreboardFor > a breaker shows its points in the call column

• scoreboardFor > a match-tiebreak decider shows its breaker in brackets once complete

• scoreboardFor > a declared set is marked as not fully watched

• coverage > counts watched games against played ones across the whole match

src/features/capture/checkpoint.test.ts — 11/11

• ScoreCheckpoint.create > accepts a well-formed checkpoint

• ScoreCheckpoint.create > rejects a missing id

• ScoreCheckpoint.create > rejects a missing matchId

• ScoreCheckpoint.create > rejects a negative or fractional sequence

• ScoreCheckpoint.create > rejects an unknown servesNext

• ScoreCheckpoint.create > rejects a non-boolean orderChanged

• ScoreCheckpoint.create > requires at least one set

• ScoreCheckpoint.create > rejects a set score that isn't two whole numbers

• ScoreCheckpoint.create > accepts a breaker score, and a set with none

• ScoreCheckpoint.create > does not check whether the score is reachable

• ScoreCheckpoint.create > round-trips through toJSON, copying the snapshot rather than sharing it

src/features/capture/capture-flow.test.ts — 84/84

• startPoint > TRACKED starts on the first serve, with the given courtSide attached

• startPoint > OPPONENT starts on the first serve — courtSide is attached the same way, not tracked-only

• serve — tracked player > in → RALLY — one tap says it landed, not where inside the box

• serve — tracked player > fault on the first serve → SERVE number 2, the fault is kept not overwritten

• serve — tracked player > fault on the second serve → READY, DOUBLE_FAULT, no rally to ask about

• serve — tracked player > ace → READY, won by the server, and carrying no placement

• serve — opponent > ace → READY, won by the opponent (our player never touched it)

• serve — opponent > fault on the second serve → READY, DOUBLE_FAULT, won by the tracked player

• serve — opponent > return in → RALLY, the serve is recorded as IN with no location

• serve — opponent > return error → READY, the no-rally shortcut — returnOutcome already says where it went

• serve — opponent > rejects a return error with no endReason — forced vs mistake must be chosen

• serve — opponent > rejects a return-in with an endReason — nothing to classify yet

• let > service let on the first serve is a no-op — nothing was recorded yet

• let > service let on the second serve keeps the first-serve fault, doesn't reset to first serve

• let > rally let resets the whole point to first serve, clearing serves and history

• let > rally let preserves courtSide unchanged — the same point, replayed

• let > rejects a let from a step other than serve or rally

• note > adds a note to the draft without advancing the step

• note > doesn't push onto history — an undo skips straight past it

• note > a second note appends onto a new line rather than overwriting the first

• note > blank or whitespace-only text is a no-op — no empty note recorded

• note > trims surrounding whitespace off a real note

• note > survives an ordinary step transition

• note > a rally let discards the note along with the rest of the draft

• rally screen > won=true endReason=FORCED → SHOT_DETAIL, length first then outcome

• rally screen > won=false endReason=FORCED → SHOT_DETAIL, length first then outcome

• rally screen > won=true endReason=MISTAKE → BALL_LOCATION, length first then outcome

• rally screen > won=false endReason=MISTAKE → BALL_LOCATION, length first then outcome

• rally screen > won=true endReason=FORCED → SHOT_DETAIL, outcome first then length — order doesn't matter

• rally screen > won=false endReason=FORCED → SHOT_DETAIL, outcome first then length — order doesn't matter

• rally screen > won=true endReason=MISTAKE → BALL_LOCATION, outcome first then length — order doesn't matter

• rally screen > won=false endReason=MISTAKE → BALL_LOCATION, outcome first then length — order doesn't matter

• rally screen > an unforced error commits with no shot recorded, which Point.create allows

• rally screen > picking a length alone is sticky — it writes into the draft without leaving RALLY or touching history

• rally screen > picking an outcome alone is sticky too — same non-advancing update, symmetric with length

• rally screen > a rally let clears a picked length along with the rest of the draft

• rally screen > a rally let clears a picked outcome too, for the same reason

• full happy path — serve through commit > a FORCED win threads through every screen and commits to a valid Point

• unusual point > from the first serve, no serve is recorded rather than one assumed

• unusual point > from the second serve, the fault already recorded survives

• unusual point > a point skipped from the serve commits with its winner and nothing else

• unusual point > reachable from the rally screen, win/loss is mandatory, reason is optional

• unusual point > an unusual point's rally length is never auto-filled, unlike ACE/DOUBLE_FAULT

• undo > undoes one step back to the exact prior draft

• undo > restores the right draft even when two different paths reach the same step

• undo > undoing with no history discards the draft

• undo > undoing twice from a two-step-deep history exhausts the stack, lands on ABORTED

• commitDraft > a valid READY draft for each endReason produces a Point that passes Point.create

• commitDraft > a return error reached via transition() commits — the no-rally shortcut is a real Point

• commitDraft > an ending nobody saw commits — endReason null is an observation hole, not a break

• commitDraft > a broken draft (READY reached with no winner) throws via Point.create, not silently

• didn't see > from the serve, no serve is recorded and the rally is still ahead

• didn't see > from the second serve, the fault already watched survives

• didn't see > from the rally it still needs the winner, and leaves the ending unknown

• didn't see > from the rally, a length already picked is kept

• didn't see > from the rally without a winner it throws rather than guessing

• didn't see > from the shot screen, the stroke is left null and the ball still asked for

• didn't see > from the ball screen, where it landed is left null and the point commits

• didn't see > undo steps back into the screen that was skipped

• didn't see > a point with every observation missing still commits with its winner

• action validity guards > rejects an action that doesn't belong on the current step

• action validity guards > rejects TAP_SERVE_IN when the opponent is serving — no such tap on that screen

• action validity guards > rejects TAP_RETURN when the tracked player is serving — no return tap on that screen

• serve — tracked player > in → BALL_LOCATION — one tap says it landed, not where inside the box

• serve — opponent > return in → BALL_LOCATION, the serve is recorded as IN with no location

• let > rejects a let from a step other than serve or final ball

• final ball > won=true is stored with the zone, then the length is asked

• final ball > won=false is stored with the zone, then the length is asked

• final ball > a rally let clears the location and winner along with the rest of the draft

• rally length and ending > the length is asked before the ending

• rally length and ending > a forced ending goes on to the shot

• rally length and ending > an unforced error skips the shot and commits with none recorded, which Point.create allows

• unusual point > reachable from the final ball, win/loss is mandatory, reason is optional

• didn't see > from the final ball it still needs the winner, and leaves the ending unknown

• didn't see > from the final ball without a winner it throws rather than guessing

• didn't see > from the shot screen, the stroke is left null and the point commits

• action validity guards > rejects an action that doesn't belong on BALL_LOCATION

• action validity guards > rejects an action that doesn't belong on UNUSUAL_OUTCOME

• action validity guards > rejects an action that doesn't belong on UNUSUAL_REASON

• action validity guards > rejects an action that doesn't belong on RALLY_LENGTH

• action validity guards > rejects an action that doesn't belong on ENDING

• action validity guards > rejects an action that doesn't belong on SHOT_DETAIL

• action validity guards > rejects any action once the point has reached READY — a double tap on a dying screen

• action validity guards > rejects any action once the point has been discarded to ABORTED

src/features/capture/match-flow.test.ts — 12/12

• startNextPoint > brand-new match: sequence, server and court side all come off the board

• startNextPoint > mid-match: the server follows the rotation and the side follows the game

• startNextPoint > the sequence is the caller's to supply, not derived from the log

• undoLast > drops the newest event and hands it back

• undoLast > drops a checkpoint just the same — the tail is the tail

• undoLast > an empty log has nothing to undo, and says so rather than throwing

• undoLast > never mutates the log it was given

• retireMatch > awards the match to the player who didn't retire

• retireMatch > rejects a player who isn't in the match

• isMatchOver > an override ends it whatever the board says

• isMatchOver > true once the sets decide it

• isMatchOver > false mid-match

Backlog item 5

Test Results Summary

Web (vitest): 380/380 passed across 29 files

Web — vitest (380/380)

src/features/capture/court-diagram.test.tsx — 7/7

• renders one tap target per zone, named for the coach

• a tap flashes before the flow advances

• a second tap during the flash does not advance twice

• reports pending while a tap's flash is in flight

• chips the net band only when labelled

• carries the full frame when zoomTo is absent, and returns to it when it goes away

• zooms into a region off-centre, uniformly scaled and centred on it

src/features/capture/match-header.test.tsx — 20/20

• marks the server of the point being captured

• shows the live game score from the folded board

• sizes the score for the whole match before a point is played

• keeps that width once sets have been played

• back greys out where there is nothing behind, rather than vanishing

• back steps the capture flow back

• tapping skip asks who won before recording the point

• the skipped point can go to the opponent

• holding skip opens the score editor instead of skipping the point

• a long-press's swallowed click doesn't leave a keyboard activation stuck

• tapping the score opens the editor — the most discoverable way in

• a set the coach declared rather than watched says so

• tapping the note button opens the text sheet where the browser can't transcribe, and adds the note

• holding back goes home instead of stepping back

• a touch long-press's contextmenu on back goes home directly, not through the click

• home is reachable by keyboard, which can't perform the hold

• the first tap on Back shows a one-time hold hint

• a second tap on Back doesn't repeat the hold hint

• the first tap on Skip shows a one-time hold hint

• a second tap on Skip doesn't repeat the hold hint

src/features/capture/match-setup-screen.test.tsx — 15/15

• names the player the match is for

• a match cannot start without an opponent

• starting hands back everything the match needs

• the opponent can serve first
• whitespace alone is not an opponent
• the coach's saved defaults are what the form opens on
• what setup was left holding wins over the saved defaults
• back steps out of setup
• says which fields are required and which can be left alone
• an event name is passed through, trimmed
• an event left blank is null, not an empty string — it never blocks starting
• offers previous opponents as suggestions on the Opponent field
• typing narrows the suggestions to matching opponents
• selecting a suggestion fills the field and closes the list
• no previous opponents means there's nothing to open

src/features/capture/serve-screen.test.tsx — 15/15

• titles the attempt and names the court the point is played from
• a tap in the live service box records a serve in, with no placement
• the live box follows the court side the point is played from
• a tap on Long faults with location LONG
• a tap on Long (ad side) faults with location LONG
• a tap on Wide (left) faults with location WIDE
• a tap on Wide (right) faults with location WIDE
• a tap on Net faults with location NET
• a tap on Ad service box faults with location WIDE
• the ace button records an ace with no placement
• ace and let are disabled while a court tap's flash is pending
• the second attempt says why the first one faulted
• a let reports itself and leaves the attempt where it was
• chips whose half is whose on the court
• marks the server's stance on the side matching the court being played from

src/features/capture/return-screen.test.tsx — 14/14

• asks where the return landed

• a return into the safe area goes straight to the rally

• a return Long asks whether it was forced before recording it

• a return Wide (left) asks whether it was forced before recording it

• a return Wide (right) asks whether it was forced before recording it

• a return Net asks whether it was forced before recording it

• a returnable ball missed is recorded as a mistake

• an ace against us carries no placement

• the opponent's fault carries no placement either

• ace, fault and let are disabled while a court tap's flash is pending

• the second attempt says a second serve is coming, without a reason

• a let reports itself and leaves the attempt where it was

• chips whose half is whose on the court

• marks the opponent's stance on the side matching the court they're serving from

src/features/capture/rally-screen.test.tsx — 12/12

• Win, forced error dispatches the outcome

• Win, unforced error dispatches the outcome

• Loss, forced error dispatches the outcome

• Loss, unforced error dispatches the outcome

• neither question is required first — the outcome cards are tappable before a length is picked

• once an outcome is already picked, the prompt asks only for the length

• once a length is already picked, the prompt asks only for the outcome

• picking a rally length reports it without ending the point

• the chosen rally length shows as chosen, and re-tapping keeps it

• the chosen outcome shows as chosen, the same way the chosen length does

• an unusual ending hands off to the flow's own unusual path

• a let replays the point from the first serve

src/features/capture/final-shot-screen.test.tsx — 17/17

• offers every stored shot on both wings, and labels the columns

• Backhand drive records that wing and shot type
• Forehand drive records that wing and shot type
• Backhand volley records that wing and shot type
• Forehand drop shot records that wing and shot type
• Backhand lob records that wing and shot type
• Forehand lob records that wing and shot type
• Backhand drive records that wing and shot type, once the exit zoom finishes
• Forehand drive records that wing and shot type, once the exit zoom finishes
• Backhand volley records that wing and shot type, once the exit zoom finishes
• Forehand drop shot records that wing and shot type, once the exit zoom finishes
• Backhand lob records that wing and shot type, once the exit zoom finishes
• Forehand lob records that wing and shot type, once the exit zoom finishes
• a shot tap zooms the court back out immediately, ahead of the dispatch it delays
• didn't see the shot also waits out the exit zoom before dispatching
• a second tap during the exit zoom does not queue a second dispatch
• renders already zoomed to the given region, with no entrance delay

src/features/capture/final-ball-screen.test.tsx — 20/20

• offers the four zones players asked for
• Deuce deep records that zone, always as a ball that landed
• Deuce short records that zone, always as a ball that landed
• Ad deep records that zone, always as a ball that landed
• Ad short records that zone, always as a ball that landed
• the zones sit in the half the ball landed in
• offers the four zones on each player's half, and names whose half is whose
• Okoye: deuce deep records that zone and the winner, always as a ball that landed
• Okoye: deuce short records that zone and the winner, always as a ball that landed
• Okoye: ad deep records that zone and the winner, always as a ball that landed
• Okoye: ad short records that zone and the winner, always as a ball that landed
• Rivera: deuce deep records that zone and the winner, always as a ball that landed
• Rivera: deuce short records that zone and the winner, always as a ball that landed

• Rivera: ad deep records that zone and the winner, always as a ball that landed

• Rivera: ad short records that zone and the winner, always as a ball that landed

• the opponent's half is beyond the net and ours this side, never flipped

• deuce is to the right of whoever stands on that half

• an unusual ending hands off to the flow's own unusual path

• a let replays the point from the first serve

• didn't see still asks who won, then records the point with its ending unknown

src/features/capture/unusual-point-sheet.test.tsx — 7/7

• asks who won first, naming both players

• /Win.*Rivera/ awards the point

• /Loss.*Okoye/ awards the point

• cancelling on the first step leaves the flow alone

• offers every stored reason once the winner is known

• the point cannot be recorded until a reason is picked

• back on the reason step steps the flow back, it does not dismiss

src/features/capture/score-flash.test.tsx — 9/9

• a point won=true flashes Point won

• a point won=false flashes Point lost

• a game won=true flashes Game won

• a game won=false flashes Game lost

• a set won=true flashes Set won

• a set won=false flashes Set lost

• it clears itself, and never blocks the screen it covers

• a game holds longer than a point

• a re-render doesn't restart the timer

src/features/capture/score-sheet.test.tsx — 18/18

• opens pre-filled from the board, so a correction only touches what is wrong

• a set each, the opponent taking the first, is enterable

• a breaker row appears at the boundary, and its score is optional
• a 7-6 can carry the breaker score it was won on
• a breaker in progress replaces the game chips with its points
• no-ad scoring offers no advantage chip
• advantage is offered where the format plays it, and stands against the other's 40
• the match-tiebreak decider shows a breaker and no games
• legacy's decider is labelled as playing no tiebreak, and takes an 8-6
• a refusal names the cell it belongs to, and holds the button
• survives 1.5 typed into a cell, refusing it per cell
• survives -1 typed into a cell, refusing it per cell
• a fractional tiebreak score is refused the same way
• a score that changes nothing is refused
• on an empty log the serving answer pins who served first, and says so
• with points already recorded, a disagreeing answer names both causes
• the order-changed escape records the deviation rather than refusing it
• ending the match still asks why and who won

src/features/capture/note-button.test.tsx — 55/55

• no SpeechRecognition on the window: tapping opens the text sheet directly
• SpeechRecognition present: tapping starts recording instead of opening the sheet
• holding opens the text sheet without ever reaching for the mic
• a hold on a touch screen opens the sheet, though its release fires pointerleave first
• a hold is spent by its own click, so the next activation dictates again
• holding while recording stops and adds the note, rather than opening the sheet
• holding while a session the service ended is pending adds the note once, not the sheet
• a touch long-press's contextmenu is prevented, so the click it precedes survives
• the typed sheet is reachable by keyboard, which can't perform the hold
• stopping a recording adds the transcript as a note
• a note goes to the caller's latest handler, not the one recording started with
• the last words before Stop are part of the note, not cut off
• a dictated note longer than the API accepts is capped, not sent whole

- the typed note field stops at the same length the API accepts
- stopping with no speech heard adds nothing
- a session the service ends by itself keeps recording, and the next tap adds the note once
- a session that ends by itself with nothing heard adds nothing
- a recognition that refuses to start falls back to the text sheet, not a stuck button
- a recognition that can't be constructed at all falls back to the text sheet
- a recognition error falls back to the text sheet, pre-filled with what was heard so far
- an errored session never auto-adds what it heard, since the sheet already holds it
- a superseded session's late error doesn't open the sheet over a live recording
- a superseded session's late end doesn't stand down a live recording
- after a recognition error the next tap records afresh, adding only the new note
- navigating away mid-dictation still adds what was heard
- navigating away after a recognition error adds nothing, since the sheet held it
- cancel closes the sheet without adding a note
- holding while recording saves the note, rather than opening the sheet
- saving a recording adds the transcript as a note
- the field fills in live as speech results arrive while recording
- typing into the field mid-recording keeps the coach's edit, not later speech
- the last words before Save are part of the note, not cut off
- saving with no speech heard adds nothing
- a session the service ends by itself says so, and the next tap adds the note once
- the panel says how to finish while the mic is live
- the panel's own Save adds the note, the same as the button above it
- the panel's Save waits for the last words the service owes
- a second Save, on either control, still adds the note once
- Save stays available before any speech arrives, and goes once the mic is off with none
- a note typed after a point boundary took the dictation still saves
- the button saves a note typed after a point boundary, rather than recording afresh
- cancelling a dictation adds nothing and stands the button down
- cancelling a live dictation releases the mic
- a cancelled session's last words never land as a note

• cancelling a session the service already ended adds nothing and doesn't stop it twice

• a cancelled dictation leaves the next one blank

• Escape from the field discards the dictation

• the typed-note escape hatch is out of the way while the panel is up

• the sheet's Save button is disabled until there's text to add

• saving a recording shows a confirmation toast

• adding a typed note from the sheet shows a confirmation toast

• the first recording ever started on this device shows a one-time hold hint

• a second recording on the same device doesn't repeat the hold hint

• no hold hint when tapping falls straight to the text sheet

• the sheet's Add note button is disabled until there's text to add

src/features/capture/result-screen.test.tsx — 9/9

• a match won in straight sets reads as a win, with the score from its points

• a lost match names the winner first and reads the score from their side

• a retirement beats the set count and says so

• a walkover with no points played drops the score, not just its numbers

• reports coverage in games, so a filled-in set never reads as watched

• offers to undo the last point only while there is one to take back

• no undo is offered when there is nothing to unwind

• finishing the match is its own explicit action

• offers the way home and the way to the match's own results

src/features/capture/live-match.test.tsx — 45/45

• setup comes first, and starting it opens the serve screen

• back on setup exits the match entirely

• the opponent serving first opens the return screen instead

• a Grass match paints the court that surface

• a Clay match paints the court that surface

• a Carpet match paints the court that surface

• a hard court keeps the default green, with nothing to override it

- a point threads serve to final ball and lands on the scoreboard
- the rally screen's two questions can be answered in either order
- an unforced error skips the shot screen
- a let after picking a rally length doesn't carry it into the replayed point
- a point the coach only half-saw is kept, not discarded
- a missed stroke still records where the ball landed
- the skip sheet points at the editor when the gap is bigger than one point
- back on the match's first screen returns to setup, still holding what was typed
- nothing is queued while setup is still reachable
- back walks the point's screens, and stops at the serve
- once the match is queued, undoing back to the serve screen leaves rather than reaching setup
- back past a recorded point offers to undo it, and asks first
- undoing takes the point back off the score and queues its delete
- a mis-tapped match point can be undone from the result screen itself
- a tab that dies mid-match reopens at the same score
- a resumed match keeps counting from where its sequence left off
- a logged event this build cannot read leaves the coach on setup, told why
- a finished match is not resumed into
- the score the coach enters is queued as its own event, not as invented points
- an out-of-turn game on court is recordable rather than a dead end
- finishing a match won on sets queues its completion, which nothing else does
- the committed point is queued behind its match, and never awaited
- a skipped point is recorded winner-only and moves the score
- recording a point as a Win flashes Point won
- recording a point as a Loss flashes Point lost
- the flash clears itself rather than sitting over the next point
- a point that takes the game flashes the game, not the point
- a point that takes the set flashes the set
- the point that ends the match doesn't flash over the result screen
- skipping ahead resumes from the score the coach entered
- ending the match shows the result, with the reason it ended

• declaring a walkover before the first point still queues the match and its outcome
• retiring after points were played still queues the outcome, as its own row
• a ball finishing on our half is a point lost
• a let from the final ball replays the point from the first serve
• holding back goes home outright, bypassing the undo prompt entirely
• a dictation still running when the point ends lands on that point, not the next
• a Save still in flight when the point ends doesn't carry the note to the next point

src/features/matches/new-match.test.tsx — 11/11

• hands the player and coach to capture, and starts at setup
• starts a match with the coach unreachable
• opens the form on the defaults the coach saved
• says so instead of starting a match no drain can send
• says not found instead of an unnamed form when the player isn't the coach's
• shows the takeover prompt when another device holds the claim
• offers to view results when the match already finished elsewhere
• stops capture and never renders it once the match is deleted elsewhere
• holding the header's back button mid-match goes straight home
• offers this player's previous opponents as suggestions
• starts a match with no previous-opponent suggestions available

src/features/outbox/match-log.test.ts — 18/18

• events > reads back in sequence order, whatever order they were written in
• events > only ever returns the match asked for
• events > rewriting an event under its own id overwrites rather than duplicating
• events > removing one leaves the rest, which is what undo needs
• events > removing one that was never there is not an error
• matches > saves and reads back the facts the events can't carry
• matches > saving again updates in place
• matches > lists a coach's unfinished matches, most recently touched first
• matches > finds the newest unfinished match for one player, and skips a finished one

• matches > another player's match is never offered for this one

• matches > removing a match takes its events with it, and leaves other matches alone

• matches > removeForPlayer takes every one of that player's matches, finished or not, with their events

• matches > removeForPlayer answers an empty list where the player has nothing logged

• watchInProgress > emits this coach's unfinished matches as the log changes, freshest first

• watchInProgress > drops a match once it's marked finished, live

• watchInProgress > never emits another coach's matches

• watchInProgressFor > emits the match the moment capture writes it, then again when it finishes

• watchInProgressFor > never emits a match logged for a different player

src/features/sync/match-outcome-payload.test.ts — 13/13

• toMatchOutcomeUpdate > produces a body the API's own contract accepts

• toMatchOutcomeUpdate > maps the tracked player winning to a win result

• toMatchOutcomeUpdate > maps the tracked player losing to a loss result

• toMatchOutcomeUpdate > does not send the reason — the API has nowhere to put it yet

• toMatchOutcomeUpdate > rejects a payload that is not an object

• toMatchOutcomeUpdate > rejects a payload that is null

• toMatchOutcomeUpdate > rejects a payload that is missing trackedPlayerWon

• toMatchOutcomeUpdate > rejects a payload that is an invalid reason

• toMatchOutcomeUpdate > rejects a payload that is a non-boolean trackedPlayerWon

• toMatchOutcomeUpdate > sends retirement as the result reason

• toMatchOutcomeUpdate > sends walkover as the result reason

• toMatchOutcomeUpdate > sends default as the result reason

• toMatchOutcomeUpdate > sends completed as the result reason

src/components/field-label-row.test.tsx — 3/3

• marks a field required or optional beside its label

• omits the note when required is left unset

• keeps the note out of the labelled control's accessible name

src/components/in-progress-banner.test.tsx — 4/4

• renders nothing with no match logged

• shows the player's name, a live score and links to resume capture

• falls back to a generic title when the player can't be read

• shows a match in progress with no network reachable at all

src/components/primary-match-action.test.tsx — 3/3

• offers to start a match with nothing in progress

• offers to resume, and resumes into the right player, once a match is under way

• offers to resume with no network reachable at all

src/features/capture/basket-icon.test.tsx — 5/5

• B1 draws 2 balls, no overflow

• B2 draws 5 balls, no overflow

• B3 draws 10 balls, no overflow

• B4 fills the basket and spills three balls out in front

• the basket itself is decorative — no accessible text of its own

src/features/capture/court-geometry.test.ts — 8/8

• zoomScale picks the smaller of the two axes' own fill scales, never cropping either

• toCourtBox turns a numeric region into the CSS-string box court-diagram.tsx takes

• subCells splits a region into an equal rows x cols grid, row-major

• subCells scales to the caller's own region, not just the whole court

• finalBallZoneRegion(won=true, { half: 'DEUCE', depth: 'DEEP' }) matches the final ball screen's own zone

• finalBallZoneRegion(won=false, { half: 'AD', depth: 'SHORT' }) matches the final ball screen's own zone

• interiorSquare shrinks to the smaller axis, minus the margin on each side, centred

• interiorSquare always matches the court's own 165:216 aspect, in real units

src/features/capture/ending-screen.test.tsx — 7/7

• forced records the ending in one tap, with no exit zoom

• unforced waits out the exit zoom before dispatching

• an unforced tap zooms the court back out immediately, ahead of the dispatch it delays

• a second tap during the exit zoom does not queue a second dispatch

• names the winner the final ball implied, so a wrong half is caught here

• and the other way round when the ball finished on our half

• renders already zoomed to the given region, with no entrance delay

src/features/capture/rally-length-screen.test.tsx — 5/5

• 1-2 records its bucket in one tap

• 3-5 records its bucket in one tap

• 6-10 records its bucket in one tap

• 11+ records its bucket in one tap

• settles on the zoomed box, not the full-court default

src/features/capture/resume.test.ts — 11/11

• restoreInProgress > returns null where capture left nothing behind

• restoreInProgress > folds the logged match and its events back into a board

• restoreInProgress > throws on a logged event this build cannot read, rather than answering half a match

• useInProgressSummary > is null with nothing logged

• useInProgressSummary > carries the match, a live score line and the log's own updatedAt

• useInProgressSummary > updates live as the log changes, without a remount

• useInProgressSummary > drops to null once the match is marked finished

• useInProgressSummary > never carries another coach's match

• useInProgressSummary > a null coachId (no owner recorded on this device yet) subscribes to nothing and stays null

• useInProgressSummary > clears to null as soon as coachId goes from a coach to null

• useInProgressSummary > a slower fold for an older emission never overwrites a newer one

src/features/matches/capture-claim.test.tsx — 12/12

• attemptClaim > does not call the server while offline

• attemptClaim > does not call the server while the match's own create is still queued

• attemptClaim > claims outright on a clean 200

• attemptClaim > sends the device id the request body contract expects

• attemptClaim > reports a conflict on a 409 whose match has no result yet

• attemptClaim > reports the match finished on a 409 whose result is already recorded

• attemptClaim > reports gone on a 404 from the claim itself

• attemptClaim > reports gone on a 404 from the follow-up read after a 409

• useCaptureClaim > claims silently once the in-progress match is known

• useCaptureClaim > re-attempts the claim once the browser comes back online

• useCaptureClaim > surfaces a conflict, then lets takeOver claim it

• useCaptureClaim > surfaces a repeat conflict from takeOver as an error, not a silent close

src/features/matches/capture-takeover-sheet.test.tsx — 5/5

• renders the takeover prompt, and Take over invokes the handler

• Cancel invokes decline

• shows the busy label and disables both buttons while taking over

• shows a repeat-conflict error

• renders nothing visible when closed

src/features/matches/start-match.test.tsx — 9/9

• lists the coach's roster to pick from

• search narrows the roster to matching names

• a search matching nobody explains how to proceed

• an empty roster shows the same guidance rather than a bare list

• picking a player carries their id into match setup

• the back control returns to Matches when that's where the coach came from

• the back control returns to Home when that's where the coach came from

• the back control falls back to Home for a direct link with no history to pop

• shows placeholders while the roster is still loading, not the empty state

src/lib/device-id.test.ts — 3/3

• mints a uuid on first access and persists it

• returns the same id on later access, without minting again

• returns an id stored by an earlier visit

Backlog item 6

Test Results Summary

API (pytest): 227/227 passed across 6 files
Web (vitest): 345/345 passed across 35 files
Total: 572/572 passed

API — pytest (227/227)

tests/routers/test_analytics.py — 204/204

- test_routes_require_a_session
- test_routes_hide_another_coachs_player
- test_overview_is_empty_before_any_data
- test_overview_records_and_form
- test_overview_point_metrics
- test_overview_ratio_without_unforced_errors_shows_the_count
- test_serve_is_empty_before_any_points
- test_serving_metrics_and_breakdown
- test_returning_metrics_and_breakdown
- test_serving_fault_modes_split_by_serve_number
- test_the_returning_side_reports_no_fault_modes
- test_serve_outcome_shares_split_by_serve_number
- test_return_outcomes_split_by_serve_number
- test_serve_trend_runs_oldest_to_most_recent
- test_serve_trend_keeps_only_the_most_recent_matches
- test_rally_is_empty_before_any_points
- test_rally_stats
- test_rally_length_buckets_are_reported_whole
- test_point_ender_styles
- test_serve_narrows_to_
- test_rally_narrows_to_one_set
- test_phase_counts_onlyp
- test_serve_narrows_to_a_phase
- test_tiebreak_phase_leher_than_zero
- test_returning_breakdown_narrows_with_the_rest_of_the_payload
- test_set_and_phase_nar
- test_overview_ignores_a_scope_it_does_not_offer
- test_filters_reject_a_
- test_opponents_lists_each_matchup_most_met_first
- test_opponents_is_empt
- test_the_opponent_axis_counts_a_match_without_a_result
- test_serve_narrows_to_
- test_rally_narrows_to_one_opponent
- test_overview_narrows_

- test_opponent_set_and_phase_narrow_together
- test_an_opponent_key_rme
- test_an_unknown_opponent_key_is_refused
- test_a_blank_opponent_
- test_match_routes_require_a_session
- test_match_routes_hide
- test_match_serve_stats_leave_the_players_other_matches_out
- test_match_rally_statshes_out
- test_squad_members_requires_a_session
- test_squad_members_is__
- test_squad_members_covers_a_player_with_no_matches
- test_squad_members_is__
- test_squad_members_counts_the_period_and_the_streak
- test_squad_members_streats
- test_squad_members_days_since_the_last_match
- test_squad_members_trabaseline
- test_squad_members_skip_a_match_a_stat_cannot_answer
- test_squad_members_omi
- test_squad_members_baseline_stops_at_ten_matches
- test_squad_members_ratnominator
- test_squad_members_baseline_skips_a_match_with_no_unforced_errors
- test_squad_members_basw_points
- test_squad_members_current_skips_a_match_of_too_few_points
- test_squad_members_sernt_floor
- test_squad_members_omit_a_stat_left_with_one_countable_match
- test_squad_members_perthe_year
- test_squad_members_period_record_ignores_a_match_without_a_result
- test_squad_members_serut_a_serve
- test_squad_trend_requires_a_session
- test_squad_trend_rejec
- test_squad_trend_answers_the_weeks_without_a_roster
- test_squad_trend_buckenday
- test_squad_trend_stops_at_the_windows_first_monday
- test_squad_trend_stops
- test_squad_trend_leaves_an_unresulted_week_null
- test_squad_trend_pools
- test_squad_trend_pools_a_weeks_points_rather_than_its_players
- test_squad_trend_has_nminator
- test_squad_trend_first_serve_pools_the_weeks_serves
- test_serve_narrows_to_one_surface
- test_rally_narrows_to_one_surface
- test_overview_narrows_to_one_surface_including_the_record
- test_surface_counts_a_match_with_no_opponent_name
- test_surface_excludes_a_match_with_no_surface_recorded
- test_surface_composes_with_opponent_set_and_phase

- test_opponents_narrows_by_surface
- test_squad_members_narrows_by_surface_and_zeroes_a_player_with_none
- test_squad_trend_narrows_by_surface
- test_routes_answer_a_player_about_itself [overview]
- test_routes_answer_a_player_about_itself [serve]
- test_routes_answer_a_player_about_itself [rally]
- test_routes_answer_a_player_about_itself [opponents]
- test_routes_hide_a_stablemate_from_a_player [overview]
- test_routes_hide_a_stablemate_from_a_player [serve]
- test_routes_hide_a_stablemate_from_a_player [rally]
- test_routes_hide_a_stablemate_from_a_player [opponents]
- test_match_routes_answer_a_player_about_their_own_match [serve]
- test_match_routes_answer_a_player_about_their_own_match [rally]
- test_match_routes_hide_a_stablemates_match_from_a_player [serve]
- test_match_routes_hide_a_stablemates_match_from_a_player [rally]
- test_squad_routes_refuse_a_player [/squad/analytics/members]
- test_squad_routes_refuse_a_player [/squad/analytics/trend]
- test_an_unwatched_serve_stays_out_of_every_serve_rate
- test_an_unwatched_second_serve_is_no_miss_in_the_trend
- test_a_player_whose_serves_all_went_unwatched_keeps_the_breakdown_and_no_rate
- test_squad_members_hold_a_lopsided_match_at_the_ratios_ceiling
- test_squad_members_report_a_capped_match_as_the_current_value_too
- test_squad_trend_draws_a_lopsided_week_where_it_really_landed
- test_squad_members_last_five_stops_at_the_fifth_newest_match
- test_squad_members_last_five_counts_matches_and_not_results
- test_squad_members_last_five_passes_over_a_match_holding_no_points
- test_squad_members_last_five_return_rate_counts_recorded_returns_only
- test_squad_members_last_five_rates_are_absent_without_their_own_points
- test_rally_length_averages_split_won_from_lost
- test_rally_length_average_is_null_for_a_side_that_bucketed_nothing
- test_squad_members_tell_a_spotless_window_from_a_ratio_that_matched_it
- test_squad_members_count_forced_errors_against_and_the_windows_points
- test_squad_trend_does_not_anchor_on_a_date_ahead_of_today
- test_squad_trend_ends_at_the_rosters_last_active_week
- test_squad_trend_answers_a_roster_that_has_never_played
- test_squad_trend_anchors_on_the_surface_it_narrows_to
- test_an_unwatched_ending_stays_out_of_every_error_rate
- test_a_match_whose_endings_all_went_unwatched_has_no_error_rates
- test_squad_trend_does_not_anchor_on_an_abandoned_match
- test_squad_trend_anchors_on_a_match_with_points_and_no_result
- test_every_scoring_system_has_a_set_count
- test_rally_trend_runs_oldest_to_most_recent
- test_rally_trend_rounds_a_half_up_to_the_longer_bucket
- test_rally_trend_leaves_out_a_match_that_bucketed_nothing
- test_rally_trend_keeps_only_the_most_recent_matches
- test_rally_trend_carries_each_matchs_own_result

- test_rally_trend_flags_a_match_that_reached_its_deciding_set
- test_rally_trend_flag_follows_the_systems_own_set_count
- test_rally_trend_has_no_flag_for_a_single_set_format
- test_rally_trend_has_no_flag_before_a_set_board_is_derived
- test_rally_trend_describes_the_whole_match_under_a_set_filter
- test_rally_trend_rounds_the_share_to_a_whole_percent
- test_overview_records_every_year_and_lists_the_last_five
- test_serving_breakdown
- test_returning_breakdown
- test_an_unwatched_ending_is_a_point_without_a_style
- test_phase_counts_only_the_points_carrying_its_stamp (8 cases)
- test_set_and_phase_narrow_together
- test_filters_reject_a_value_outside_the_contract (10 cases)
- test_opponents_is_empty_without_a_named_opponent
- test_serve_narrows_to_one_opponent
- test_overview_narrows_to_one_opponent
- test_an_opponent_key_reaches_any_spelling_of_the_name
- test_a_blank_opponent_key_is_refused [overview, serve, rally]
- test_match_routes_hide_another_coachs_match [serve, rally]
- test_match_rally_stats_leave_the_players_other_matches_out
- test_squad_members_is_empty_without_a_roster
- test_squad_members_is_the_coachs_own_roster_by_name
- test_squad_members_streak_runs_while_the_result_repeats
- test_squad_members_tracked_stats_measure_their_own_baseline
- test_squad_members_omit_a_stat_without_a_baseline
- test_squad_members_ratio_stands_in_for_a_missing_denominator
- test_squad_members_baseline_skips_a_match_of_too_few_points
- test_squad_members_series_counts_a_match_at_the_point_floor
- test_squad_members_period_record_and_serve_stop_at_the_year
- test_squad_members_serve_percentage_is_absent_without_a_serve
- test_squad_trend_rejects_an_unknown_metric
- test_squad_trend_buckets_the_six_weeks_monday_to_sunday
- test_squad_trend_pools_the_roster_and_nobody_elses
- test_squad_trend_has_no_stand_in_for_a_missing_denominator

tests/routers/test_matches.py — 12/12

- test_list_matches_for_a_player_is_their_own
- test_list_matches_filtered_by_another_player_is_empty_for_a_player
- test_get_match_answers_a_player_about_their_own
- test_get_match_hides_a_stablemates_match_from_a_player
- test_match_routes_a_player_cannot_reach [post /matches, get /matches/in-progress, patch /matches/{id}, delete /matches/{id}, post /matches/{id}/capture, post /matches/{id}/capture/release, post /matches/{id}/points, get /matches/{id}/points, post /matches/{id}/score-checkpoints, get /matches/{id}/score-checkpoints, delete /matches/{id}/points/{id}, delete /matches/{id}/score-checkpoints/{id}]
- test_a_coach_cookie_decides_when_both_are_in_the_jar

- test_a_coach_cookie_that_authenticates_nothing_steps_aside
- test_a_coach_cookie_that_authenticates_nothing_is_no_session_of_its_own
- test_a_drained_coach_budget_refuses_rather_than_narrowing
- test_ingest_stores_an_unwatched_serve_and_ending_as_null
- test_update_match_records_how_the_match_ended_beside_the_result
- test_update_match_can_correct_a_completed_matchs_result_reason

tests/routers/test_players.py — 3/3

- test_a_player_session_cannot_read_its_own_player_row
- test_a_player_session_cannot_read_its_own_player_row [/notes]
- test_a_player_session_cannot_read_the_roster

tests/schemas/test_point.py — 1/1

- test_an_omitted_serve_or_ending_reads_as_unwatched

tests/models/test_match.py — 5/5

- test_database_rejects_a_result_reason_outside_the_enum
- test_database_accepts_every_result_reason [completed]
- test_database_accepts_every_result_reason [retirement]
- test_database_accepts_every_result_reason [walkover]
- test_database_accepts_every_result_reason [default]

tests/schemas/test_match.py — 2/2

- test_coerces_a_result_reason_string_to_the_enum
- test_rejects_a_result_reason_outside_the_enum

Web — vitest (345/345)

src/components/breakdown-bars.test.tsx — 2/2

- renders a zero-width fill, not a dropped style, when the group's ceiling is zero
- still fills proportionally against a real ceiling

src/components/court-heatmap.test.tsx — 5/5

- renders the court diagram and the heat legend
- labels each zone with its share of the total
- labels the single Wide zone once, not on both sidelines
- handles every zone at zero without dividing by zero
- flip swaps net and long between the top and bottom bars

src/components/rally-tab-error.test.tsx — 1/1

- shows a load-failure message instead of an indefinite skeleton

src/components/rally-tab.test.tsx — 17/17

- shows the point-ender style cards, none open by default
- tapping a style card opens its drill
- tapping the open card again collapses it
- tapping a different style card swaps the drill
- switching You won / They won leaves the drill closed
- switching sides while a drill is open closes it, rather than silently swapping its data
- switching sides changes the radar's title between Your and Their
- the Forced/Unforced explanation defines them by pressure, not ball placement
- shows the rally-length distribution, uncaptioned in the middle
- Unforced's drill omits Wing and Shot type entirely, but still shows Final ball
- switching You won / You lost leaves the drill closed
- the winning-shot heading stays the same across the Won/Lost toggle
- the forced:unforced stat names the opposite side from the Won/Lost toggle, and updates with it
- the winning-shot radar shows each shot type's real count, summed across both wings
- shows the rally-intensity hypnogram
- Unforced's drill has no Wing or Shot type heading, only Final ball
- Wing sums forehand/backhand across every style for the current side, with no card open

src/components/serve-tab-error.test.tsx — 1/1

- shows a load-failure message instead of an indefinite skeleton

src/components/serve-tab.test.tsx — 18/18

- both serve cards render, headline and stack visible, drills collapsed
- the headline % in includes aces, not just the plain-in outcome bucket
- tapping a card's header opens its own drill without affecting the other card
- both cards' drills can be open at the same time
- their serve's drill also shows your return outcome
- switching 1st/2nd serve changes the fault-mode numbers shown
- switching 1st/2nd serve also updates the headline and outcome stack above it
- both serve cards render under their new titles
- Your serve shows a 1st serve bar directly above a 2nd serve bar, with visible percentages
- the 1st serve row appears before the 2nd serve row in the document
- there is no drill-down control and no 1st/2nd serve toggle
- the trend graphs are gone
- Your serve shows two fault-mode courts, one per serve number
- Return of opponent serve shows one return-in-vs-error bar, combined across both serve numbers

- Return of opponent serve shows one fault-mode court, combined from both serve numbers' return-outcome breakdown (not the always-null fault_mode field)
- shows an empty state for a serve number that's never been attempted
- shows an empty state when the opponent card has no returns recorded at all
- the opponent card's fault court holds at zero when no return faulted

src/components/stat-hero.test.tsx — 1/1

- renders the value and label beside the icon

src/features/home/home.test.tsx — 34/34

- renders the welcome heading
- greets the coach by name once loaded
- links to the profile page
- links to the settings page
- shows the signed-in coach's name once loaded
- the profile link shows the coach's initials once loaded
- the profile link falls back to a generic icon before the coach loads
- shows no welcome-back line when the coach can't be loaded
- shows the install nudge for a signed-in coach
- shows no install nudge before the coach is known
- shows the add-photo nudge for a signed-in coach with no photo
- shows no add-photo nudge once the coach has a photo
- shows no results list until the user types
- filters players as the user types
- shows an empty state when no player matches
- renders a bubble with initials for each player
- clicking a player bubble offers viewing them or starting a match
- choosing View player opens the profile and records them as last-viewed
- 'See all' links to the players roster
- 'Add' opens the add-player sheet directly
- shows no continue-with box when nothing has been viewed yet
- shows the last-viewed player once one has been viewed
- 'View full stats' opens the last-viewed player's profile
- tapping the continue-with player row opens the view/start-match sheet, not straight to stats

- shows the squad's player count
- ranks best and worst form, and shows nothing at all for a legitimate empty needs-attention state
- shows a dismissible card when a player is flagged
- dismissing the attention card hides it
- best/worst form show last-5 and forced:unforced as icon+number pairs, like Continue with
- tapping a squad-snapshot row goes straight to that player's stats page
- renders the bottom nav with Home active
- a player bubble links straight to their profile
- tapping a player bubble opens the profile and records them as last-viewed
- tapping the continue-with player row opens their profile directly

src/features/matches/match-list.test.tsx — 13/13

- renders the matches the API returns
- falls back to a placeholder for a match whose player can't be resolved
- falls back to a placeholder for a match with no opponent name
- marks a finished match with its result, as an icon rather than a hard-to-read letter
- shows an empty state when there are no matches
- reports a failed load instead of rendering an empty list
- tapping a match row navigates to its stats page
- the edit icon hands the match to the caller instead of navigating
- the delete icon hands the match to the caller instead of navigating
- the edit icon is hidden once the match has a result
- filters by opponent, surface, result and date range
- filters by which of the coach's players played the match
- clearing filters restores the full list

src/features/matches/matches-page.test.tsx — 11/11

- renders the page heading and the match list
- offers starting a match when none is in progress
- tapping a match row navigates to its stats page
- the edit icon is hidden once the match has a result
- the edit icon opens the edit sheet, pre-filled, and PATCHes only what changed

- Cancel closes the delete sheet without deleting
- Delete match asks for confirmation before sending anything
- confirming deletes the match and it leaves the list
- a failed delete says so and keeps the match
- the delete names this device, so its own capture claim cannot block it
- a match being captured on another device offers to release and delete anyway

src/features/players/player-details-card.test.tsx — 10/10

- shows date of birth, dominant hand and notes
- shows a dash for every field the player hasn't filled in
- Edit lives in this card and fires the handler
- shows username, date of birth and dominant hand
- shows no empty-state illustration when there are no notes yet
- shows a dash for date of birth and dominant hand when unset
- shows a player's own details with no Edit button and no notes section
- lists an existing note with its focus area and date
- Quick note opens a composer, and saving creates the note
- Edit note opens the composer prefilled, and Delete asks for confirmation first

src/features/players/player-list.test.tsx — 17/17

- renders the players the API returns
- flags a player whose transfer code is still out
- the pending-transfer flag is part of the row, not a control of its own
- requests the roster from the API's own origin
- groups players under their surname initial
- filters the roster by the search term
- reports a failed load instead of rendering an empty roster
- prompts to sign in again on a 401, instead of the generic error
- tapping a player hands it to the caller instead of navigating
- the pending-transfer flag sits beside the name, not as a control of its own
- each card offers a link to view stats and a separate link to start a match
- tapping the header navigates to the player's profile and records last-viewed
- tapping start match reaches new-match without touching the profile route
- a player with no logged matches shows no metrics section
- shows a real attention flag, and renders the metrics section once matches exist
- the header is not a link, only View stats and Start match are
- View stats navigates to the player's profile and records last-viewed

src/features/players/player-notes-sheet.test.tsx — 16/16

- lists the player's notes, theme shown as an icon rather than a colored tag
- renders nothing when the sheet is closed
- Quick note opens a composer, not a disabled button
- saving a quick note adds it to the list with its chosen theme
- Save note is disabled until something is typed
- deleting a note asks for confirmation before removing it
- Cancel on the delete confirmation keeps the note
- shows an empty state with no notes yet
- shows a real error when the list fails to load
- Quick note opens a composer defaulted to today, not a disabled button
- saving a quick note sends body, recorded_on and focus_areas, then re-lists
- Save note is disabled — shows an inline error — for an empty body
- a future date blocks submit and sends nothing
- Edit prefills the note and PATCHes only what changed
- clearing every focus area on edit sends an empty list, not nothing
- a failed save surfaces the server's reason

src/features/players/player-profile.test.tsx — 23/23

- renders the player's name and avatar
- says not found for a player id that isn't the coach's
- sends the coach back to sign in on a 401, not to the roster
- falls back to a generic message for an unexpected failure
- announces a load failure as an alert, for every branch
- links back to home
- links to the match setup screen
- shows the all-time record and player details side by side
- the Notes button opens the notes sheet
- opens on the Serve tab by default, with the scope panel and recent form above it
- switches to Rally when its toggle is clicked, scope panel and recent form stay put
- Edit opens the sheet pre-filled with the player's current values
- saving Edit PATCHes only the field that changed
- saving Edit with nothing changed just closes, no request sent
- Delete player asks for confirmation before sending anything
- Cancel on the delete step returns to the form without deleting
- confirming Delete removes the player and returns to the roster

- a player already flagged as pending transfer skips straight to the explanation, no request sent
- a code created since the page loaded is caught by the request's own 409, same explanation
- an unexpected delete failure shows the server's own message
- back returns to wherever the coach actually came from
- back falls back to home for a direct link with no history to pop
- shows player details above the all-time record, stacked

src/features/players/player-record-card.test.tsx — 11/11

- shows the all-time win/loss record, no season row
- shows recent form in the same card, all-matches by default
- each result dot is announced as Won or Lost, not just shown by fill
- scopes recent form to the chosen opponent
- hides recent form for an opponent with no fixture, rather than mislabelling the all-matches record
- sends the surface axis when one is scoped, and omits it for 'all'
- omits the surface param entirely for the 'all' sentinel
- shows the all-time win/loss record as wins and losses, no season row
- names the opponent in the caption once one is scoped
- falls back to all matches when the opponent key 404s
- shows a real error for a failure that isn't a stale opponent key

src/features/players/players-page.test.tsx — 12/12

- renders the page heading and the roster
- tapping a player offers viewing them or starting a match
- choosing View player opens the profile and records them as last-viewed
- filters the roster as the coach searches
- renders the bottom nav with Players active
- the labelled row opens the redeem-transfer-code sheet
- the floating action adds a player rather than starting a match
- submitting the add-player form posts every required field
- refuses to submit until date of birth and gender are given
- rejects a date of birth in the future before sending it
- each player's card offers a link to view stats and a separate link to start a match
- choosing View stats opens the profile and records them as last-viewed

src/features/players/stats-scope-panel.test.tsx — 13/13

- starts collapsed, showing the current filters as a summary
- tapping the trigger expands the panel, not a popup
- selecting chips doesn't commit until Apply
- Apply commits the draft, updates the summary, and closes the panel
- Apply commits both Set and Phase, and the summary shows both together
- the opponent picker only shows once scope is 'vs opponent', grouped by most played
- Apply is disabled for 'vs opponent' until an opponent is actually picked
- searching the opponent list filters it
- picking an opponent and applying sets opponentId and echoes their name
- the surface chip row is absent until the player has a match with a surface logged
- the surface chip row only offers surfaces the player has actually played on
- committing a surface updates the summary and calls onApply with it
- picking an opponent and applying sets opponentKey and echoes their name

src/lib/analytics/fixtures.test.ts — 9/9

- the Rally tab's style-card counts sum to won and lost
- the hero ratio matches the combined forced and unforced counts
- the rally-length histogram sums to won plus lost
- rallyStatsFor returns the base fixture for the default filter
- rallyStatsFor scales won/lost/hero together, so they stay reconciled
- rallyStatsFor never produces a negative count from a small scale
- serveStatsFor returns the base fixture for the default filter
- serveStatsFor's outcome triples still sum to 100 after scaling
- serveStatsFor is deterministic per opponent, not random

src/features/players/player-surface-record-card.test.tsx — 6/6

- shows a win/loss row per surface actually played, newest enum order
- excludes a match missing either a surface or a result
- renders nothing rather than an empty card with no matches at all
- shows a real error on a failed load
- paints each surface's panel with its own real-court-colour texture
- draws a simplified court-line diagram on every panel

src/features/stats/stats-page.test.tsx — 10/10

- the surface filter offers Hard/Clay/Grass but never Carpet
- picking a surface re-fetches both the members and the trend with it applied
- shows the squad headline with an icon per stat, split forced/unforced counts, and the period's match count

- leaderboard ranks every player, no minimum-matches threshold or dimming
- switching the leaderboard metric chip reorders the ranked rows
- the Spread histogram is gone
- the trend chip row still offers Forced : Unforced; the leaderboard's own row no longer does
- Resilience excludes a player with no forced or unforced errors logged yet
- shows an alert instead of an indefinite skeleton when the roster fails to load
- shows an alert on the trend card alone when only the trend query fails

src/features/stats/player-insights-card.test.tsx — 12/12

- shows a loading skeleton, then the real grid
- renders an avatar only for players with a computable aggression and resilience score
- positions an avatar via aggression (x) and mistake resilience (y), scaled to the plotted group's own range
- sizes each avatar by its own average rally length, relative to the plotted group's range
- each avatar is a real link to that player's stats page
- notes how many players are excluded for having no rally data yet
- explains all three encodings as bullet points: position (both axes) and size (rally length)
- shows an empty state instead of a blank grid when no player has computable scores yet
- zooms into a crowded plot on tap, and back out again
- pinching off-centre zooms into that spot, not the box's own centre
- zooming all the way in disables further zoom-in
- zooms into a crowded plot on tap, and back out again, without growing the avatars

src/lib/analytics/squad-stats.test.ts — 34/34

- squadHeadline sums wins/losses and raw forced/unforced counts, and weighs first-serve % by match count
- squadHeadline returns a zeroed record, zeroed counts and a null rate for an empty squad
- buildLeaderboard ranks every player by the selected metric, no threshold
- buildLeaderboard puts a null metric value last, displayed as an em dash
- buildLeaderboard filters by search
- fillSquadTrendGaps carries a single interior null forward from the preceding value
- fillSquadTrendGaps carries the same preceding value across a run of nulls, not the following one
- fillSquadTrendGaps leaves leading nulls untouched — nothing to carry forward yet
- fillSquadTrendGaps leaves an all-null trend unchanged
- fillSquadTrendGaps fills every null after a single real value, and none before it
- fillSquadTrendGaps passes a trend with no nulls through unchanged
- fillSquadTrendGaps preserves each point's week label, only overwriting the value
- fillSquadTrendGaps marks exactly the carried positions, not real or leading-null ones
- mistakeResilience reads the tracked player's own errors from styles.opponents (points they lost)
- mistakeResilience is null when the player has lost no points via a recorded error yet
- mistakeResilience is 100 when every own error was forced, 0 when every one was unforced
- aggressionScore reads the opponent's own errors from styles.you (points the tracked player won)

- aggressionScore is null when the player has won no points via a recorded error yet
- aggressionScore is 100 when every opponent error was forced, 0 when every one was unforced
- overallAverageRallyLength weights won/lost averages by their own point counts
- overallAverageRallyLength falls back to whichever side is non-null
- overallAverageRallyLength is null when neither side has a value yet
- overallAverageRallyLength falls back to a plain mean when point counts are both zero
- rallyLengthSizeBucket maps a value to one of four sizes, relative to the group's own range
- rallyLengthSizeBucket maps every value to md when the group has no spread
- scaleToGroupRange maps the group's min/max to the 10-90 padded band, not raw 0-100
- scaleToGroupRange centers every value at 50 when the group has no spread
- metricDisplay signs a positive Pressure Index and leaves zero/negative unsigned
- resiliencePercent is a forced share of a player's own errors, null when neither has happened
- tacticalEfficiencyPercent is forcing shots induced over points won, null with no points won yet
- buildLeaderboard excludes a player with no computable metric value entirely
- buildLeaderboard excludes a null firstServe value the same way as the newer metrics
- metricValue and metricDisplay resolve each metric key
- buildLeaderboard reorders when the metric key changes

src/components/add-player-sheet.test.tsx — 11/11

- dominant hand reaches the request
- a typed creation note seeds the player's notes list
- leaving the creation note blank sends no notes request at all

src/lib/player-form.test.ts — 1/1

- buildPlayerUpdateBody trims names before comparing and sending

src/components/fault-mode-court.test.tsx — 9/9

- renders one accessible label naming all three shares
- renormalises shares that don't sum to 100
- labels the top out-zone Long and the bottom Net
- prints the wide share exactly once
- a fault mode with no clear majority still renders without throwing
- renders one accessible label naming all three raw counts
- colours by percentage even when the raw counts alone would suggest otherwise
- prints the wide count exactly once
- never prints a percentage sign anywhere on the court

src/components/serve-rally-toggle.test.tsx — 3/3

- shows Serve and Rally, and marks the current value as pressed
- clicking the other option reports it, without changing which one is marked active here
- the active option is styled with the primary color, not the muted default

src/components/final-ball-court.test.tsx — 7/7

- percentages are a share of the four in-court zones only, ignoring net/long/wide

- draws no net/long/wide markers
- a style scoped to points won colors its majority zone with the win pattern
- a style scoped to points lost colors its majority zone with the loss (vermillion) pattern
- renders without throwing when there are no in-court zones at all
- prints each zone's raw count, ignoring net/long/wide
- the net graphic moves to the top when not flipped, matching the row order

src/components/rally-hypnogram.test.tsx — 7/7

- shows the card heading and caption
- shows a legend for won, lost, no result yet, and went the distance
- draws one bar per match
- a won bar is a plain solid fill, a lost bar carries the diagonal-stripe pattern
- marks only the match that went the distance
- shows an empty state when there are no matches to plot
- falls back to the empty state rather than crashing when trend is missing entirely

src/components/rally-intensity-chart.test.tsx — 5/5

- numbers games continuously across the match, not restarting each set
- a game's bar height reflects the share of its points running past 5 shots
- a game with no rally-length data gets an honest 'no data' bar, not a false 0%
- shows an empty state when the match has no points yet
- shows a real error on a failed load

src/components/shot-radar.test.tsx — 5/5

- shows the real count next to a shot's label when the caller provides one
- draws a labelled axis per shot
- a series named in prose still emits a parseable custom property
- an overlay keeps one property per series
- renders nothing when there is no series to draw

src/features/matches/momentum-card.test.tsx — 7/7

- shows nothing below the chart for a match with no notes at all
- lists a note under the chart, labelled by exactly where it was captured
- multiple notes in the same game each get their own labelled line
- a note in one game carries that game's own label, not a different game's
- explains what the break-of-serve dot means
- shows a real error on a failed load
- explains what the bars encode, behind the header's info button

src/lib/momentum.test.ts — 6/6

- derives a set's score and per-game winner from the raw point log
- collects a game's own notes only, in sequence order, oldest first
- a game with no notes gets an empty array, not undefined or null
- longRallyShare is the percentage of a game's bucketed points past 5 shots
- longRallyShare is null, not 0, when a game recorded no rally-length bucket at all
- blockHeight scales within the given max/min, defaulting to the small strip's own scale

src/components/rally-tab-empty-drill.test.tsx — 2/2

- an unforced style with a real count but no breakdown says so instead of showing bare headings
- an unusual style with no wing and no reasons says so, singular count included

src/features/matches/match-stats-page.test.tsx — 8/8

- shows the match's own summary and its result
- names the player whose match it is, with the opponent alongside the rest
- falls back to the opponent as the title where the roster has no such player
- falls back to the matches list for a direct link with no history to pop
- returns to Home when that's where the coach came from
- opens on the Serve tab, with the same Serve/Rally toggle the player page uses
- switches to Rally when its toggle is clicked
- shows a load error instead of an indefinite skeleton

src/lib/color-scale.test.ts — 4/4

- weakest end matches the vermillion anchor
- midpoint matches the orange anchor
- strongest end matches the jade green anchor
- clamps values outside 0-1

src/lib/format-record.test.ts — 2/2

- marks which number is wins and which is losses
- handles a shutout in either direction

Backlog item 7

Test Results Summary

Web (vitest): 180/180 passed across 11 files

Web — vitest (180/180)

src/features/outbox/outbox.test.ts — 42/42

- stamps the payload version rows are written with
- append > stores the entry as pending, stamped with the payload version
- append > keeps the payload opaque — it is stored, not inspected
- append > hands out increasing enqueue order in call order
- append > keeps enqueue order increasing after earlier rows are drained
- append > keeps the first payload for an id already queued
- append > leaves a parked row parked when its entity is appended again
- append > survives concurrent appends of the same id
- append failures > surfaces a quota error as a typed failure instead of rejecting
- append failures > finds a quota error Dexie has wrapped in an aborted transaction
- append failures > distinguishes any other write failure from a quota one
- append failures > reports a payload the browser cannot clone as a write failure
- append failures > keeps its promise to the tap when a listener throws
- append failures > stops notifying a listener that has unsubscribed
- readPending > returns only this coach's pending rows, oldest first
- readPending > is empty for a coach with nothing queued
- readByMatch > returns one match's rows in enqueue order, failed ones included
- remove > drops only the given ids
- remove > ignores an id that is not queued
- remove > drops nothing when given no ids
- removeByMatch > clears one match's rows, parked ones included, and counts them
- removeByMatch > answers zero for a match with nothing queued
- markFailed > parks a match and its dependent points in one call
- markFailed > records why the rows were parked
- markFailed > parks nothing when given no ids
- retryFailed > returns this coach's failed rows to the queue and counts them
- retryFailed > clears the recorded failure along with the status
- deleteOutboxDatabase > takes every coach's rows with it and leaves the queue writable
- watchCounts > emits this coach's pending and failed counts as the queue changes
- watchCounts > ignores writes belonging to another coach
- persistent storage > asks for eviction exemption once, on the first append
- persistent storage > appends fine where the browser offers no storage manager
- persistent storage > appends fine when the request is refused
- persistent storage > appends fine when the storage manager throws instead of returning
- hasQueuedCreate > is true while the match's own create row is pending
- hasQueuedCreate > is still true once the create row is parked
- hasQueuedCreate > is false once the create row is drained

- hasQueuedCreate > is false for a match with nothing queued at all
- hasQueuedCreate > a queued point with no matching create row doesn't count as one
- watchQueuedCreate > flips to false the moment the drain acknowledges the create
- watchCounts > counts checkpoint and eventDelete rows like any other pending kind
- watchCounts > counts checkpoint and eventDelete rows like any other failed kind

src/features/sync/drain.test.ts — 41/41

- flushing > creates the match first, then flushes its points in enqueue order
- flushing > groups interleaved rows by match, matches in first-queued order
- flushing > flushes points straight away for a match created on an earlier run
- flushing > chunks a long queue at the batch ceiling
- flushing > closes a batch on the body limit before the count ceiling
- flushing > tolerates a row deleted while the drain is mid-flight
- flushing > sends nothing for a match purged while the drain is mid-flight
- refusals > parks a refused match and its points, with the reason, without blocking others
- refusals > parks a refused point batch with the 422's own words, never retrying it
- refusals > skips every batch of a match whose earlier rows are parked
- refusals > parks a payload this app version cannot map, keeping its cause
- refusals > parks an unmappable match with its own points, before any request
- rows that can never land > discards a match the server says was deleted, without parking anything
- rows that can never land > discards the points of a deleted match and flushes the next one
- rows that can never land > parks a match whose player the server will not accept
- rows that can never land > parks the points of a match the server cannot find
- halting and retrying > halts everything on a 401 and keeps the queue intact
- halting and retrying > halts on a 401 from point ingest too, with no match row ahead of it
- halting and retrying > retries a network failure on the backoff schedule, then succeeds
- halting and retrying > gives up after the schedule and leaves the rows pending
- halting and retrying > gives up on a failing match create and leaves the match and its points pending
- halting and retrying > waits out a 429's Retry-After when it outlasts the backoff step
- halting and retrying > retries the sweep when another tab's upgrade closes the database
- the drain lock > skips, sending nothing, while another drain holds the lock
- score checkpoints > sends a checkpoint after the match's points, batched like a point
- score checkpoints > maps a checkpoint off the match log once its own create has already landed
- score checkpoints > parks a checkpoint the drain has no logged match to map it against
- event deletes > deletes a point and a checkpoint by the id undo queued
- event deletes > parks one refused delete and still sends the next, independent, delete
- event deletes > holds the outcome back when an event delete is refused
- event deletes > discards every queued delete together once one of them turns out to be gone
- match outcome > sends the outcome once a checkpoint and an event delete both land
- match outcome > sends the outcome as its own request, right after the match create

- match outcome > sends the outcome for a match whose create already landed on an earlier run
- match outcome > parks a refused outcome without touching the match's points
- match outcome > parks an outcome payload this app version cannot map
- match outcome > discards an outcome for a match the server says was deleted
- refusals > parks a refused match's checkpoint and eventDelete rows alongside its match and points
- rows that can never land > forgets a deleted match's log, so capture cannot resume into it
- rows that can never land > discards a deleted match's checkpoints alongside its points
- halting and retrying > halts on a 401 from the outcome PATCH too

src/features/sync/flush-triggers.test.ts — 13/13

- registerFlushTriggers > drains at launch, for the recorded owner as this device
- registerFlushTriggers > never drains while the device has no recorded owner
- registerFlushTriggers > drains when connectivity returns
- registerFlushTriggers > drains when the app becomes visible, not when it hides
- registerFlushTriggers > a queue write while online nudges a drain
- registerFlushTriggers > a queue write while offline waits for the online event
- registerFlushTriggers > a burst of triggers coalesces into one drain
- registerFlushTriggers > re-arms after a drain that found the lock held
- registerFlushTriggers > unregistering removes every trigger, the pending debounce included
- retryLostAppend > re-appends an entry lost to a closed database
- retryLostAppend > leaves other write failures alone
- retryLostAppend > gives up once the schedule runs out
- retryLostAppend > a repeat failure for an entry already mid-retry starts no second chain

src/features/sync/match-payload.test.ts — 18/18

- toMatchCreate > produces a body the API's own contract accepts
- toMatchCreate > sends the id the match was minted with
- toMatchCreate > maps the tracked player, not the opponent, to player_id
- toMatchCreate > passes played_on through unchanged — Match already stores it date-only
- toMatchCreate > sends a concrete surface through unchanged
- toMatchCreate > maps an unspecified surface to null
- toMatchCreate > maps a free-text opponent name through unchanged
- toMatchCreate > maps a rostered opponent, which has no name, to null
- toMatchCreate > sends the traditional scoring system through unchanged
- toMatchCreate > sends the mens_grand_slam scoring system through unchanged
- toMatchCreate > sends the womens_grand_slam scoring system through unchanged
- toMatchCreate > sends the no_ad_scoring scoring system through unchanged
- toMatchCreate > sends the match_tiebreak_10_points scoring system through unchanged
- toMatchCreate > sends the short_sets scoring system through unchanged
- toMatchCreate > sends the pro_set scoring system through unchanged

- toMatchCreate > sends the legacy scoring system through unchanged
- toMatchCreate > sends the event through unchanged
- toMatchCreate > maps no event to null

src/features/sync/point-payload.test.ts — 28/28

- toPointCreate > throws until the capture model carries the score-context stamps
- toPointCreate > produces a body the API's own contract accepts for a struck point
- toPointCreate > produces a body the API's own contract accepts for an ace
- toPointCreate > produces a body the API's own contract accepts for a second serve in after a fault
- toPointCreate > produces a body the API's own contract accepts for a double fault
- toPointCreate > produces a body the API's own contract accepts for a lone fault whose second serve went unwatched
- toPointCreate > produces a body the API's own contract accepts for an unwatched serve
- toPointCreate > produces a body the API's own contract accepts for an unwatched ending
- toPointCreate > produces a body the API's own contract accepts for only the winner seen
- toPointCreate > produces a body the API's own contract accepts for a struck point whose final ball went unwatched
- toPointCreate > produces a body the API's own contract accepts for a final ball out
- toPointCreate > produces a body the API's own contract accepts for an opponent serve returned in
- toPointCreate > produces a body the API's own contract accepts for an opponent serve missed on the return
- toPointCreate > produces a body the API's own contract accepts for an opponent double fault
- toPointCreate > produces a body the API's own contract accepts for an unusual ending
- toPointCreate > produces a body the API's own contract accepts for a skipped point
- toPointCreate > produces a body the API's own contract accepts for a noted point
- toPointCreate > sends the id and sequence the point was minted with
- toPointCreate > lowercases every enum onto its API member
- toPointCreate > sends the side the point was played from
- toPointCreate > splits two attempts across the two serve columns
- toPointCreate > records no placement on an opponent's serve
- toPointCreate > sends an unwatched serve and ending as null
- toPointCreate > splits a final ball that stayed in over half and depth
- toPointCreate > sends a final ball that went out as its failure mode alone
- toPointCreate > copies all nine score stamps
- toPointCreate > sends the coach's general note
- toPointCreate > truncates a note to exactly the API's own bound, never past it

src/components/sync-status.test.tsx — 8/8

- renders nothing once everything has synced
- shows a quiet pending count while points are waiting to sync
- pluralizes the pending count

- shows a reassurance note instead of the pending badge while offline
- the offline note still shows with nothing queued
- shows the failed count, taking priority over a pending one
- tapping the failed badge returns the rows to pending and runs the drain
- the badge keeps its count and spins until the retry's drain has finished

src/features/outbox/db.test.ts — 5/5

- isDrainableKind > every kind now has a request the drain can send
- the schema > builds the three tables the outbox and the match log need
- the schema > keys the queue on enqueue order, with one row per entity id
- the schema > indexes a match's own log by the sequence it is read back in
- upgrading a database written by an older app version > keeps the rows already queued and adds the log's two tables

src/features/sync/checkpoint-payload.test.ts — 7/7

- toScoreCheckpointCreate > produces a body the API's own contract accepts
- toScoreCheckpointCreate > sends the id, sequence and lowered enums the checkpoint was recorded with
- toScoreCheckpointCreate > leaves the set in progress without a winner
- toScoreCheckpointCreate > credits a completed set's winner and opens the next set
- toScoreCheckpointCreate > carries no trailing set once the match is decided
- toScoreCheckpointCreate > carries a completed set's breaker score
- toScoreCheckpointCreate > throws on a snapshot that does not interpret against its match

src/features/sync/event-delete-payload.test.ts — 7/7

- toEventDeleteTarget > recovers the target id from the row's own id
- toEventDeleteTarget > reads a checkpoint delete the same way
- toEventDeleteTarget > rejects a payload that is not an object
- toEventDeleteTarget > rejects a payload that is null
- toEventDeleteTarget > rejects a payload that is an invalid kind
- toEventDeleteTarget > rejects a payload that is missing kind
- toEventDeleteTarget > rejects a row id missing the :delete suffix

src/lib/error-status.test.ts — 7/7

- reads a status attached to the error object
- returns undefined when there is no status, or nothing recognisable at all
- a thrown 401 carries its status, via error-interceptor.ts's error interceptor
- a network failure passes through the interceptor untouched, still an Error
- gives up immediately on a 4xx other than 429
- keeps retrying a 429 — that's what its own backoff is for
- keeps retrying a 5xx or a status-less network failure, up to the default budget

src/lib/query-persist.test.ts — 4/4

- round-trips a snapshot through a database of the read cache's own
- drops the snapshot outright when sign-out clears the cache
- busts on a real app version, not the library's empty default

- keeps a paused write out of the read cache's snapshot

Backlog item 8

Test Results Summary

Web (vitest): 414/414 passed across 52 files

Web — vitest (414/414)

src/components/add-photo-nudge.test.tsx — 4/4

- shows the nudge for a coach with no photo
- shows nothing once the coach already has a photo
- hides itself once dismissed, and stays hidden across a remount
- choosing a photo uploads it and closes the picker

src/components/add-player-body.test.ts — 4/4

- the body the form builds satisfies the generated contract
- rejects a date of birth that isn't a calendar date
- rejects names over 100 characters and notes over 2000
- rejects an unknown gender

src/components/add-player-sheet.test.tsx — 3/3

- adds a player with no photo
- adds a player with a photo, uploaded after creation
- a photo-upload failure still creates the player and closes the sheet
- says which fields are required and which can be left alone
- adds a player with no photo, then reveals credentials before closing
- reveals the generated username and password after creating a player
- Done clears the credentials and returns to the form
- a photo-upload failure still creates the player and reaches the reveal step
- a refused creation keeps the sheet open and shows the server's reason
- dismissing the sheet clears what was typed
- a pending photo can be removed again before the player exists

src/components/bottom-nav.test.tsx — 5/5

- renders all four tabs
- marks the current tab active and no other
- switches tabs on click
- renders all four coach tabs by default
- renders the player's three tabs instead, when given PLAYER_TABS

src/components/breakdown-bars.test.tsx — 2/2

- renders a zero-width fill, not a dropped style, when the group's ceiling is zero
- still fills proportionally against a real ceiling

src/components/install-app-nudge.test.tsx — 4/4

- shows the nudge when not dismissed and not installed
- hides itself once dismissed, and stays hidden across a remount
- never shows once the app is already installed
- 'See how' links to the full install guide

src/components/not-found-page.test.tsx — 2/2

- renders for a URL that matches no route
- links back to home

src/components/photo-avatar.test.tsx — 6/6

- renders initials when no photo is set
- renders a circle for the profile shape
- renders a square for the default (roster) shape
- renders the photo, sized and shaped the same way, once one is set
- shows no camera badge by default
- shows a camera badge when editable, over initials or a photo alike

src/components/photo-picker-sheet.test.tsx — 4/4

- offers Remove photo only when one is set
- Choose photo opens the file picker, and a picked file reaches onChoose
- Remove photo calls onRemove
- Cancel closes the sheet without choosing or removing anything

src/components/player-avatar.test.tsx — 6/6

- renders the player's initials
- picks the same color for the same id
- renders the photo instead of initials once one is set
- shows no camera badge by default
- shows a camera badge when editable
- falls back to initials if the photo fails to load

src/components/root-layout.test.tsx — 3/3

- supplies a QueryClient every route can use, with no provider of its own

- renders a read from the persisted cache with no connection
- discards a snapshot an earlier app version persisted

src/features/home/home.test.tsx — 45/45

- renders the welcome heading
- greets the coach by name once loaded
- links to the profile page
- links to the settings page
- shows the signed-in coach's name once loaded
- the profile link shows the coach's initials once loaded
- the profile link falls back to a generic icon before the coach loads
- shows no welcome-back line when the coach can't be loaded
- shows the install nudge for a signed-in coach
- shows no install nudge before the coach is known
- shows the add-photo nudge for a signed-in coach with no photo
- shows no add-photo nudge once the coach has a photo
- shows no results list until the user types
- filters players as the user types
- shows an empty state when no player matches
- renders a bubble with initials for each player
- clicking a player bubble offers viewing them or starting a match
- choosing View player opens the profile and records them as last-viewed
- 'See all' links to the players roster
- 'Add' opens the add-player sheet directly
- shows no continue-with box when nothing has been viewed yet
- shows the last-viewed player once one has been viewed
- 'View full stats' opens the last-viewed player's profile
- tapping the continue-with player row opens the view/start-match sheet, not straight to stats
- shows the squad's player count
- ranks best and worst form, and shows nothing at all for a legitimate empty needs-attention state
- shows a dismissible card when a player is flagged

- dismissing the attention card hides it
- best/worst form show last-5 and forced:unforced as icon+number pairs, like Continue with
- tapping a squad-snapshot row goes straight to that player's stats page
- renders the bottom nav with Home active
- offers starting a match from the home screen too
- a player bubble links straight to their profile
- tapping a player bubble opens the profile and records them as last-viewed
- tapping the continue-with player row opens their profile directly
- 'View stats' opens the last-viewed player's profile
- shows the squad's player count, with a matching row of tally icons
- caps the player tally at 8 icons and shows the overflow count
- shows the seven tendency rows with their scope-labelled captions and headline players
- a tied metric shows only the best side, not a duplicated worst
- ranked rows go straight to that row's own player's stats page
- the squad snapshot and continue-with headings are large and bold
- tapping a row's info icon explains what that stat measures
- shows an alert instead of a false empty roster when the players query fails
- roster tiles read the win-rate underline off the squad query, firing no per-player overview request

src/features/matches/match-list.test.tsx — 27/27

- renders the matches the API returns
- falls back to a placeholder for a match whose player can't be resolved
- falls back to a placeholder for a match with no opponent name
- marks a finished match with its result, as an icon rather than a hard-to-read letter
- shows an empty state when there are no matches
- reports a failed load instead of rendering an empty list
- tapping a match row navigates to its stats page
- the edit icon hands the match to the caller instead of navigating
- the delete icon hands the match to the caller instead of navigating
- the edit icon is hidden once the match has a result
- filters by opponent, surface, result and date range
- filters by which of the coach's players played the match
- clearing filters restores the full list
- renders the matches the API returns, both names inline at the same size
- crowns the tracked player once they've won the match
- an unfinished match crowns nobody and says so

- derives set scores and a per-set win/loss mark from the raw point log
- a game still being played (no game-end point yet) is left off the timeline
- the edit menu item hands the match to the caller instead of navigating
- the delete menu item hands the match to the caller instead of navigating
- the actions menu is still offered once the match has a result, with both Edit and Delete
- a finished match's last set still gets a winner mark even without its own is_set_end point
- filters by opponent name
- filters by surface
- filters by result
- filters to matches that were never finished
- filters by a date range, inclusive at both ends

src/features/matches/match-setup.test.tsx — 4/4

- renders the setup form for the player
- the start match button is disabled — nowhere for it to go yet
- cancel returns to the player profile
- says not found instead of an unnamed form when the player isn't the coach's

src/features/matches/matches-page.test.tsx — 12/12

- renders the page heading and the match list
- offers starting a match when none is in progress
- tapping a match row navigates to its stats page
- the edit icon is hidden once the match has a result
- the edit icon opens the edit sheet, pre-filled, and PATCHes only what changed
- Cancel closes the delete sheet without deleting
- Delete match asks for confirmation before sending anything
- confirming deletes the match and it leaves the list
- a failed delete says so and keeps the match
- the delete names this device, so its own capture claim cannot block it
- a match being captured on another device offers to release and delete anyway
- the edit menu item is still offered once the match has a result

src/features/players/edit-player-body.test.ts — 5/5

- a body with a single changed field satisfies the generated contract
- an empty body — nothing changed — is still valid
- rejects a date of birth that isn't a calendar date
- rejects a name over 100 characters and notes over 2000
- rejects an unknown gender

src/features/players/edit-player-sheet.test.tsx — 18/18

- shows initials when the player has no photo
- choosing a photo uploads it and shows it in place of initials
- removing a photo restores initials
- no Remove photo option when there is no photo to remove
- generating a transfer code shows it once, with a copy button
- shows 'Manage transfer code' and lets the coach cancel a pending one
- generating a new code while one is pending
- OK returns to the form from the can't-delete-yet step
- deleting a player clears its logged matches and queued outbox rows on this device
- Cancel backs out of the transfer explanation without minting a code
- Done clears the minted code rather than leaving it on screen
- a refused save keeps the sheet open and shows the server's reason
- pressing the picked dominant hand again clears it, and the save sends null
- a failed photo upload says so and leaves the initials showing
- a failed photo removal explains itself rather than appearing to succeed
- regenerating a password warns it signs the player out, then reveals the new one
- Cancel backs out of the regenerate-password confirmation without generating one
- Done clears the new password rather than leaving it on screen

src/features/players/player-list.test.tsx — 16/16

- renders the players the API returns
- flags a player whose transfer code is still out
- the pending-transfer flag is part of the row, not a control of its own
- requests the roster from the API's own origin
- groups players under their surname initial
- filters the roster by the search term
- reports a failed load instead of rendering an empty roster
- prompts to sign in again on a 401, instead of the generic error
- tapping a player hands it to the caller instead of navigating
- shows a real attention flag instead of the form figure when one fires
- the pending-transfer flag sits beside the name, not as a control of its own
- each card offers a link to view stats and a separate link to start a match
- tapping the header navigates to the player's profile and records last-viewed
- tapping start match reaches new-match without touching the profile route
- a player with no logged matches shows no metrics section
- shows a real attention flag, and renders the metrics section once matches exist

src/features/players/player-profile.test.tsx — 21/21

- renders the player's name and avatar
- says not found for a player id that isn't the coach's
- sends the coach back to sign in on a 401, not to the roster
- falls back to a generic message for an unexpected failure
- announces a load failure as an alert, for every branch
- links back to home
- links to the match setup screen
- shows the all-time record and player details side by side
- the Notes button opens the notes sheet
- opens on the Serve tab by default, with the scope panel and recent form above it
- switches to Rally when its toggle is clicked, scope panel and recent form stay put
- Edit opens the sheet pre-filled with the player's current values
- saving Edit PATCHes only the field that changed
- saving Edit with nothing changed just closes, no request sent
- Delete player asks for confirmation before sending anything
- Cancel on the delete step returns to the form without deleting
- confirming Delete removes the player and returns to the roster
- a player already flagged as pending transfer skips straight to the explanation, no request sent
- a code created since the page loaded is caught by the request's own 409, same explanation
- an unexpected delete failure shows the server's own message
- tapping the header photo opens the same edit sheet

src/features/players/players-page.test.tsx — 12/12

- renders the page heading and the roster
- tapping a player offers viewing them or starting a match
- choosing View player opens the profile and records them as last-viewed
- filters the roster as the coach searches
- renders the bottom nav with Players active
- the labelled row opens the redeem-transfer-code sheet
- the floating action adds a player rather than starting a match
- submitting the add-player form posts every required field

- refuses to submit until date of birth and gender are given
- rejects a date of birth in the future before sending it
- each player's card offers a link to view stats and a separate link to start a match
- choosing View stats opens the profile and records them as last-viewed

src/features/players/redeem-transfer-code-sheet.test.tsx — 6/6

- redeeming a valid code shows the transferred player and links to their profile
- shows the server's message when the code is unknown or already used
- shows the server's message when the player is already on this coach's roster
- a 429 shows a retry countdown
- Done dismisses the sheet from the success step, without visiting the player
- reopening after a refused code starts from a clean form

src/features/profile/profile.test.tsx — 12/12

- renders the coach's name and email
- shows initials when no photo is set
- links back to home
- editing the name PATCHes only display_name and updates live
- saving with nothing changed closes without a request
- choosing a photo uploads it and shows it in place of initials
- Remove photo is offered only once one is set, and clears it
- no Remove photo option when there is no photo to remove
- a failed photo upload says so, in the server's own words
- a failed photo removal leaves the photo and explains itself
- a refused name change keeps the sheet open with the typed name
- dismissing the edit sheet discards the typed name

src/features/pwa/install-guide.test.tsx — 7/7

- shows iOS steps on an iPhone
- shows Android steps on an Android phone
- shows a generic note on a device that isn't iOS or Android
- shows a confirmation instead of steps once already installed
- always shows what happens if signal is lost
- links back to settings
- links back to the player's own settings when given that backTo

src/features/settings/settings.test.tsx — 6/6

- links back to home
- the Match Defaults row links to its category page
- the Account & Security row links to its category page
- the App row links to its category page
- the Support row links to its category page
- offers Sign Out at the root, for fast single-tap access

src/features/stats/stats-page.test.tsx — 8/8

- shows the squad headline with an icon per stat
- leaderboard defaults to Record order, sub-threshold player dimmed at the bottom
- switching the leaderboard metric reorders the ranked rows
- search filters the leaderboard
- the trend card's metric selection is independent of the leaderboard's
- shows the trend caption
- shows the squad headline with an icon per stat, and the period's match count
- shows an empty state instead of a blank chart when every week is null

src/lib/date-of-birth.test.ts — 6/6

- accepts an ordinary date of birth
- requires a value
- rejects a future date
- accepts today
- rejects more than 120 years ago
- holds the boundary in a timezone ahead of UTC

src/lib/device-platform.test.ts — 9/9

- detects an iPhone from its user agent
- detects an iPad from its user agent
- detects an iPad browsing in desktop mode by its touch support
- does not mistake an actual Mac (no touch) for an iPad
- detects Android
- falls back to other for a desktop browser
- isRunningInstalled is false when neither platform check matches
- isRunningInstalled is true when display-mode is standalone

- isRunningInstalled is true on iOS Safari's navigator.standalone

src/lib/downscale-image.test.ts — 4/4

- returns a smaller file when encoding succeeds at the preferred type
- falls back to JPEG when the canvas cannot encode WebP
- keeps the original file when the canvas cannot get a 2d context
- keeps the original file when the re-encode is not actually smaller

src/lib/last-viewed-player.test.ts — 4/4

- returns null when nothing has been viewed yet
- returns the id most recently set
- records when the player was set as last-viewed
- a pre-timestamp plain-string value reads as null rather than throwing

src/lib/player-form.test.ts — 8/8

- playerFormValuesFrom maps every field, falling back for nullable ones
- validatePlayerForm requires a name, a valid date of birth, and a gender
- buildPlayerUpdateBody is empty when nothing changed
- buildPlayerUpdateBody includes only the fields that changed
- buildPlayerUpdateBody trims names and notes before comparing and sending
- buildPlayerUpdateBody sends null for notes cleared back to empty
- buildPlayerUpdateBody can include several changed fields at once
- buildPlayerUpdateBody sends null for dominant hand cleared back to unpicked

src/lib/pwa-install-prompt.test.ts — 2/2

- has not been dismissed until dismissInstallPrompt is called
- stays dismissed once set

src/route-tree.test.tsx — 29/29

- /players renders the roster
- /players/$playerId renders that player's profile, not the roster
- redirects to sign-in on a confirmed 401, instead of rendering the protected route
- renders the protected route through a non-401 session-check failure
- renders the protected route when the session check is rate-limited
- renders a protected route from the persisted cache with no connection
- a confirmed 401 leaves no persisted snapshot on the device

- a confirmed session claims a device nothing has claimed yet
- a confirmed 401 keeps the points that never synced
- / redirects straight to /home with a live session, rather than showing the form
- / shows the sign-in form with no session
- / goes to /home for a recorded owner with no connection, not the form
- / shows the form, not /home, with no connection and no owner recorded
- / shows the form on a confirmed 401, even with an owner recorded from before
- /me renders the player's own view
- redirects a player session to sign-in on a confirmed 401
- / redirects straight to /home with a live coach session, rather than showing the form
- / redirects straight to /me with a live player session, rather than showing the form
- / shows the sign-in form with no session at all
- / shows the form on a confirmed 401 from both checks, even with an owner recorded from before
- / goes to /home for a recorded coach with no connection, not the form
- / goes to /me for a recorded player with no connection, not the form
- / shows the form, not the app, with no connection and no owner recorded
- /me/settings renders as its own page, not the Me screen
- /me/matches renders as its own page, not the Me screen
- /me/install renders as its own page, not the Me screen
- /privacy renders with no signed-in session
- /terms renders with no signed-in session
- /settings/faq renders as its own page, not nested under Support

src/lib/analytics/attention.test.ts — 18/18

- Tier 1 states a player has never logged a match, distinctly from stale data
- Tier 2 can still fire just after the calendar year turns over (#238)
- rankForm ranks by this period's win rate, not the forced:unforced ratio
- rankForm passes through a null forced:unforced ratio rather than crashing
- Tier 1 no longer hard-flags on elapsed days alone, even well past two weeks
- Tier 1 states a player has never logged a match
- no flag when nothing crosses either tier
- Tier 1 fires on a losing streak at the threshold
- a losing streak below the threshold does not fire
- Tier 2 requires the minimum severity before it can fire
- Tier 2 fires on a genuine decline vs. the player's own baseline, in points for a percent stat
- Tier 2 formats a ratio stat's decline as a decimal, not points
- Tier 2 picks the single worst-declining stat, not the first one listed
- a stat with zero spread can't be deviated from, and is skipped
- Tier 1 always outranks Tier 2 across the squad
- within Tier 2, the larger |z| wins
- ties break by recency, most recent match wins
- nobody flagged is a legitimate result, not a fallback to the least-bad player

src/lib/analytics/squad-stats.test.ts — 15/15

- metricDisplay shows an em dash for a metric that isn't computable yet

- squadHeadline sums wins/losses, and weighs each player's rate by their own match count
- squadHeadline excludes a null rate from the average rather than counting it as zero
- squadHeadline returns a zeroed record and null rates for an empty squad
- pickBestWorst returns the two ends of the entries, ranked by value
- pickBestWorst excludes entries with a null value entirely
- pickBestWorst returns both null when nothing is eligible
- pickBestWorst never duplicates a single eligible entry as both best and worst
- pickBestWorst never duplicates a tied group as both best and worst
- periodWinPercent reads wins out of a player's period record
- periodWinPercent is null for a 0-0 record, not a fabricated 0%
- periodWinPercent is 100 for an all-win record, 0 for an all-loss one
- periodResiliencePercent reads a member's own period error counts as a forced share of total
- periodResiliencePercent is null when neither count has happened yet
- periodResiliencePercent is 100 when every own error was forced, 0 when every one was unforced

src/features/players/player-record-card.test.tsx — 3/3

- shows recent form in the same card
- recent form isn't yet scoped by the chosen opponent
- hides recent form for a player with no matches at all

src/features/auth/sign-in.test.tsx — 13/13

- renders the identifier and password fields
- navigates to /home on a successful coach sign-in
- sends the credentials to /api/auth/login when the identifier has an @
- shows the server's error message on a failed coach sign-in
- falls back to an explicit message when the coach response has no detail to show
- wipes what the previous owner left when a different coach signs in
- refuses a coach 200 that names no coach rather than signing in as nobody
- navigates to /me on a successful player sign-in
- sends the credentials to /api/auth/player/login when the identifier has no @
- shows the server's error message on a failed player sign-in
- falls back to a username-specific message when the player response has no detail
- refuses a player 200 that names no player rather than signing in as nobody
- wipes what a previous coach left when a player signs in on the same device

src/features/matches/player-match-list.test.tsx — 5/5

- renders the player's own matches, with no Edit or Delete controls
- shows an empty state with no matches yet
- reports a failed load instead of rendering an empty list
- renders the player's own matches, with no actions menu
- tapping a match row navigates to the player's own stats route, not the coach one

src/features/players/player-details-card.test.tsx — 1/1

- shows a player's own details with no Edit button and no Notes row

src/features/players/player-self-view.test.tsx — 4/4

- renders the player's own name and avatar
- shows the record and details cards, with no Edit button
- has no Start match button, no Notes button, and no coach-only affordance
- opens on the Serve tab, and switches to Rally

src/features/players/player-settings.test.tsx — 8/8

- renders the change-password fields and the sign-out action
- blocks submit until the new password and its confirmation match
- shows a mismatch error and blocks submit when the confirmation differs
- changes the password and shows a success message
- shows the server's message when the current password is wrong
- shows the server's message when the new password fails its policy
- signs out through playerLogout, then lands on sign-in
- links to the player's own install guide route

src/lib/local-data.test.ts — 5/5

- no owner kind is recorded until one signs in or a guard confirms one
- records the owner's kind alongside its id on sign-in
- records the owner's kind on a route guard's first confirmed session, same as the id
- never overwrites a recorded kind any more than it overwrites the id
- purging forgets the owner's kind along with everything else

src/components/authenticated-layout.test.tsx — 3/3

- renders the persistent mark above whatever the route renders
- the mark links to /home
- the mark links to /me

src/features/matches/match-stats-page.test.tsx — 2/2

- includes the match's event in the subtitle when one was recorded
- shows no stray separator in the subtitle when a match has no event

src/features/settings/settings-app.test.tsx — 2/2

- has no Notifications row
- has no Appearance row

src/lib/theme.test.ts — 1/1

- resolves light even with a stale stored 'dark' value

src/features/legal/legal-page.test.tsx — 3/3

- shows the title and body
- Back pops real history back to wherever the coach came from
- Back falls back to sign-in for a direct link with nothing to pop

src/features/legal/privacy-policy.test.tsx — 1/1

- renders a placeholder, not the final policy text

src/features/legal/terms-of-service.test.tsx — 1/1

- renders a placeholder, not the final terms text

src/features/pwa/pwa-config.test.ts — 4/4

- prompts for an update instead of reloading the page itself
- keeps the API out of the offline navigation fallback
- ships an icon file at every size the manifest declares
- keeps the masked icon opaque and the full-bleed ones transparent

src/features/pwa/update-prompt.test.tsx — 4/4

- reloads only when the user asks for it
- asks to be dismissed without reloading
- takes focus as a labelled dialog
- hands focus back to the screen behind it when the user dismisses it

src/features/settings/settings-faq.test.tsx — 2/2

- answers the questions this app's own design raises
- links back to support

src/lib/relative-time.test.ts — 7/7

- under a minute reads as just now
- minutes ago
- just under an hour still reads in minutes
- hours ago
- just under a day still reads in hours
- days ago
- a future timestamp doesn't go negative

Backlog item 9

Test Results Summary

API (pytest): 76/76 passed across 1 file

API — pytest (76/76)

tests/scripts/test_import_sample_data.py — 76/76

- test_blank_to_none_strips_and_empties (4 cases)
- test_parse_date_reads_an_iso_date
- test_parse_date_treats_a_blank_cell_as_unset
- test_parse_date_refuses_a_malformed_date
- test_parse_bool_reads_every_spelling_of_true [true, TRUE, True, 1, yes, YES]
- test_parse_bool_reads_every_spelling_of_false [false, FALSE, False, 0, no, NO]
- test_parse_bool_refuses_anything_else (5 cases)
- test_a_coach_row_becomes_a_coach
- test_a_coach_email_joins_case_insensitively
- test_a_player_row_carries_its_optional_columns
- test_a_players_blank_columns_land_as_null
- test_a_player_gets_the_username_the_route_would_mint
- test_players_whose_names_fold_alike_get_distinct_usernames
- test_a_match_row_links_to_its_player_and_carries_its_columns
- test_a_matchs_blank_columns_land_as_null
- test_a_match_row_with_no_played_on_names_its_line
- test_a_match_row_with_a_malformed_date_names_its_line
- test_a_match_row_with_an_unknown_enum_value_names_its_line
- test_a_match_row_naming_an_unknown_player_names_its_line
- test_a_match_with_no_match_key_still_imports
- test_a_point_row_becomes_a_point_on_its_match
- test_a_point_row_carries_the_columns_the_other_shape_leaves_blank
- test_a_point_row_breaking_a_schema_rule_names_its_line
- test_a_point_row_with_an_unreadable_boolean_names_its_line
- test_a_point_row_with_an_unknown_match_key_names_its_line
- test_match_sets_are_recomputed_only_for_matches_with_points
- test_an_import_with_no_points_file_writes_no_points
- test_a_points_file_with_only_a_header_writes_no_points
- test_the_summary_counts_what_it_wrote
- test_the_same_csvs_twice_write_one_copy_of_everything
- test_a_re_run_keeps_each_players_username
- test_a_re_run_leaves_a_stored_point_untouched
- test_a_point_whose_sequence_is_taken_is_skipped_not_rewritten
- test_a_re_run_adds_only_the_rows_the_csvs_gained
- test_a_re_run_refreshes_a_corrected_player_and_match
- test_a_re_run_rehashes_the_coaches_it_reused
- test_two_matches_one_day_apart_only_by_opponent_stay_two

- test_a_re_run_restates_the_set_scores_of_a_match_that_gained_points
- test_a_repeated_row_inside_one_points_file_lands_once
- test_more_points_than_one_statement_can_carry_still_land
- test_main_imports_from_the_csv_dir_and_prints_the_password
- test_main_says_nothing_about_points_when_there_is_no_points_file
- test_a_photo_cell_stores_the_file_under_a_key
- test_a_photo_that_is_not_an_image_fails_the_import
- test_a_re_run_keeps_the_photo_already_stored
- test_a_point_row_can_leave_the_serve_and_the_ending_unwatched
- test_a_checkpoint_row_becomes_a_checkpoint_on_its_match
- test_a_checkpoint_is_the_base_the_points_after_it_add_to
- test_a_checkpoint_row_breaking_the_snapshot_shape_names_its_line
- test_a_checkpoint_row_with_an_unknown_match_key_names_its_line
- test_a_re_run_leaves_a_stored_checkpoint_untouched
- test_a_note_row_becomes_a_note_on_its_player
- test_a_note_with_no_focus_areas_lands_with_none
- test_a_note_row_with_an_unknown_focus_area_names_its_line
- test_a_note_dated_in_the_future_names_its_line
- test_a_note_row_naming_an_unknown_player_names_its_line
- test_a_re_run_refreshes_a_notes_focus_areas_rather_than_duplicating_it
- test_a_demo_password_is_words_a_reader_can_relay
- test_demo_passwords_differ_between_runs